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Svetlana Masgutova Educational Institute®for
Neuro-Sensory-Motor and Reflex Integration, LLC

MNRI® NeuroStructural Reflex Integration



S. Masgutova, Ph.D.
D. Masgutov, MA.

Orlando 2012

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The authors of this program use the Masgutova Method® concept of reflex integration. The key elements of this concept are based on the understanding of: a reflex as the unit of the nervous system functioning and as a unit of the protective strategy of the brain in stress and danger (I. Pavlov, I. Setchenov, C. Sherington, A. Ukhtomsky, N. Bernstein, P. Anokhin, L. Vygotsky), the functioning of the brain on level of integrative processes and mechanisms (A. Ukhtomsky, A. Luria, P. Anokhin, H. Selye, A. Asratian), the essential role of the primary motor system maturation for motor control development (N. Bernstein, F. Lesgaft); the importance of movement-based learning for non-verbal and cognitive experience (P. Dennison, G. Dennison); the concept of the 3-D body structure (its axis and planes), its motor coordination and movement freedom point with correlated brain functioning (R. Sperry, R. Ornstein, N. Bernstein, J. Thie, P. Dennison, G. Dennison), the concept of muscle-tendinous system correspondence to brain organization (N. Bernstein, J. Thie), neurostructural basis of reflex patterns integration (S. Masgutova, D. Masgutov).

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- TSI NY – Therapeutic Services Inc, NY (2013)
- NCBTMB – National Certication Board for Therapeutic Massage and Bodywork (2010)
- PTSD ISTSS – Membership in International Society for Traumatic Stress Study
- Florida Board of Speech-Language Pathology and Audiology
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manual® Yanuar Adi Sanjaya 10252026:12227 Neurosensorimotor Integration

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– our greatest teachers – the children and adults – who have taken part in this program of NeuroStructural Reflex Integration and Repatterning. We are grateful to them for being open to receive the Program and to positive changes.

– our own children who inspired us for creativity and finding the new ideas on reflex integration. We are grateful for their love, support, and inspiration.

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ABOUT THE AUTHORS


Svetlana Masgutova, Ph.D

Dr. Svetlana Masgutova received her Doctorate in Developmental and Educational Psychology in 1988 and Associate Professor degree in 1992, in Russia. She worked as a lecturer at various universities and as a scientific researcher at the Russian Educational Academy. Dr. Masgutova has completed three years of medical studies and massage therapy school (Russia). She also received a post-graduate degree in Clinical Neuro-Speech Development at the Medical Academy (Poland).

She also was a lecturer on the Early Intervention Faculty of the Wroclaw Medical University (Poland), Director of the International Dr. Svetlana Masgutova Institute for Reflex Integration, and Director of the Svetlana Masgutova Educational Institute® for Neuro-Sensory-Motor and Reflex Integration, LLC. Dr. Masgutova has authored over 170 published works on psychology, education, Neurosensorimotor Reflex Integration, and sensory-motor based development.

Dr. Masgutova has been leading research since 1989 for the MNRI® Institutes, and has studied the influence of primary sensory-motor patterns on different aspects of neuro-development and learning. Her work focuses on the concepts of Reflex Integration processes to facilitate sensory processing, motor-physical and sensory-motor rehabilitation, emotional recovery, PTSD, as well as learning and developmental enrichment.

Dr. Masgutova has been teaching her MNRI® programs in USA, Poland, Canada, Russia, Germany, France, Netherlands, Austria, Australia, Sweden, Switzerland, Singapore, Belgium, Hong-Kong, Argentina, Denmark, Indonesia, Slovenia, Vietnam, and Norway.

She is the organizer of over 250 International Neurosensorimotor Reflex Integration and Rehabilitation Family Conferences for children and adults with neurodeficits, post-traumatic stress experience, and learning challenges.


Denis Masgutov

Denis Masgutov, MA (1983-2016), was the cofounder of the International Dr. Svetlana Masgutova Institute in Poland. He served as an International Specialist of the Masgutova Method®; he received his MA degree in the Psychology of Special Education and General Psychology from the Moscow Educational State University of Russia, accomplished Post-Graduate studies at Krakow University (1½ years; Polish the language) and massage therapy school (Russia). Denis also has studied psychology for 4½ years at the Warsaw University of Poland. He had significant experience in working with adults and children with developmental challenges such as: cerebral palsy, motor development deficits, ADHD, ADD, autism/Aspergers syndrome, FAS, intellectual delay, aggressive behavior, and PTSD.

Denis was the organizer of the International Camps for children with neurodeficits in Poland and the PT and OT specialists' education program in MNRI® for camps and clinics. He taught MNRI® programs including *Dynamic and Postural Reflex Integration, Lifelong Reflex Integration, NeuroTactile Integration, Oral-Facial Reflex Integration, NeuroStructural Reflex Integration*, and

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ABOUT THE AUTHORS

Dysfunctional and Pathological Reflex Repatterning in Poland. Denis Masgutov was also a co-researcher during the conferences and workshops in Poland. He traveled extensively with Dr. Svetlana Masgutova in the United States, Canada, Singapore, Australia, Israel, Netherlands, France, and Indonesia.

Denis' primary interest focused on the psychological and motor aspects of the neurosensorimotor integration of primary motor and reflex patterns serving emotional and cognitive development. In Poland he used EMG/BTS equipment to obtain objective data on the correlation of reflex integration with neurostructural changes in over 300 children with CP and autism. Through this research he developed a deeper understanding of and many practical techniques for working with stress, the Withdrawal response, Crawling, Automatic Walking, Leg Cross Flexion-Extension, and Breathing Reflex patterns. Children and their parents loved Denis for his understanding, support, and enthusiasm. Professionals appreciated his bright personality, the depth of his knowledge, and his ability to facilitate development within a holistic context. Denis researched the exercises of this NeuroStructural Reflex Integration program in complete using the BTS/EMG Nortex.

Denis had an intuitive understanding of MNRI®, immersed himself in it, and was a constant innovator of the techniques – refining old ones and developing new ones. He developed the new program of NeuroTactile Integration (Level 2) that was later successfully taught by the MNRI® Instructors. His imprint on the development of the MNRI® System and MNRI® Instructors Education independently of his young age had a big impact on the MNRI® Program.

He is greatly missed by all who knew him.

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OBJECTIVES: Goals, Tasks, and Processes

The objectives for this course include to:

1. Describe the process of using core flexion-extension mechanisms to describe the protection (freezing response) and survival (fight and flight response) strategies of the brain.
2. Explore the use of these strategies to release non-productive protection tendencies which keep people from reaching their goals, potential, and from making productive decisions.
3. Describe the importance of optimizing the body's symmetry mechanisms affecting binocular vision and binaural hearing and to release past stress and traumas.
4. Describe the innate nature of the sensory-motor reflex system and general principals of the Masgutova Method ® for facilitation of changes in the physical, emotional, social, and cognitive functions of a person.
5. Describe the role of the Red Light Tendon Guard Reflex (H. Selye, 1974) and its effect on the abdominal, shoulder, and neck muscles when preparing the body to deal with either unexpected situations or a situation that requires intense focus or a quick reaction.
6. Explore the importance of a matured Red Light Tendon Guard Reflex for the process of being able to pause and calm oneself down vs becoming over focused on unimportant details and shutting down.
7. Describe the Green Light Tendon Guard Reflex and its effect on muscles around the spine which prepare the body for action.
8. Describe the importance of a matured Green Light Tendon Guard for a person to see life's big picture in order to learn and advance vs becoming hyperactive, acting without forethought, and misreading situations resulting in 'fight or flight' reactions.
9. Describe how body symmetry and gravity line are used to determine if the Core Tendon Guard is matured and appropriately integrated, or if it is dysfunctional which may lead to inappropriate responses to everyday life challenges.
10. Explore techniques using neurosensorimotor biomechanical points to release tension in the Core Tendon Guard and activate the proprioceptive system to optimize regulation of a person's behavior, emotions, and self.
11. Describe how the activation of reflex patterns facilitate neurodevelopmental mechanisms for proper physiological functions which provide for easier learning for sensory-motor abilities and skills necessary for daily life functions.
12. Describe how the integration of the primary motor patterns support cognitive development.
13. Describe how improvement of the Core Tendon Guard response determined by activity of the brain stem aids in the development of appropriate and healthy protection.
14. Explore how the Tactile System with the activation of receptors of deep touch, pressure, and low vibration create the kinesthetic feeling and state of physical orientation within one's body and confidence in own sensory sensations and movements.
15. Describe how the correct regulation of specialized skin receptors increase tolerance for inadequate hypersensitivity to life's daily occurrences.

Explore how the activation of the proprioceptive system increases regulation of muscular-tendoneous tone which works with any negative somatic anchors caused by stress.

17. Describe how the diaphragm mobilization technique releases stress and improves

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breathing.

18. Explore how the activation of biomechanics of reflex patters offer correction of non- productive, dysfunctional, or pathological motor automaticity.
19. Describe how the activation of reflex patterns offer optimization of motor automaticity.
20. Describe how activation of reflex patterns offer support for the coordination of the neurostructural aspects of the body, its reactions and overall cognitive development.
21. Participate in direct hands-on instruction for the techniques of Tactile-Proprioceptive Greeting of the front of the body and core rectangle.
22. Participate in direct hands-on instruction for the techniques of the Pendulum and other sternum areas to optimize functions of breathing, vision, and auditory and articulation systems.
23. Participate in direct hands-on instruction for Fear Paralysis and Moro biomechanical points to organize the response of freezing and fight or flight; balance self-preservation instincts and activate self-regulation mechanisms in the brain stem and diencephalon (sub-cortical structures).
24. Participate in direct hands-on instruction for the activation of the lower limbs and feet to release negative Core Tendon Guard response, and activate the self-regulating responses of the brain stem and balancing and adjusting mechanisms of the cerebellum.
25. Participate in direct hands-on instruction for the activation of the back of the body for optimizing the functions of the Spinal Perez as a detoxifying mechanism, optimize the regulation of stress hormones of adrenalin and cortisol, and optimize the Spinal Galant Reflex pattern to increase Beta and Alpha waves in the brain wave spectrum.
26. Explore the MNRI ® NeuroStructural assessment.
27. Describe different techniques for creating an individual program for clients and the development of home practice activities.

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MNRI® IN RESEARCH

The Masgutova Method® approach is based on the results from research of the effect of the MNRI® exercises and techniques for motor-cognitive development of children with neurodeficits, neurodevelopmnetal and learning challenges and problems. Research studies, both internal and independently done, have been carried out by Dr. S. Masgutova and teams of scientific colleagues at different times since 1989 to present and in different countries.

The MNRI® program has been utilized by Dr. S. Masgutova and her international team for therapeutic work with more than 300,000 children (0-19 yrs) and 80,000 adults* with different diagnoses including: cerebral palsy, traumatic brain injury, ASD and Asperger's, ADD & ADHD, genetic disorders, Post-Traumatic Stress, birth stress, fetal alcohol syndrome, hypoxic-ischemic encephalopathy (HIE), non-fatal drowning, etc.

The MNRI® Reflex Integration Assessment was conducted for 49,000* individuals.

With children, for example, having a diagnosis of ASD, Dr. Masgutova and her Team have worked with an estimated 54,000* individuals aged 2-19 throughout 40 countries including the USA, Poland, the Netherlands, France, Belgium, Russia, Australia, Sweden, Slovenia, Indonesia, and Canada.

MNRI® Specialists have worked with over 6,000* newborns and infants in the above various countries.

Dr. Masgutova's Organizations has also funded scientific research on the effectiveness of MNRI® on different symptoms with over 6,929* children and adults based on metrics of: Reflex Assessment, neurophysiological markers, QEEG results, neurotransmitters analysis, ATEC-test and other PT, OT and psychological tests, ATEC and other (* - statistical data from 2005 to 2020).

For a current listing of scientific studies involving MNRI® please see our website, www.MasgutovaMethod.com

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FOUNDATIONS OF MNRI® NEUROSTRUCTURAL REFLEX INTEGRATION

MNRI® NeuroStructural Reflex Integration Program is based on the results of:

- Long-term clinical work with children and adults and observations of the results of this intervention method (1990-2011- Russia, Poland, USA, Canada, and other countries)
- Research using the BTS EMG tool – the highest quality computerized equipment for muscle tone regulation control with use of the (84 electrodes) by D. Masgutov (ISMI, Poland)
- Evoked brain potentials (in 2008-2011, Wroclaw Medical University of Poland)

This Program uses the concept of reflex pattern integration evolving the neuro-structuralaspect of our body-brain system, particularly the manual techniques working with neuromuscular-tendoneous organization of reflex patterns, such as: Core Tendon Guard, Moro, Fear Paralysis, Spinal Perez, Spinal Galant, Trunk Extension, Babinski, and Foot Tendon Guard; and includes diaphragm mobilizing activities. It addresses the 3-dimentional symmetry of the body, and the protective response of the whole body causing the withdrawal of the body center and the stress response of the limbs and neck area.

The MNRI® NeuroStructural Reflex Integration Program can be used for a variety of challenges that range from light to severe dysfunctions in:

- Motor development and sensory-motor integration (general and specific delays in motor development, dyspraxia, neurodeficits of motor system, CP, and other)
- Emotional development and behavior patterns (tendency for overprotection, emotional instability, poor emotional maturation, infantile behavior, ADHD, aggression, fears, and phobias)
- Cognitive development (ADD, dyslexia, dyscalculia, delay of language development, poor memory, and other)

This Program can be used in cases of normal development for prevention of challenges, optimizing developmental mechanisms, for support of the learning process, and creativity.

Goals of the MNRI® NeuroStructural Reflex Integration Program:

The Neurosensorimotor process involves work with the:

- Core flexion-extension mechanisms correlating with the protection (freezing response - flexion) and survival (fight and flight - extension) strategies of the brain
- Release of non-productive protection tendencies
- Optimization of the body symmetry mechanism
- Primary movements such as body righting (Stretch Reflex), homologous movements (to support bilateral reflex patterns – Hands Pulling, Hands Grasp), homolateral movements (to support reflex patterns activating one-sided motions), and cross-lateral movements (to support reflex patterns activating movements crossing the body midline)
- Activation of the reflex patterns for building the neurosensorimotor basis for motor programming and motor control
- Activation of reflex patterns for facilitating the neurodevelopmental mechanisms for proper use of physiological functions for easier and productive learning of motor skills
- Integration of the primary motor patterns for support of cognitive development
- Optimization of physical/kinesthetic aspects of

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bonding to one's own body

- Activation of awareness of body processes and feeling of safety for interaction with the world and for communication
- Improvement of the protection response and tendon guard reaction

The NeuroStructural Reflex Integration Program and other MNRI® Programs were used with over 35,000 individuals – children and adults, with neurodeficits and general delays in motor development and performance, and with overall learning challenges. Research protocols are complete for over 3,560 of individuals with neurodeficits – CP, Autism, ADHD, and Dyspraxia.

The preliminary studies using the scientific studies listed and carried out by the Masgutova Institute (Poland and USA) in 2008-2011 demonstrate positive results in children, namely seen by improvement of:

- Muscle tone regulation and in symmetry in tension of bilateral muscles (research using BTS and EMG)
- Functioning the biomechanics of motor performance and motor coordination (research using BTS and EMG, balance system)
- Postural control (research using EMG and balance)
- Functions of auditory and visual systems (evoked brain potentials, research of visual skills)
- Brain wave spectrum (EEGR, EEG)
- Reflex patterns functioning (Reflex Parameter Assessment by S. Masgutova based on statistic analysis by A. Krefft – "NewKrefft" system)

Clinical observations show that these children have demonstrated progress in tactile and proprioceptive system functioning, which means improvement in the functioning of the sensory system and motor programming. Speech development improved: 7 children out of 10 nonverbal children started developing expressive language.

Parents, MNRI® Specialists, and others noticed an increase in children's feelings of safety and being bonded. Based on these reports the children improved the traits of matured behavior and communication.

The MNRI® Reflex patterns of Spinal Perez, Spinal Galant, Tonic Labyrinthine, Spinning, Trunk Extension, Leg Cross Flexion-Extension, Moro, and Fear Paralysis improved significantly.

Based on positive results of this research, these parents are requested to use this Program at home three times a week for CP and cases of strong deformities; two times for children with autism, and in other cases, one time a week. This program usually takes 40-45 minutes of work for parents. The use of the Core Tendon Guard techniques can be used individually from the NeuroStructural Reflex Integration session. It is recommended that MNRI® NeuroStructural Reflex Integration protocol be continued to:

- Activate the neuro-muscular-tendoneous system tone regulation to improve patterns of certain reflexes for optimal functioning of the early automatic motor system
- Optimize the work of the body flexors and extensors as the mechanisms correlating with strategies of protection (freezing response - flexion), and survival (fight and flight - extension)
- Release the non-productive protection tendencies at the somatic level, and to make new possibilities for experiencing positive protection, supplying the feelings of comfort and being bonded
- Activate functioning of the body symmetry mechanism
- Activate primary motor biomechanics, such as body righting, rotation in joints, homolateral and homologous movements
- Facilitate reflex patterns formation for building the neurosensorimotor basis for easier and more productive learning of motor skills
- Activate the links and integration processes for the primary motor patterns that support cognitive development
- Optimize the physical/kinesthetic aspect of bonding (to one's own body, and physical/emotional "Self") and awareness of body processes and feelings of safe interaction with the world and communication with others

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The main techniques of MNRI® NeuroStructural Reflex Integration are focused on the neurosomatic reflex pattern facilitation of the:

1. Tactile system aimed at activation of receptors of deep touch, and low vibration in order to create:

a) The kinesthetic feeling and state of physical orientation within own body

b) The basis for proper motor programming and control, coordination of body parts for postural organization and motor realization, goal oriented movements

c) The proper regulation of specialized skin receptors – to increase tolerance for inadequate hypersensitivity, and activate receptors that are hyposensitive

d) The feelings of comfort, emotional release, and support of the Bonding Reflex pattern

2. Proprioceptive system using deep, firm and zone (flat) touch, for activation of the receptors of deep touch and proprioceptors:

a) Muscle-tendoneous activation to increase regulation of tone control in muscles and tendons that participate in reflex pattern presentation (work with negative somatic anchors caused by stress)

b) Work with intermediate space between muscle fibres and tendons for release of nonproductive protection on somatic level

3. Mobilization of the diaphragm for:

a) Stress release and improvement of breathing

b) Coordination of the work of short postural muscles (postural triangle: diaphragm –> psoas muscles –> abdominal rectus muscles)

4. Activation of biomechanics of reflex patterns orienting at correction of non-productive, dysfunctional or pathological motor automatism:

a) Flexion-extension mechanisms activation

b) Lengthening-shortening mechanisms activation

c) Increase in rotational abilities

d) Activation of homologeous and homolateral motor patterns

e) Activation of cross-lateral motor patterns

5. Activation of reflex patterns for:

a) Optimization of motor automatisms (prevention work)

b) Correction of dysfunctional and pathological patterns

c) Development, maturation, and integration with controlled movements and motor skills

d) Support of coordination of neuro-structural aspects of the body, its reactions, and cognitive functions

6. Peripheral nerve system activation to support all other systems of the body and limbs:

a) Circulation

b) Lymphatic and other

c) Sensitivity and motor mobility

Application of main techniques:

1. Tactile-proprioceptive stroking along the limbs and their segments, and whole body

2. Circular activation in intermediate spaces of the core and limbs

3. Stretch across the lines of connection in intermediate spaces/lines of the core and limbs

4. Activation of Golgi receptors in the intermediate space between the muscle fibers and tendons (below the origin and above the insertion) where there are fewer receptors, thus, protection/overprotection is less

5. Peripheral nerve system activation through stimulation of the nerve net system of certain muscles:

a) Fascia/muscle across/perpendicular stretch (muscle away from bone) for activation of peripheral nerve system, and activation of muscle-bone connections

b) Quick release/pulsating stimulation of the areas along the nerve pathway (sciatic nerve)

6. Fascia activation regulation techniques for opti-

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mizing the links between neuromuscular structure and the reflex patterns functioning of:

a) Fascia/muscle compression for tone regulation and stretch effect (automatic flexibility)

b) Fascia/muscle lengthening for tone regulation and contraction effect (automatic flexibility)

7. Activation of gross motor system for the release of automatisms and freedom for development of specialized motor-cognitive functions

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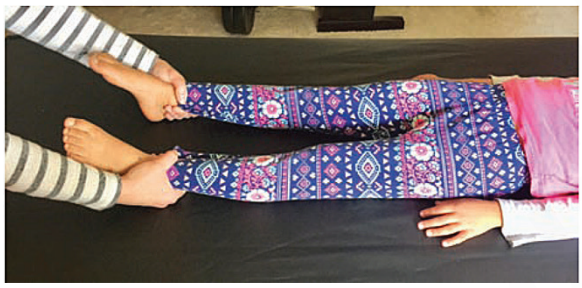
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MNRI® - MASGUTOVA TENDON GUARD REFLEX REGULATION TECHNIQUE

The purpose is to:

- Activate the neuro-structural links and mechanisms of core flexors and trunk extensors for optimal motor functioning and development
- Activate the neuro-structural aspect of Tendon Guard and symmetry
- Activate the work of the flexors and extensors for optimal protection and survival
- Coordinate the work of flexors for antigravity and grounding
- Coordinate the work of extensors for antigravity and balancing
- Activate the peripheral nerve system for Tendon Guard regulation and protection/ survival strategy of the brain stem
- Revive the links between the work of flexors and visual-auditory reflex patterns
- Revive the links between the work of extensors and visual-auditory reflex patterns
- Release the muscular-tendinous hyper-tension in the body as the result of improper functioning of the Tendon Guard Reflex
- Activate the proprioceptive system for optimizing the self-regulation mechanism of postural control
- Activate the work of the receptors of deep touch and pressure for optimal work of protection, survival and Bonding Reflex pattern.





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TENDON GUARD REFLEX PATTERN, SYMMETRY, AND THE BODY'S RESPONSE TO STRESS

Trigger of the Tendon Guard Reflex Pattern – check for protection and symmetry

Position of the client: Person is lying on their back (prone).

Action: Stretch the limbs bilaterally to find out how the person is reacting to interference of the stretch mechanism of the muscular-tendinous system. Hold the extremities above the joints firmly and do traction gradually "inviting for stretching":

- Upper limbs
- Lower limbs



The bilateral traction of the limbs shows the tendency of shortening of the limbs on the right side.

1. Find out if limbs were stretching easily with no contraction/tendency for protection. Note if the right or left limb demonstrated a protective response.

2. While doing the traction of the limbs, stop at the moment when the opposing response/refusal for stretch is felt. Next compare also the length of the limbs: for upper – at the wrist joint or thumbs and for lower – at the end of heel bone or ankle joints. The shortened side usually stands for over-reactive contraction of the muscle and tendons and overprotection.

3. Do the traction of the limbs twice to determine, if the same protective response is triggered or not (response is stable/not stable).

Notice improper asymmetrical contractions of the arm or leg. The response by contraction indicates the presence of the hyperactive Tendon Guard Reflex pattern.

Discuss the results of bilateral traction of the limbs. The possible responses are:

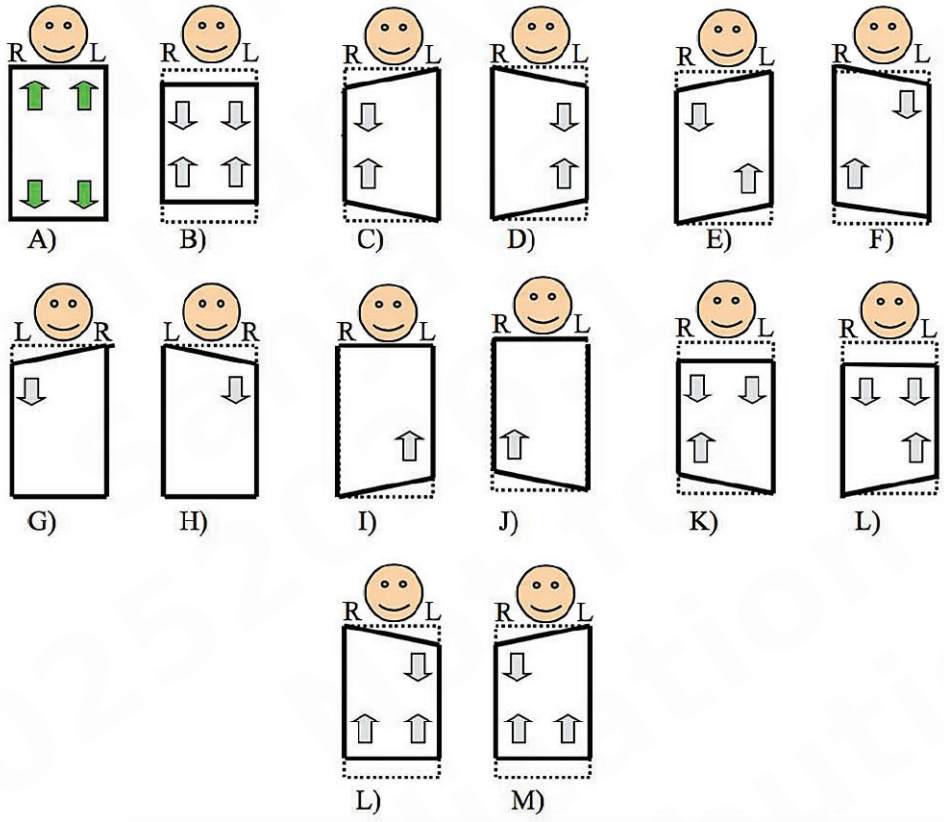
- Proper – no overprotection, length of the limbs is even
- Upper limbs – right or left - shows overprotection and response is shortening for stretch
- Lower limbs – right or left - shows overprotection and response is shortening for stretch

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TENDON GUARD REFLEX

Variations of the Tendon Guard responses are:

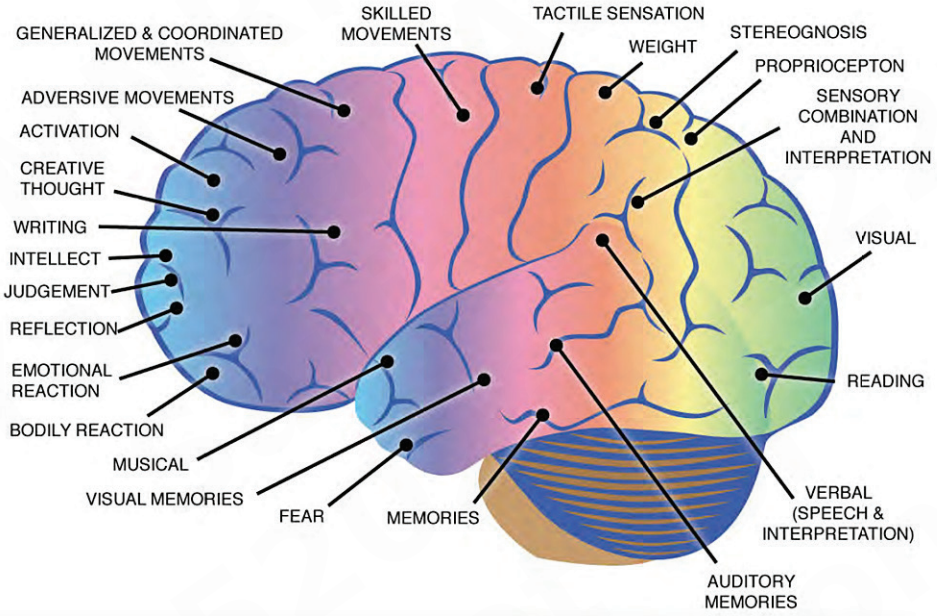


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TENDON GUARD REFLEX

Brain Area Functions



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Possible Correlation of the Tendon Guard Response and Pre- and Post-Motor Cortex Functions

	Right Side ⇨ Left Hemisphere	Left Side ⇨ Right Hemisphere
Upper limbs ⇨ Premotor cortex	1. Conscious control of skills and coordination systems 2. Fine motor coordination for manual skills 3. Manual fine motor programming 4. Hand ↔ eye coordination 5. Hands/fingers ↔ tongue coordination 6. Logic: analytical, linear 7. Operative and short-term memory 8. Expressive language 9. Articulation control 10. Conscious space-time orientation	1. Automatic and reflexive response 2. Medium and gross motor coordination for manual skills 3. Medium and gross manual motor programming 4. Hand ↔ eye reflexive responses 5. Arms ↔ jaw (TMJ) coordination 6. Logic: gestalt, whole, structure 7. Long-term memory 8. Receptive language 9. Symbolic language 10. Automatic space-time orientation 11. Spontaneous regulation of feelings and emotions
Lower limbs ⇨ Post-motor cortex	1. Conscious control of gross motor coordination 2. Conscious control of motor multitasking 3. Conscious gross motor coordination of lower limbs 4. Conscious gross motor programming 5. Core ↔ upper-lower limb conscious control 6. Core ↔ upper-lower limb conscious coordination 7. Conscious control of motor action in closer space (short distance) 8. Operative and short-term gross motor memory 9. Body-space orientation control	1. Automatic sensory-motor integration mechanism 2. Automatic/spontaneous and reflexive responses 3. Automatic/spontaneous gross motor coordination 4. Automatic/spontaneous gross motor programming 5. Core ↔ limbs automatic/spontaneous coordination 6. Reflexive/impulsive action or reactive fight-flight 7. Automatic control of motor action in larger space (longer distance) 8. Long-term gross motor memory 9. Body-space orientation spontaneous, non-conscious

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TENDON GUARD REFLEX

General Guidelines for MNRI® NeuroStructural Core Tendon Guard

1. Start at upper corners of the rectangle – at the more relaxed shoulder area (the longer arm in the traction), then work on other side shoulder/arm.
2. Next continue at lower corners of the rectangle - hip area (the leg that stretches longest), then work at other side hip area.
3. Work with one corner of the rectangle at a time in order not to inhibit the body structure and its dynamics.
4. Work with right and left shoulders, and right and left hips.
5. When you work in a team – one specialist works with the shoulder area and the other with hip area on opposite side of the body.
6. Work deeply, and be aware not to evoke a physiological defense response. (All procedures must create a feeling of safety, relaxation, and pleasure.) MNRI® procedures are directed to deep proprioceptors and do not allow pain.

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GENERAL GUIDELINES FOR MNRI® NEUROSTRUCTURAL

PROCEDURAL STEPS

1. Tactile-Proprioceptive Greeting for the Front of the Body

Procedure

Position of the client: Lying down supine (on back). Client can be sitting/half-sitting.

Technique: Deep tactile-proprioceptive stroking along the limbs and front (anterior plane of the body) from the chest down the arms and legs - distally.

1) Upper limbs

a) Stroking along the arms. Put both your hands symmetrically at the area of the connection of the pectoralis major and sternum clavicular (midway point), press firmly for 2-3 seconds with flat hands (360–800gr/sm^2) to stimulate proprioceptive awareness and giving the feeling of comfort. Next move along both arms distally stroking them simultaneously with the same amount of pressure one time.

b) Complete with arms traction. When reaching the wrists complete stroking with light traction of the arms.

c) Pressure on joints ('koala' and 'leverage'). While repeating the second and third time, press with a pleasant and embracing touch ('koala embrace') on the joints (shoulder, elbows, wrists) for 5-7 seconds. Finish with an accent applying it for 2-3 seconds or longer on sides of the limbs following the 'Leverage' manner: while doing an embrace of a joint press a bit more with your fingers at the same time – thumb moving a bit to upward direction (on insertion of main muscle of the limb segment above the joint) and other fingers to downward direction 'leveraging' with the origin of muscle below the joint. Particularly do leverage accenting for:

A) shoulder joint (articulatio humeri; the deltoid m. is covered) – thumb pushes in area of insertion of anterior part of deltoid muscle moving slightly upwards, and other fingers press in area of posterior part of deltoid muscle closer to insertion of *supraspinatus m.* (and also origin of *triceps m.*) and move downwards;

B) elbow joint *(articulatio cubiti)* – thumb pushes in area of origin of *brachioradialis m.* moving slightly upwards, and other fingers press in area of origin of flexor *carpi ulnaris m.* and move downwards;

C) wrist joint *(articulatio radiocarpea)* – thumb pushes in area of insertion of slightly upwards, and other fingers press in area of insertion of *brachioradialis m.* and insertion of flexor *carpi ulnaris m.* and move moving downwards.

Repeat step 1-b) 'Completing by arms traction' each time.

2) Trunk and lower limbs

a) Stroking along the trunk and legs. Continue from the *pectoralis major clavicular,* move along the trunk and then both legs distally stroking them simultaneously with firm pressure (360–800gr/sm^2) one time.

b) Completing by legs traction. When reaching the ankles complete stroking with light traction of the legs.

c) Pressure on joints. While repeating the second and third time, press with a pleasant and embracing touch ('Koala Embrace') on the joints (hip, knees, ankles) during 5-7 seconds. Finish

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by an accent applying it during 2-3 seconds or longer on sides of the limbs following the 'Leverage' manner: while doing an embrace of a joint press a bit more with your fingers at the same time – thumb moving a bit to upward direction (on insertion of main muscle of the limb segment above the joint) and other fingers to downward direction 'leveraging' with the origin of muscle below the joint. Particularly do leverage accenting for:

A) Hip joint (*articulatio coxae; the tensor facia lata m.* is covered): thumb pushes in area of front aspect *tensor facia lata m.* moving upwards, and other fingers press on back aspect of *tensor facia lata m.* and move downwards;

B) Knee joint *(articulatio genus)* - thumb push in area of *vastus medialis* and *sartorius mm.* moving upwards, and other fingers press in area of origin of *peroneus m. anterior tibialis m.* and move downwards;

C) Ankle joint *(articulatio talocruralis)* for:

- Foot Tendon Guard response mobilization: thumb pushes in area of *tibialis anterior m. and extensor digitorum longus m.* moving upwards, and other fingers press in area *surae tendon Achillas* and move downwards;
- Babinski Reflex pattern mobilization: thumb pushes in area of *ligamentum collaterale mediale* moving upwards, and other fingers press in area of *ligamentum talofibulare anterius (closer to peroneous m. line)* and move downwards.

Repeat step 2-b) each time.

This technique allows the body to receive the next techniques for regulation of the muscular-tendinous tone in the upper and lower limbs and trunk for functioning of these reflex patterns: Trunk Extension, Spinal Galant, Spinal Perez, Segmental Rolling, Spinning, Automatic Gait, Leg Cross Flexion/Extension, and Crawling (lower part). This exercise also regulates the links and differentiation of the biomechanics of reflex patterns serving for flexion and extension, coordination of the trunk motions with upper and lower limb reflex patterns, and activates the peripheral nerve system of the posterior plane of the upper and lower parts of the body.

The joint mobilization techniques (koala and leverage) are aiming at the guard response regulation for muscles, tendons, and ligaments. The greeting-stroking techniques also serve to regulate the skin-muscle spindle cells guard response.

Position of specialist: Standing at the side close to the upper body where the technique application starts, and moving down the body when work is applied on the lower part of the body; facing the center of the client's body.

Position of the hand/palm: Palms and fingers of both hands are touching and stroking the anterior plane of the body (palms are open and fingers more together). For this exercise use the deep and slow pressure and pulling with fingers/palms (touch # 2, 3 and palms).

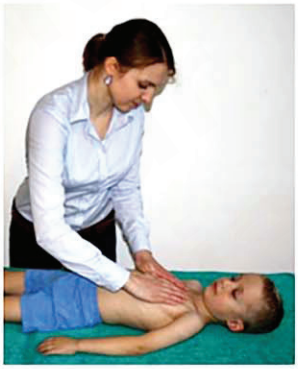
Team Work. When you work with a second person – NS therapists or a parent - these techniques can be done. In two ways: one after another (4 reps all together for upper limbs and 4 repetitions for core-lower limbs) or work on sides bilaterally 2 times and change sides and repeat the same.

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GREETING FRONT OF THE BODY

Chest and Upper Limbs

1A 

Pic. 1A) Light pressure on upper chest and move along shoulder.

1B 

Pic. 1B) Close view.

2A 

Pic. 2A) Continue to stroke along shoulder.

2B 

Pic. 2B) Close view.

3A 

Pic. 3A) Continue to stroke along elbows.

3B 

Pic. 3B) Close view.

4A 

Pic. 4A) Continue to stroke along wrists and do light traction.

4B 

Pic. 4B) Close view.

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Chest & Core

5A 

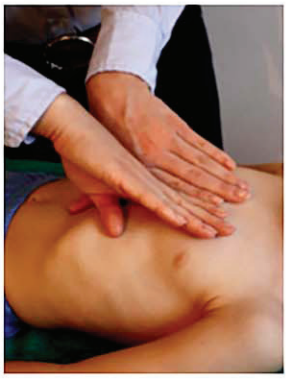
Pic. 5A) Light pressure on upper chest and move to sternum.

5B 

Pic. 5B) Close view.

6A 

Pic. 6A) Continue stroking along sternum.

6B 

Pic. 6B) Close view.

7A 

Pic. 7A) Light pressure on upper chest and move to sternum.

7B 

Pic. 7B) Close view.

8A 

Pic. 8A) Continue stroking along the rib cage line.

8B 

Pic. 8B) Close view.

9A 

Pic. 9A) Continue stroking along sides of the hips.

9B 

Pic. 9B) Close view.

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Lower Limbs

10A 

Pic. 10A) Continue stroking along sides of the upper legs up to 1/3 of the femor bone and change to anterior surface.

10B 

Pic. 10B) Close view.

11A 

Pic. 11A) Continue stroking along anterior surface of the upper legs and move to knees.

11B 

Pic. 11B) Close view.

12 

Pic. 12) Continue stroking along anterior surface of the ankle and move to its posterior plane.

13 

Pic. 13) Hold behind ankle joints and give light traction for 2-3 seconds.

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CORE RECTANGLE ACTIVATION

Goals

- Release the negative effect of the Core Tendon Guard (Red Light).
- Activate the regulation mechanism of muscle-tendon-ligaments tension in the core/limbs.
- Activate the dynamic of upper and lower body gates.
- Activate the dynamic and stability mechanisms of the core.
- Activate coordination between homologous, homolateral and cross-lateral limbs.
- Activate coordination of core-limb rotation.
- Activate self-regulating mechanisms of back brain as well as activate the processes and mechanisms of the cerebellum (because of its rhythmical movements).
- Increase nerve innervations and blood circulation.
- Activate lymph and oxygen supply.
- Activate symmetry mechanism and body midline functioning and core dynamics and stability.



Symmetry mechanisms with its links with the Tendon Guard Reflex (presented in Rectangle) are responsible for coordination of different systems— body structure, postural control, and movement. This brings the body structure and reflexes into integrated functioning. It helps to activate the self-regulating process for organization of the neurostructural aspect of reflex patterns functioning.

Upper Gate Area

Location: *Regio infraclavicularis*

Bone area: *Clavicula, I-III costae*

Muscles of the upper gate

- *Pectoralis Major Clavicular (m. pectoralis major pars clavicularis –> n. pectorales medialis et lateralis (CVII- Th1)*
- *Pectoralis Minor (m. pectoralis minor–>n.pectorales medialis et lateralis (CVII- Th1)*
- *Deltoid (m. deltoideus pars clavicularis –> n. axillaris)*
- *(m. subclavius –> n. subclavius (CV)*
- Also involves the *Pectoralis Major Sternal and Latissimus Dorsi muscles*

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FRONT: CORE TENDON GUARD

Lower Gate Area

Location: *Regio abdominalis, and Regio femoralis facies anterior et medialis*

Bone area: *os ilium*

Muscles of the lower gate (hip area)

- Abdominals - *(m. obliguus externus abdominis)*
- Abdominals - *(m. obliguus internus abdominis)*
- Abdominals - Transverse *(m. transversus abdominis)*
- Also involves these muscles:

– Psoas – *Illiopsoas*
– Iliacus – *Inguinal Ligament*
– Quadriceps – *Fascia Lata*

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Procedures

2.1. Rocking – Bouncing at Rectangle Corners

Position of the client: Lying supine (on back)

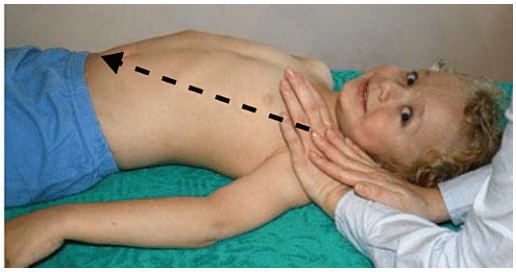
Technique: Pushing in a rhythmical motion at each corner of the rectangle of the body at:

A) Upper gate corners **B) Lower gate corners**

Repeat 5-7 times depending on therapeutic needs and client's preference. Stay in appropriate communication and distance to client.

Tasks:

- Activate the regulation mechanism of muscle tendon- ligaments tension in the core/limbs
- Organize the coordination in links: body center and midline –> joint of limbs
- Focus the activation of the whole body symmetry mechanism



A) Upper gate corners

Position of the hand/ palm: Middle of the palm of one hand is over shoulder joint and fingers and palm of other hand are resting on the top of initial hand.

Action: Push with a rhythmical motion slowly on the area of the shoulder joint with soft touch around the joint. Create bouncing effect by your pushing. Push the shoulder with your palm softly and deeply in 1-2-3 seconds, reaching the optimum of compression in the direction of the body's center and midline to opposite hip. Next, quickly release your palm/hands away using the quick release technique within one minute. Repeat 5-7 times for each upper corner of the rectangle.

Direction of applying the technique: From shoulder to the middle of the opposite hip joint/pelvic area.

B) Lower gate corners: Rocking





Position of the specialist: Stand at the lower corner of the rectangle of the body

Position of the hand/palm: Middle of the palm of one hand is over lateral plane of hip/pelvic area and palm/ fingers of other hand are placed on top of initial hand.

Action: Push with a rhythmical motion slowly on the area of the hip/pelvic area (higher than the joint – in the area of upper fascia lata) with soft cupping touch around the bone – create bouncing effect by your pushing. Push with your palm softly and deeply reaching the optimum of compression in the direction of the body center and midline to rib cage line between ribs 5-6.

Next, quickly release your palm/hands away using the quick release technique. Repeat 5-7 times for each lower corner of the rectangle.

Direction of applying the technique: From the middle of the upper hip/pelvic area (middle of upper part of *fascia lata tensor m.)* to middle of the lateral side of rib cage on the level of ribs 5-6.

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FRONT: CORE TENDON GUARD

**** Note - Contraindications:***

1. Physical conditions
2. Health conditions (broken bones, dislocated joints, new injuries, burned skin/tissue)
3. Over sensitivity to rhythmical motions
4. Condition of the vestibular system
5. Hyperactive physiological nystagmus, oculovestibular and oculokinetic responses
6. Pathological Core Tendon Guard response such as in quadropleagic cerebral palsy.
7. Refusals (dislike, motivational reasons) In all cases of contraindication not concerned with health (broken bones, dislocated joints, new injuries, burned skin/tissue), slow introduction of the techniques is suggested.

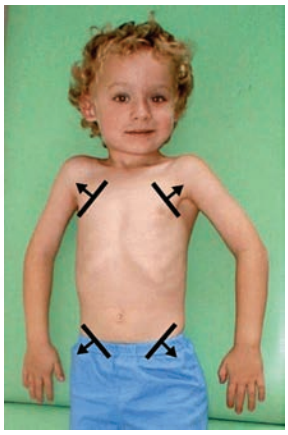
Note: When the technique is not accepted (because of physical/health conditions, oversensitive vestibular system, dislike, no place to lie down, inability to lie down still – hyperactivity) skip it, and continue with next step.

Team Work. When you work with a second person – NS therapists or a parent – these techniques can be done in a reciprocal/cross lateral way – one person does work pushing on one shoulder, and next other person works on the hip working sequentially, then they repeat the same on other opposite shoulder and hip.

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2. 2. Circular Activation at Lines of the Rectangle Corners



Position of the client: Lying down supine (on back). Client can be sitting/standing if more comfortable.

Technique: Tactile-proprioceptive activation in every corner of rectangle of the body using deep circular motions at:

A) Line at upper gate corners

B) Line at lower gate corners

This technique allows the regulation of the muscular- tendinous tone in the upper and lower and whole body, and optimizes the proprioceptive system and body structure links.

For this exercise two basic techniques are done:

1. Circular motion repeated three times.

2. Stretch in three-dimensional directions:

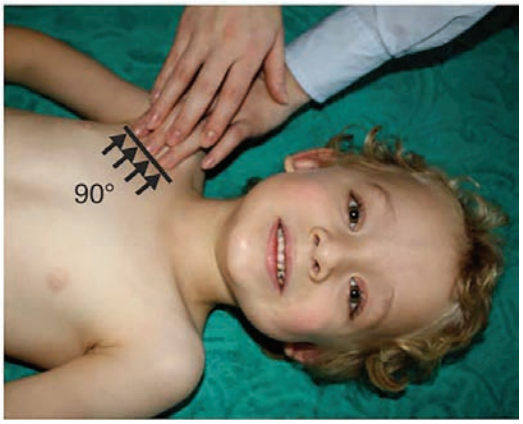
- Upper gate – simultaneous lateral, posterior, and superior micro-movements
- Lower gate - simultaneous lateral, posterior, and inferior micro-movements

Use finger tips and pads. Repeat 3-7 times depending on therapeutic needs and client's preference. Stay in appropriate communication and distance to client.









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FRONT: CORE TENDON GUARD

A) Line at upper gate corners: circular

Position of hands: Fingers of both hands (with even touch) are in a raglan line in the intermediate space between the shoulder joint and top of the rib cage.

Action:

1) Circular motions. Do three slow (7 second) circular motions in this area (along the line of connection of the shoulder and top of the rib cage) with firm and caring touch (soft, yet pleasant at the same time) across the connective line. Reach the deeper level of "corner" increasing pressure gradually.

Direction of applying the technique: Move fingers in circular motion starting downward from the inside of the body, moving in to the midline and then up towards the head and away from midline. Note: Always circle up the midline or the right and left sides.

2) Stretch across the intermediate line in the upper corner. Place your hands with fingers in the middle of the circles, press, 'hook' and stretch this area that you work with. Keep fingers apart a bit (not too tight) to keep them in a normal active state with no over-tension. Stretch, pulling 90 degrees from the line in the upper corner in the three-dimensional direction: – simultaneous lateral, posterior, and superior micro-movements. Do one stretch after three circular motions. Repeat 3-7 times for each upper corner of the rectangle.

B) Line at lower gate corners: circular

Position of the hands: Fingers of both hands (with even touch) are on the line in the intermediate space between the hip/pelvic bone area and transverse and rectus abdominal muscles.

Action:

1) Circular motions. Do three slow circular motions in this area with firm and caring touch (soft, pleasant) across the connective line. Reach the deeper level of "corner" increasing pressure gradually.

Direction of applying the technique: Move fingers in circular motion starting from the inside of the place where your hands are located, moving in to the midline and then up towards the head and away from the center and complete circle and continue.

2) Stretch across the intermediate line in the lower corner. Place your hands with fingers in the middle of the circles, press, 'hook' and stretch this area that you are working with. Keep fingers apart a bit (not too tight) to keep them in normal active state with no over-tension and like embracing the hip joint; stay not too close to hip bone not to cause sensitivity and discomfort. Stretch, pulling 90° from the line in the lower corner in the three-dimensional direction: – simultaneous lateral, posterior, and inferior micro-movements. Do one stretch after three circular motions. Repeat 3-7 times for each upper corner of the rectangle.

Note: This technique is not essential for the lower gate in cases of hypersensitive or hyperactive response.

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2. 3. Deep Waves at Rectangle Corners

Position of the client: Lying down supine (on back). Client can be half-sitting/sitting/standing if more comfortable.

Technique: Tactile-proprioceptive activation in every corner of rectangle of the body using deeper intervention motions at:

A) Line at upper gate corners

B) Line at lower gate corners

This technique allows regulation of the muscular-tendinous tone in the upper and lower and whole body, and optimizes the proprioceptive system and body structure links. It also is aimed at lengthening of tendons and ligaments in the area of connection of limbs and core, activates the receptors of deep touch and pressure, proprioceptors at the origin and insertion of muscles and ligament receptors.

For this exercise a deeper intervention of slow stretch and quick release technique is proposed using straight motion with fingers/hands (touch #1-2-3) during 1-2-3-4-5 seconds going deeper into soft tissues during next 6-7 seconds, and coming out gently on 8-9. Repeat 2-3 times or more depending on therapeutic needs and client's preference. Stay in appropriate communication and distance to client (not to close and not too away – for territorial perception).

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FRONT: CORE TENDON GUARD

A) Upper gate corners – deep wave

Position of the specialist: Stand at the upper corner of the rectangle where the technique is applied (against to the opposite hip).

Position of the hands: Fingers of both hands (with even touch) are along the line in the intermediate space between the shoulder joint and top of the rib cage (at raglan at the free and softer spot).

Action:

Slow deep pressure stretch – quick release in the intermediate line area in the upper corner.

Stretch in the intermediate line along the line in the area of the upper corner in posterior direction– into body/tissues. Locate your fingers in the line and do:

1) A deep slow stretch into soft tissues (1-2-3-4-5 seconds)
2) Next, stretch even more in "the bottom" (seconds 6-7)
3) Quickly release (seconds 8-9). Do gently at the same time.

Repeat 2-3 times or more for each upper corner of the rectangle.

Direction of applying the technique: Move fingers in the line as above – with deep touch (posterior direction) to reach the upper rib cage, and after the pause for more stretch at "the bottom", do quick release away towards the body center –median-inferior direction.

Note: This technique is not essential in cases of hypersensitive or hyperactive response.







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B) Lower gate corners – deep wave

Position of the specialist: Stand at the lower corner of the rectangle where the technique is applied.

Position of the hands: Fingers of both hands (with even touch) are along the line in the intermediate space between the hip/pelvic bone area and transverse and rectus abdominal muscles.

Action:

Slow deep pressure stretch – quick release in the intermediate line area in the lower corner.

Stretch in the intermediate space between the hip/ pelvic bone area at the transverse and rectus abdominal muscles in the area of the lower corner in the posterior direction – into body/tissues. Locate your fingers in the line and do:

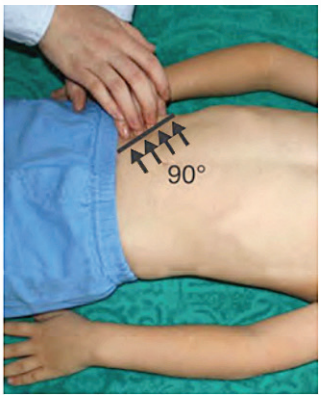
1) A deep slow stretch (1-2-3-4-5 seconds)
2) Next, stretch even more in "the bottom" (seconds 6-7)
3) Quickly release (seconds 8-9), Do gently at the same time.

Repeat 2-3 times or more for each upper corner of the rectangle.

Direction of applying the technique: Move fingers in the line as above from the upper corner – with deep touch (posterior direction) to reach the rib cage, and after the pause for more stretch at "the bottom", do quick release away towards the body center – to median- superior direction.







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Exercise Effects: Deep Wave Upper Gate

Nervous system

Innervation for muscles:

1) *m.pectoralis major pars clavicularis–> n.pectorales medialis et lateralis (CVII- Th1)*
2) *m. pectoralis minor–>n.pectorales medialis et lateralis (CVII- Th1)*
3) *m. deltoideus pars clavicularis –> n. axillaris*
4) *m. subclavius –> n. subclavius (CV): –> n. subclavius (CV)*

Skin innervation:

1) *m. pectoralis major pars clavicularis–> n. pectorales medialis et lateralis (CVII- Th1)):–> n. supraclaviculares*
2) *m. pectoralis major pars clavicularis –> n. pectorales medialis et lateralis (CVII- Th1)): –> n. intercostales*

Circulatory system

1) *m. pectoralis major pars clavicularis –> n. pectorales medialis et lateralis (CVII- Th1): –> a. thoracoacromalis r.pectoralis, a.thoracica lateralis, rr.intercostales anteriores (exits from a. thoracica interna)*
2) *m. pectoralis minor–> n.pectorales medialis et lateralis (CVII- Th1): –> a. thoracoacromalis r.pectoralis, rr. intercostales anteriores (exits from a. thoracica interna)*
3) *m. deltoideus pars clavicularis –> n. axillaris: –> a. circumflexa humeri anterior et posterior (rr. a. axillaris)*
4) *m. subclavius –> n. subclavius (CV): –> a.thoracoacromialis r.subclavius*

Lymphatic system

Lymphatic vessels flow into lymphatic glands in armpit area:

1) *m. pectoralis major pars clavicularis –> n. pectorales medialis et lateralis (CVII- Th1): –> nodi lymphatici axillaris pectoralis* –> the vessels from lateral and frontal surface of the chest and abdomen flow in to this gland
2) *m. pectoralis minor–> n.pectorales medialis et lateralis (CVII- Th1): –> nodi lymphatici apicales* –> the vessels from upper part of breast are flowing into these glands
3) *m. deltoideus pars clavicularis –> n. axillaris –> nodi lymphatici subclaviculares*

Respiratory system

Muscles:

1) *m. pectoralis major pars clavicularis*
2) *m. pectoralis minor:* –> raise the ribs and sternum and serve as additional muscles participating in action of breathing

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FRONT: CORE TENDON GUARD

Lower Gate: Deep Wave Upper Gate

Nervous system

Innervation for muscles:

1) m. obliguus externus abdominis

2) m. obliguus internus abdominis

3) m. transversus abdominis: –> nn.intercostales Th-VThXII), n. iliohypogastricus (ThXII-L1), n. ilioinguinalis (LI)

Circulatory system

1) m. obliguus externus abdominis: –> aa. intercostales posteriores, a. thoracica lateralis, a. circumflexa iliaca superficialis

2) m. obliguus internus abdominis

3) m. transversus abdominis: –> aa. intercostales posteriores, aa.epigastricae inferior et superior, a. musculophrenica

Lymphatic system

1) m. obliguus externus abdominis: –> Nodi lymphatici inguinalis superficialis

Team Work. When you work with a second person – NS therapists or a parent – these techniques can be done in a reciprocal/cross lateral way – one person does work pushing on one shoulder, and next other person works on the hip working sequentially, then they repeat the same on other opposite shoulder and hip.

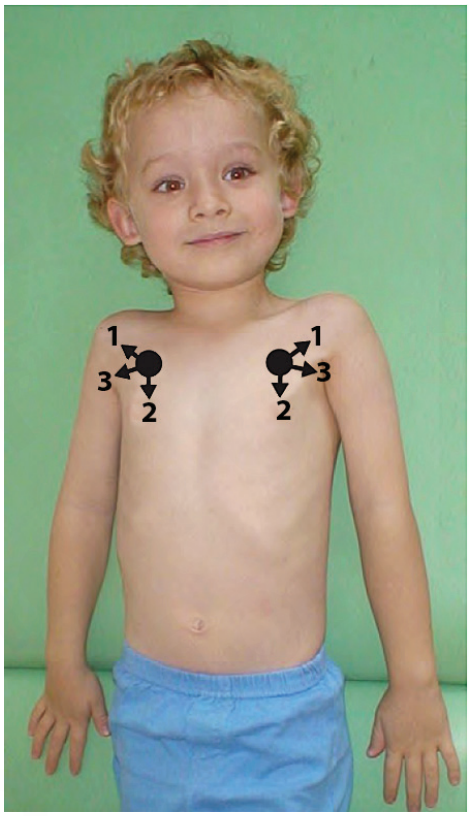
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3. 'Pendulum' under mid clavicle

Goals:

- Deeply release the tension or activate a low tone Tendon Guard Reflex.
- Regulate the tone of the tendons in the upper part of the core.
- Activate the proprioceptive system and body muscle-skeleton structure links for optimizing the automatisms of the upper limbs reflex patterns: Hands Pulling, Hands Supporting, Hands Grasp.
- Optimize the functioning of Babkin Palmomental, Head Righting, Chewing, Swallowing, Ocular Convergence/ Divergence, Ocular-Vestibular, and Optokinetic reflex patterns.
- Activate the receptors of deep touch and pressure.
- Stimulate breathing reflex patterns and oxygen supply.
- Stimulate the whole body symmetry mechanism functioning.
- Optimize the functioning of vision, hearing, and articulation systems.
- Increase peripheral nerve innervations, lymphatic and blood circulation.
- Activate Upper Tendon Guard (neck flexors and extensors) to increase the functioning of visual, auditory, and vestibular systems.



Location: *Regio infraclavicularis*

Bones: *Clavicula, I-III costae*

Muscles:

1) Pectoralis Major Clavicular (m.pectoralis major pars clavicularis)
2) (m. pectoralis minor)
3) (m. deltoideus pars clavicularis)
4) (m. subclavius)
Also influences the muscles:
- *Intercostal muscles (m. intercostales)*
- *Sternalclidomastoid*
- *(caput breve m. bicipis)*
- *(m. coracobrchialis)*
- Anterior neck flexors

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FRONT: CORE TENDON GUARD – PENDULUM

3.1. Pendulum

Position of the client: Lying down supine (on back). Client can be sitting/standing if more comfortable.

Technique: Tactile-proprioceptive circular activation in the area of the upper corner of rectangle of the body in the middle of pectoralis major clavicular muscle (under clavicula, and in area of I-III costae) using deeper pressure.

This technique allows regulation of the muscular-tendoneous tone in the upper and whole body through optimizing the proprioceptive system, body structure links for upper limbs reflexes to function properly. It is aimed at activation of the receptors of deep touch and pressure, and proprioceptors and ligament receptors.

For this exercise the deep and slow pressure with fingers/hands (touch # 2-3) is proposed.

Position of specialist: Stand at the upper corner of the rectangle where the technique is applied; facing the client.

Position of the hands: Fingers – first and second fingers of one hand for children and all fingers for adults - touch deeply and evenly the area of the middle of the *pectoralis major clavicular* – at the top of the rib cage (at softer spot), between the sternum and shoulder; the other hand is on the top of the first one provides the motions described below.

Action:

A) Slow deep circular motions (in the area of the pectoralis major clavicular)

Do three circular motions moving your hands in circular motion away from the center of the chest laterally – to the sides (12 o'clock at the clavicle bone) in the area of the upper corner on level of the pectoralis major clavicular (reach softly the "bottom" – the skeleton of the rib cage).

B) Slow deep stretch in three directions (see page 50)

Continue with slow pressing into the rib cage *(I-III costae)* during 7-10 seconds, and do micro-stretch in direction of:

- Shoulder
- Down
- Armpit area

Do the exercise 2-3 times on one side, then on other.

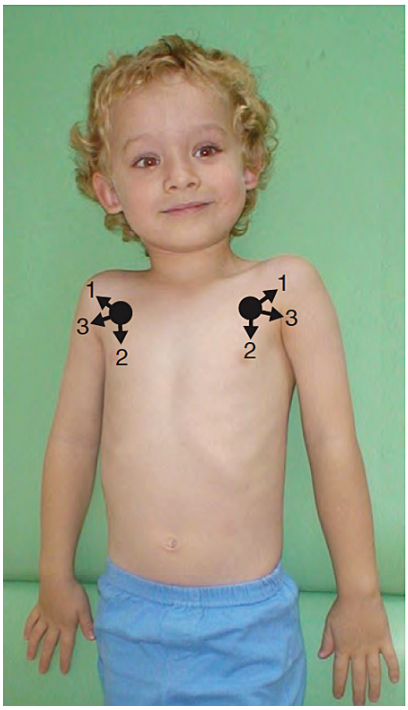
Direction of applying the technique:

A) Slow deep circular motions – start at the level closer to clavicle bone and do circular motions away from body center – laterally, inward, up

B) Slow deep stretch in three directions:

- Shoulder - posterior, lateral, superior
- Down - posterior, inferior
- Armpit area - posterior, lateral, inferior

Note: This technique is essential in cases of improper functioning of the Red and Green Tendon Guard and upper limbs reflex patterns: Hands Pulling, Hands Supporting, Hands Grasp, Babkin Palmomental, Head Righting, Ocular Convergence/Divergence, Ocular-Vestibular, and Optokinetic reflex patterns.



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FRONT: CORE TENDON GUARD – PENDULUM

Exercise Effects: Pendulum

Nervous system

Muscle innervation:

1) *m. pectoralis major pars clavicularis and*
2) *m. pectoralis minor: –> n.pectorales medialis et lateralis (CVII- Th1)*
3) *m. deltoideus pars clavicularis: –> n. axillaris*
4) *m. subclavius: –> n. subclavius (CV)*

Skin innervation:

1) *m. pectoralis major pars clavicularis: –> n. supraclaviculares*
2) *m. pectoralis minor: –> n. intercostales*

Circulatory system

1) *m. pectoralis major pars clavicularis: –> a. thoracoacromalis r.pectoralis, a.thoracica lateralis, rr.intercostales anteriores (from a. thoracica interna)*
2) *m. pectoralis minor: –> a. thoracoacromalis r.pectoralis, rr. intercostales anteriores (a. thoracica interna)*
3) *m. deltoideus pars clavicularis: –> a. circumflexa humeri anterior et posterior (rr. a. axillaris)*
4) *m. subclavius: –> a.thoracoacromialis r.subclavius*

Lymphatic system

Lymphatic vessels flow into glands in the armpit area:

1) *m. pectoralis major pars clavicularis –> nodi lymphatici axillaris pectoralis*
2) *m. pectoralis minor: –> nodi lymphatici apicales –> vessels from upper part of breast lymph system flow in to this gland*
3) *m. deltoideus pars clavicularis: –> nodi lymphatici subclavicularis*

Respiratory System

Muscles

1) *m. pectoralis major pars clavicularis*
2) *m. pectoralis minor:* –> in fixed state of the upper shoulder raise the ribs and sternum; they are additional muscles participating in the action of breathing.

Note: This exercise can be done sequentially by two repetitions on every side changing the sides for giving some pause, and creating the ease in the perception of the depth of the pressure.

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FRONT: Pendulum

Procedure: Pendulum At-a-Glance

1. Use the second and third position of finger tips (stimulation #2 and 3 by fingers).

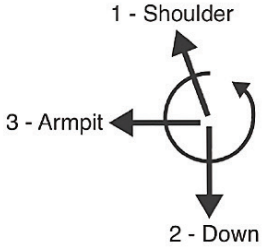
2. Find the central the deepest area under the clavicle half way between the shoulder and sternum. Do a larger circle starting with lighter proprioceptive touch and increasing the depth. You can activate this area under the clavicle bone with one circular motion - stopping each time and stretching in three directions:

A. Circle – move in the direction of the shoulder – 45°

B. Circle – move straight down

C. Circle – move in direction of armpit – 45°. This soft, steady and deep proprioceptive stimulation regulates the ligaments and tendons tone - loosens the tension and at the same works for strengthening them. Work first with one area first then the other (also when you work with helper or second NS specialist). Repeat the whole cycle of techniques 3-5 times.





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4. Rib Cage and Sternum Area Activation: Under Clavicle Line, Rib 5-6 Line, Rib Cage Bow, Sternal Dominos



Goals

- Releases the negative effect of the Tendon Guard and open the core.
- Activates symmetry mechanism and body midline and horizontal lines functioning.
- Regulates breathing reflexes.
- Sternum serves as a pendulum for the body and takes part in gravity line and gravity center organization.
- Establishes a clear sense/feeling of the midline of the body and correlation with gravity center.
- Activates symmetry between sternum and sacrum = structural and dynamic symmetry for up/down and front-back.
- Activates antigravity mechanisms and postural control through activation of balancing, grounding and stability reflexes (in brain stem and cerebellum responses).
- Assists with natural breathing reflex patterns: a) natural cycle – inhale-exhale, b) inhale and hold, c) exhale and hold.
- Increases nerve innervations and blood circulation through activation of peripheral nerve system of muscles.
- Activates lymph and oxygen supply.

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4.1. Under Clavicle Line

Location: *Regio infraclavicularis*
Bones: *Clavicula, I-III costae*
Muscles:
1) (m. pectoralis major pars clavicularis)
2) (m. pectoralis minor)
3) (m. deltoideus pars clavicularis)
4) (m. subclavius)

Procedure

Position of the client: Lying supine (on back). Client can be sitting/standing if more comfortable.

Technique: Tactile-proprioceptive circular activation following with stretch in the area below clavicle bone (*Clavicula, and in I-III costae*). Do three circular stimulations. Next, keeping your fingers in the middle of the "circles", do a deeper 90° stretch from line of clavicle with some pressure. This technique allows regulation of the muscular-tendinous tone in the upper part of the body, and for the proper functioning of the links of upper limb and core reflexes. It is aimed at activation of the receptors of deep touch and pressure, and proprioceptors and ligament receptors. For this exercise, deep and slow pressure with fingers/hands (touch # 1 and 2) is proposed.

Position of specialist: Stand at the upper part of the body where the technique is applied; facing the client. Position of the hands: Fingers – tips and pads of both hands - touch evenly and deeply in the area under clavicle bone – at the top of the rib cage (at softer spot).

Action:

A) Slow deep circular motions (in the area under clavicle bone)

Do three circular motions moving your hands in circular motion away from the center of the body laterally – to the sides and inward (12 o'clock at the clavicle bone; reach softly the "bottom" – the skeleton of the rib cage).

B) Slow deep stretch down - in inferior direction.

Continue with deeper stretch with increasing pressure into the rib cage *(I-III costae)* during 7-10 seconds, and do micro-stretch in down direction. Do the exercise 2-3 times on one side, then on other.

Direction of applying the technique:

A) Slow deep circular motions

Start at the level closer to clavicle bone and do circular motions away from body center – laterally inward-up.

B) Slow deep stretch in down directions

Stretch with pressure in three-dimensional direction: posterior-lateral-inferior.

Note:

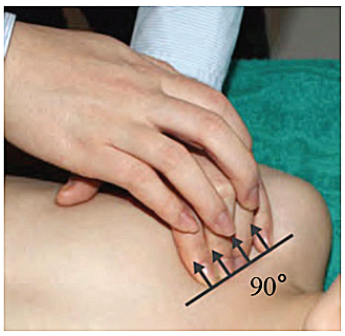
1) This technique is essential in cases of improper functioning of the Red and Green Tendon Guard, Babkin Palmomental, chewing, biting, swallowing, Head Righting, Ocular Convergence/Divergence, Ocular- Vestibular, and Optokinetic reflex patterns.

2) This exercise can be done sequentially by two repetitions on every side changing the sides for giving some pause, and creating the ease in the perception of the depth of the pressure.

Work in a Team. When you work with second person – NS therapists or a parent - this technique can be done on both sides at the same time. After 2 reps change sides for symmetry.







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FRONT: Clavicle

Exercise Effects: Under *Clavicle* Line

Nervous system

1) CN X - Vagus,

2) Phrenic nerve functions

Muscle innervation:

1) *m. pectoralis major pars clavicularis and*

2) m. pectoralis minor: –> n.pectorales medialis et lateralis (CVII- Th1)

3) m. deltoideus pars clavicularis: –> n. axillaris

4) m. subclavius: –> n. subclavius (CV)

Skin innervation:

1) m. pectoralis major pars clavicularis: –> n. supraclaviculares

2) m. pectoralis minor: –> n. intercostales

Circulatory system

1) m. pectoralis major pars clavicularis: –> a. thoracoacromalis r.pectoralis, a.thoracica lateralis, rr.intercostales anteriores (from a. thoracica interna)

2) m. pectoralis minor: –> a. thoracoacromalis r.pectoralis, rr. intercostales anteriores (a. thoracica interna)

3) m. deltoideus pars clavicularis: –> a. circumflexa humeri anterior et posterior (rr. a. axillaris)

4) m. subclavius: –> a.thoracoacromialis r.subclavius

Lymphatic system

Lymphatic vessels flow into glands in the armpit area:

1) m. pectoralis major pars clavicularis –> nodi lymphatici axillaris pectoralis

2) m. pectoralis minor: –> nodi lymphatici apicales –> vessels from upper part of breast lymph system flow in to this gland

3) m. deltoideus pars clavicularis: –> nodi lymphatici subclavicularis

Respiratory system

1) m. pectoralis major pars clavicularis

2) m. pectoralis minor: –> in fixed state of the upper shoulder raise the ribs and sternum; they are additional muscles participating in the action of breathing.

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4.2. Rib 5-6 Line Activation

Location: *Thorax*

Bones: *Thorax, costae V-VI, angulus infrasternalis, corpus sterni,*

Muscles:

1) (m. pectoralis major)
2) (m. pectoralis minor)
3) (m. intercostales externus, also m. intercostales internus and intimi)

Also activates muscles:

- (m. transversus thoracis)
- (mm. subcostales)
- (m. serratus anterior)

* This exercise is done in the area of ribs 5-6 as it is most dynamic and flexible part in the rib cage.

Procedure

Position of the client: Lying supine (on back). Client can be sitting/standing if more comfortable.

Technique: Tactile-proprioceptive circular activation following with stretch in the area between ribs 5-6. Do three circular stimulations, then keep your fingers in the middle of the 'circles' do a deeper 90° stretch with some pressure. This technique allows regulation of the tone in muscular-tendinous system in between the ribs and whole rib cage, it supports breathing and provides proper functioning for the Fear Paralysis, Moro, and Core Tendon Guard reflexes. It is aimed at activation of the receptors of deep touch and pressure, and proprioceptors in the tiny muscular system around the rib cage and ligament receptors. Deep and slow pressure with fingers/hands (touch # 1 and 2) is proposed.

Position of specialist: Stand on the side of the body close to rib cage where the technique is applied; facing the client.

Position of hands: Fingers – tips and pads of both hands - touch evenly and deeply in the area under clavicle bone horizontally – at the top of the rib cage.

Action:

A) Slow deep circular motions (in the area under ribs 5-6)

Do three circular motions moving your hands in circular motion away from the center of the body laterally, to the sides (12 o'clock at the rib bone 5; reach softly the line between rib 5-6). Do circles wider than the line between ribs 5-6, and more surface way and gently (as bones in this area are more superficial).

B) Slow deep stretch down - in inferior direction.

Continue with deeper stretch with increasing pressure into the *intercostal* muscles between ribs 5 and 6 (*costae 5-6*) during 7-10 seconds, and do microstretch in down direction. Do the exercise 2-3 times on one side, then on other.

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FRONT: Ribs 5 - 6

Direction of applying the technique:

A) Slow deep circular motions

Start at the level between ribs 5-6 and do wider circular motions laterally-inward-up touching the space above and under ribs 4-5, 5-6, and 6-7.

B) Slow deep stretch down - in inferior direction.

Stretch with pressure in two-dimensional direction: posterior-inferior 90° against of the rib line 5-6.

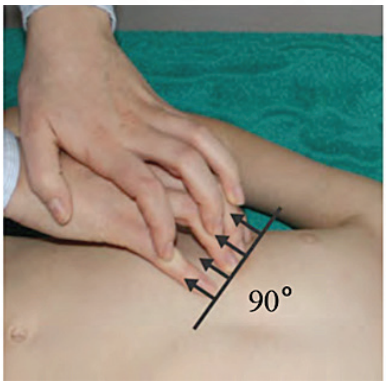
Note: This technique is most essential in cases of improper functioning of the Upper and Core Tendon Guard, and breathing reflexes. It can be used in cases of reflux and other digestive challenges. This exercise can be done sequentially by two repetitions on every side changing the sides for giving some pause, and creating the ease in the perception of the depth of the pressure.

Additional Note: You can work with the area in between ribs 7-8, and also all others.

Team Work. When you work with second person – NS therapists or a parent – this technique can be done on both sides at the same time. Do two reps and change sides for symmetry.







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FRONT: Ribs 5 - 6

Exercise Effects: Ribs 5-6 Line Activation

Nervous system

1) CN X - Vagus,
2) Phrenic nerve functions

Muscle innervation:

1) *m. pectoralis major*
2) m. pectoralis minor: –> n.pectorales medialis et lateralis (CVII- Th1)
3) m. intercostales externus, also m. intercostales internus: –> nn. intercostales (ThI - ThXI)

Skin innervation:

1) m. pectoralis major: –> rr. cutanei nn. thoracici I - XI

Circulatory system

1) m. pectoralis major: –> a. thoracoacromalis r.pectoralis, a.thoracica lateralis, rr.intercostales anteriores(a. thoracica interna)
2) m. pectoralis minor: –> a. thoracoacromalis r.pectoralis, rr. intercostales anteriores (a. thoracica interna)
3) m. intercostales externus, also m. intercostales internus: –> aa. Intercostales posteriors, a thoracica interna (r. anterior), a. musculophrenica

Lymphatic system

Lymphatic vessels of this area flow to next groups of glands:

1) m. pectoralis major: –> nodi lymphatici axillaris
2) m. pectoralis minor:–> nodi lymphatici parasternalis
3) m. intercostales externus, also m. intercostales internus: –> nodi lymphatici subclaviculares
4) nodi lymphatici mediastinalis

Respiratory system

When contracted, these muscles raise the ribs and sternum; they serve as additional muscles participating and helping in the action of breathing.

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4.3. Rib Cage Bow

Location: *Regio abdominalis, regio thorax, arcus costales, angulus infrasternalis*

Bones: *Costae X, arcus costales, angulus infrasternalis, corpus sterni*

Muscles:

1) m. obliguus externus abdominis

2) m. obliguus internus abdominis

3) m. transversus abdominis

4) m. rectus abdominis

Also activates muscles of the diaphragm:

- (m. latissimus dorsi)

- (m. serratus anterior and m. serratus posterior inferior)

- (m. sulcus inguinalis)

- (m. sulcuspubicus)

Procedure

Position of the client: Lying supine (on back). Client can be sitting/standing if more comfortable.

Technique: Tactile-proprioceptive circular activation following with stretch in the area rib cage line (costae X). Do three circular stimulations with your finger tips and first level of fingers, giving slightly stronger stretch when reaching the boundary of the line. Next, touch in an embracing way with your fingers and palm the rib cage line/bow, and do stretching of the whole bow in the direction of the shoulder on the same side of the body. This technique allows for regulation of the tone in muscular-tendinous system in whole rib cage, activates the diaphragm, supports breathing and core reflex patterns (Fear Paralysis, Moro, and Core Tendon Guard). It is aimed at activation of the receptors of deep touch and pressure, and proprioceptors around the whole rib cage and ligament receptors. Deep and slow pressure with fingers/hands (touch # 1, 2, and 3) is proposed.

Position of specialist: Stand on the side at the rib cage where the technique is applied.

Position of hands: Fingers – tips, pads, first fingers and palms of both hands - touch evenly and deeply in the area of the rib cage and do microstretch of 90° away from the rib cage line.

Action:

A) Slow deep circular motions (in the area of rib cage)

Do three circular motions moving your hands away from the center of the body inward and laterally – to the sides (12 o'clock at the rib cage line). Stretch slightly at the end of every circle against of the rib cage line giving the effect of micro-lifting the whole rib cage.

B) Slow deep stretch in upward direction.

Embrace the whole rib cage and do slow deep stretch in upward direction – anterior-lateral-superior.

Continue with deeper and micro-stretch increasingly lifting the rib cage line during 7-10 seconds intending to activate the diaphragm. Do the exercise 2-3 times on one side, then on other.

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FRONT: Rib Cage Bow

Direction of applying the technique:

A) Slow circular motions

Start at the line of the rib cage, and do circular motions laterally starting from inward-up-to sides holding this line stably and safely from internal part of it (under rib cage).

B) Slow deep leverage stretch in lateral-upward direction

Holding the rib cage line embracing the ribs on edge with your fingers from the external and internal part of it, stretch with pressure in three-dimensional direction: anterior-lateral-inferior – keeping it flat; micro stretch is done in shoulder direction with projection of above it.

Note: This technique is beneficial in cases of improper functioning of the Red Core Red Tendon Guard, and in cases of poor diaphragm functioning it helps with regulation of breathing. It can be used in cases of reflux and other digestive system dysfunctions. This exercise can be done sequentially by two the perception of the depth of the pressure.

Additional Note: This technique can work with the area of the lower rib cage line with the diaphragm itself. Place your fingers in the top of the abdomen under the rib cage line, press deeply in and then in upward direction intending to stimulate the diaphragm itself. Be aware of the hyperactive Fear Paralysis response. Keep pressure for 7-10 seconds and then do quick release. Work gently and safely. Be aware of feeding tube or other complications.

Team Work. When you work with second person – NS therapists or a parent - this technique can be done on one side at a time only. You can work with helping person sequentially - 2-3 times on one side and next changing for the sides for symmetry.







Variant

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Exercise Effects: Rib Cage Bow

Nervous system

1) CN X - Vagus

2) Phrenic nerve functions

Muscle innervation:

1) m. obliguus externus abdominis

2) m. obliguus internus abdominis

3) m. transversus abdominis: & n. intercostales, n. ilioinguinalis, n.iliohypogastricus 4) nn. intercostales (ThVI-ThXII), n. iliohypogastricus (ThXII-ThXII)

Skin innervation:

1) m. obliguus externus abdominis

2) m. obliguus internus abdominis

3) m. transversus abdominis: & r. cutanei anterior et lateralis nn. intercostalium et iliohypogastrici

Circulatory system

1) m. obliguus externus abdominis: "a. circumflexia ilica superficialis, a. intercostales posteriors. a. thoracica lateralis

2) m. obliguus internus abdominis

3) m. transversus abdominis: & a. intercostales posteriors, aa. epigastricae superior et inferior, a. musculophrenica

4) m. rectus abdominis: " aa.epigastricae superior et inferior, aa. intercostales posteriores

Lymphatic system

On the surface of the body vessels that flow in:

1) m. obliguus externus abdominis: " n. lymphatici axillaris, epigastrici et inguinalis superficialis

In deeper layer of the body vessels flowing in:

1) n. lymphatici epigastrici, mediastinalis anterior et thoracicus

Respiratory system

When contracted, these muscles raise the ribs and sternum; they serve as additional muscles participating and helping in the action of breathing.

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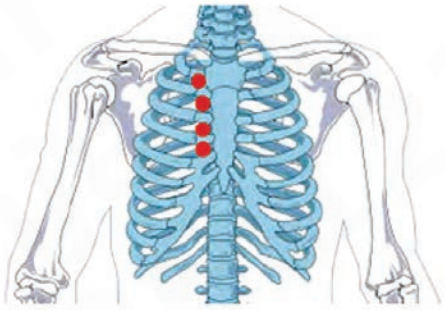
4.4. Sternal Dominos: Intercostals and Pectoralis Major Sternal

Location: *Thorax, facies* anterior

Bones: *1) Sternum (upper part - manubrium sterni, and lower - corpus sterni), 2) Costae II-V*

Muscles:

1) (m. pectoralis major)
2) (m. pectoralis minor)
3) (m. intercostales externus, also m. intercostales internus)



Helps to coordinate the links of the insertions of all the basic muscles from the thoracic area to the arm at *cristatuberculi maioris* and holds the structure of the front part of the thorax/rib cage: *os. clavicularis, os sternum, m. abdominalis rectus.*

* This exercise is done in the area of lateral sides of the *sternum,* as it is located in the midline of the body and work with it can help establish the links between the gravity and midline of the body and postural control.

Procedure

Position of the client: Lying supine (on back). Client can be sitting/standing if more comfortable.

Technique: Tactile-proprioceptive circular activation following with stretch in the lateral area in the interspace between the sternum and all the ribs. Touch with tips of your fingers in the soft areas between the connections of the side of the sternum and ribs. Do three circular stimulations. Next, keeping fingers in the middle of the "circles" (indentations) do a deeper micro- stretch using light and stable pressure. This technique allows for regulation of the tone in muscular-tendinous system in between the sternum and whole rib cage. It supports breathing, upper body (ocular and auditory reflex patterns, patterns concerned with eating, also smiling) and core reflex patterns (Fear Paralysis, Moro, and Core Tendon Guard). It is aimed at activation of the receptors of deep touch and pressure, and proprioceptors in the tiny muscular system around the sternum, rib cage, and ligament receptors. Deep and slow pressure with fingers/hands (touch # 1 and 2) is proposed.

Position of specialist: Stand at the upper part of the body close to sternum/rib cage where the technique is applied; facing the client.

Position of the hands: Fingers – tips and pads of both hands - touch evenly and stably in the intermediate connection spots between the sternum and ribs vertically – on the side of the rib cage and pull laterally. The areas of touch and light pressure are between ribs 1-2, 2-3, 3-4, 4-5.

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FRONT: STERNAL DOMINOS

Action:

A) Slow soft circular motions (intermediate connection spots between the sternum and ribs)

Do three circular motions moving your hands in circular motion away from the center of the body laterally – to the sides (12 o'clock at the sternum bone; reach softly the intermediate spots in line between the sternum and all connected ribs).

B) Slow stable stretch in lateral direction holding in the intermediate spots between the sternum and ribs)

Continue stretch with light increasing pressure into the rib cage (costae 1-6) for 7-10 seconds, and do a micro-stretch in the lateral direction. Do the exercise 2-3 times on one side, then on other.

Direction of applying the technique:

A) Slow circular motions

Do circular motions inward-up-laterally touching the softer intermediate spots between the sternum and ribs in the area of ribs 1-2, 2-3, 3-4, 4-5.

B) Slow stable stretch in lateral direction

Stretch in a two-dimensional direction: posterior-lateral 90° from the sternum line.

*Note: T*his technique is most essential in cases of improper functioning of the postural control, vestibular deficits, Upper and Core Red Tendon Guard, fears, tendency for freeing or fight-flight responses, and also breathing reflexes (like asthma, long term coughing). It can also be used in cases of reflux and other digestive challenges.

Additional Note: This exercise can simultaneously be done with the area on the back in between the spine opposite the sternum/ribs area.

Work in Team. When you work with second person – NS therapists or a parent - this technique can be done on one side at a time. You can work with helping person sequentially – 2-3 times on one side and next changing for the sides for symmetry. You also can work on both sides simultaneously.







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FRONT: STERNAL DOMINOS

Exercise Effects: Sternal Dominos

Nervous system

1) CN X - Vagus
2) Phrenic nerve functions

Muscle innervation:

1) *m. pectoralis major*
2) m. pectoralis minor: –> n.pectorales medialis et lateralis (CVII- Th1)
3) m. intercostales externus, also m. intercostales internus: –> nn. intercostales (ThI - ThXI)

Skin innervation:

1) m. pectoralis major: –> rr. cutanei nn. thoracici I - XI

Circulatory system

1) m. pectoralis major: –> a. thoracoacromalis r.pectoralis, a.thoracica lateralis, rr.intercostales anteriores(a. thoracica interna)
2) m. pectoralis minor: –> a. thoracoacromalis r.pectoralis, rr. intercostales anteriores (a. thoracica interna)
3) m. intercostales externus, also m. intercostales internus: –> aa. Intercostales posteriors, a thoracica interna (r. anterior), a. musculophrenica

Lymphatic system

Lymphatic vessels of this area flow to next groups of glands:

1) m. pectoralis major: –> nodi lymphatici axillaris
2) m. pectoralis minor:–> nodi lymphatici parasternalis
3) m. intercostales externus, also m. intercostales internus: –> nodi lymphatici subclavicularis
4) nodi lymphatici mediastinalis

Respiratory system

When contracted, these muscles raise the ribs and sternum; they serve as additional muscles participating and helping in the action of breathing.

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5. Abdominal Points (Navel Dominoes): Fear Paralysis and Moro Points

Goals:

- Release the negative effect of the Tendon Guard (Red Light).
- Release the negative effect of Fear Paralysis, Moro and withdrawal responses.
- Organize links between the upper and lower parts of the body.
- Activate the center of gravity.
- Activate the biomechanics and dynamics of the core work for Fear Paralysis (organizes the response on the level of "freezing" of the structure in the solar plexus), and Moro (organizes the response on the level of fight and flight of postural triangle – psoas, diaphragm, and abdominal muscles).
- Balance self-preservation instincts.
- Increase the peripheral nerve innervations and blood circulation.
- Activate lymph and oxygen supply.
- Activate self-regulating mechanisms in the brain stem and diencephalon level.
- Activate the processes and mechanisms in the cerebellum
- Support the CN X - Vagus, also Phrenic nerve functions.



Location: *Regio abdominalis*

Bones: *Costae X*

Muscles:

1) m. obliguus externus abdominis
2) m. obliguus internus abdominis
3) m. transversus abdominis
4) m. rectus abdominis

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5.1. Fear Paralysis Points

Position of the client: Lying down supine (on back). Client can be sitting/standing if more comfortable.

Technique: Deep tactile-proprioceptive activation of three paired points on both sides of the navel in the area of rectus abdominal muscles:

1) One inch above the navel

2) On the level of navel

3) One inch below the navel

Hold all the points as one structure and move them stably as you want project the shape of the spine as you do circular motions up to the left (clockwise).

This technique allows regulation of the muscular-tendinous tone in the abdomen area, diaphragm, support the CN X - Vagus & Phrenic nerve functions, and whole body serving to trigger the biomechanics of the Fear Paralysis reflex pattern. It is aimed at: strengthening the gravity center; activation of the biomechanics of flexion of the body forward; balance of Fear Paralysis with Moro, Core Tendon Guard, and Trunk Extension patterns. For this exercise use the deep and slow pressure with fingers/hands (touch # 1, 2, 3).

Position of specialist: Stand at the side close to the abdomen area where the technique is applied.

Position of the hands: Using the fingers and palms of both hands, hold deep in the area of the six points and do circular motions to the left side.

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FRONT: Fear Paralysis Points

Action:

Slow deep circular motions (on the both sides of the abdominal muscles involving the midline of the core (linea alba)

Do circular motions moving your hands in one line in the shape of the spine to the left (clockwise), then for 7 circles, then right (counter clockwise) for 5 circles and back left (clockwise) for 3 circles.

(Note: It is possible do these motions to right finishing again to left, when the goal is to activate peristaltic movements of the intestine system). These points can be activated one level after another (level one – first two upper points, level two – next two middle points, level three – last two lower points).

Direction of applying the technique: Start with more pressure on the right side of abdomen intending to move in circular motion to the left. Press inward posteriorly– laterally inferiorly, and then up – superiorly finishing the circle.

Note: This technique is essential in cases of improper functioning of the Red and Green Tendon Guard and other core reflex patterns: Moro, Core Tendon Guard, ATNR, Babkin Palmomental, Head Righting, Ocular Convergence/Divergence, Ocular-Vestibular, and Optokinetic.





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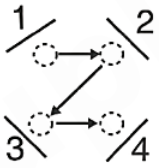
5.2. Moro Points

Position of the client: Lying supine (on back). Client can be sitting/standing if more comfortable.

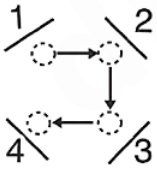
Technique: Deep tactile-proprioceptive activation of four points in the area of abdominal muscles *(m. rectus abdominis)* - around the navel. Points are found by: project a line connecting opposite side of the rib cage line (midpoint of right side rib cage) and hip bone (midpoint of the left hip line). Divide the part of the projected line from these bone lines to navel into three parts. The points are located one/third away from bone midpoint (and 3/4 away from navel in each direction). These points have links with Moro effect and predisposition for worries, fears, and phobias (this is based on observations, and is not a scientifically studied fact):

1) Moro Point-1 is on the upper part of the right *rectus abdominis* (observation shows its link with: fear of instability, changes, new decisions, transformation, adaptation to new environment).

2) Moro Point-2 is on the upper part of the left *rectus abdominis* (links with: fear/phobia of diseases, pain, suffering, fears created by scared mind, fear of unreal objects, heroes of scaring fairy tales, stories and movies, of 'unknown future', secondary beneficial fears/phobias, 'manipulative fears').

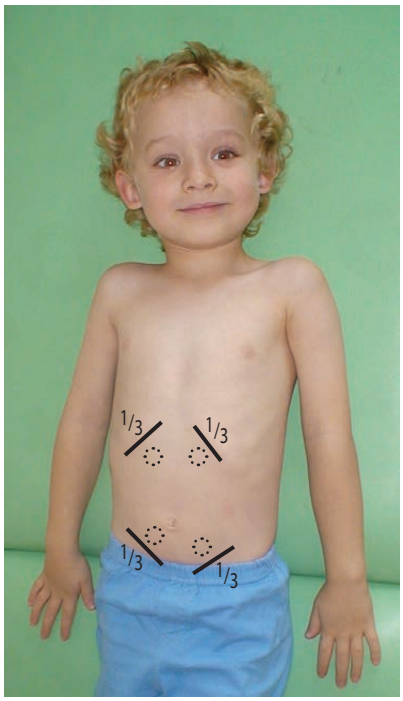


3) Moro Point-3 is on the lower part of the right *rectus abdominis* (link with: fear of letting go old information/relations and accepting new, fear of loss, instability, transformation).



4) Moro Point-4 is on the lower part of the left *rectus abdominis* (link with: fear of letting go and loss, fear of the end, also secondary beneficial fears/phobias).

Touch the point and press very deeply (with no pain) all the points one after another holding them 10-30 seconds each. Go deep increasing the pressure as though you want to project your pressure on the origin of the *psoas* muscle (upper Moro Points), and the direction of the lower Moro Points are projected towards the middle half of the *psoas*.



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FRONT: Moro Points

This technique allows regulation of the muscular-tendinous tone in the abdomen area, spine, and diaphragm, and whole body serving for triggering the biomechanics of the Moro Reflex pattern. This exercise also regulates the links and differentiation of the biomechanics of the Moro and Fear Paralysis Reflex patterns. It is aimed at: strengthening the gravity center and diaphragm; activation of the biomechanics of flexion of the body forward and extension backwards; balance of Fear Paralysis Reflex with Moro, Core Tendon Guard, and Trunk Extension patterns. For this exercise use the deep and slow pressure with fingers/ hands (touch # 1, 2, 3).

Position of specialist: Standing at the side close to the abdomen area where the technique is applied; facing the client.

Position of the hands: Using the fingers and palms of both hands, hold deep stably in the area of the four Moro Points. Hold fingers parallel to the closest bones.

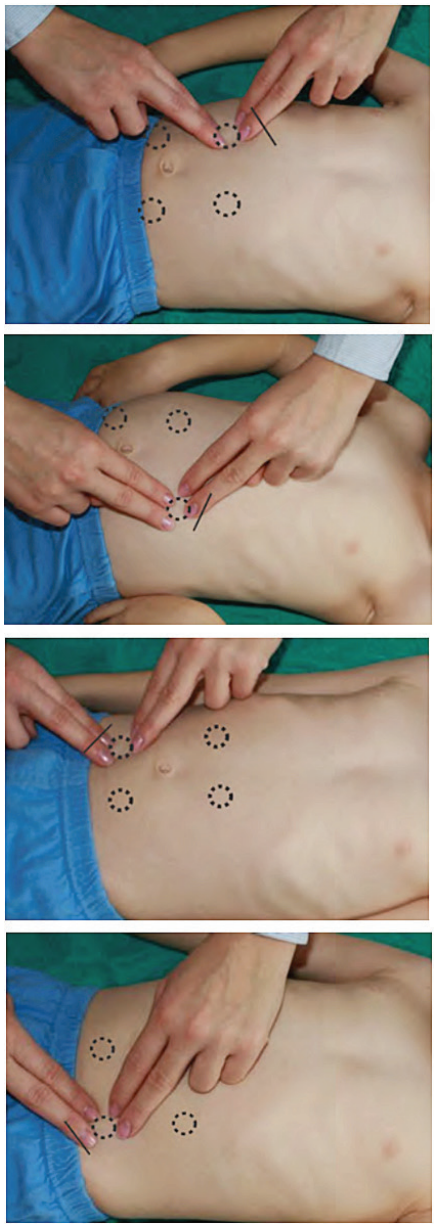
Action:

Moro points - slow deep pressure (on all four points on the abdominal muscles)

Press deeply very slowly increasingly with straightened fingers of your both hands in certain sequence: 1) Moro Point-1 (right upper side), 2) Moro Point-2 (left upper side), 3) Moro Point-3 (right lower side), and 4) Moro Point-4 (lower left side). Keep your pressure adequate to feelings of the client during 7-30 seconds. Repeat two times if needed. Be as delicate as possible when doing this intense technique.

Direction of applying the technique: Press very slowly with straightened fingers projecting to the line of the psoas muscle: 1) Moro Point-1 – in direction of the origin of the right side *psoas* muscle, 2) Moro Point-2 – in the direction of the origin of the left *psoas*, 3) Moro Point-3 – in the direction of half of the belly of the right *psoas* muscle in direction of growth to insertion, and 4) Moro Point-4 - in the direction of half of the belly of the left *psoas* muscle in direction of growth to insertion.

Note: This technique is essential in cases of improper functioning of the Moro Reflex (when Moro is dysfunctional/ pathological - becomes the emotional response instead of being the physiological fear as the result of the loss of the stable ground), also Red and Green Tendon Guard and other core reflex patterns: ATNR, Babkin Palmomental, Head Righting, Hands Supporting, Ocular Convergence/Divergence, Ocular- Vestibular, and Optokinetic.



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Exercise Effects: Moro Points

Nervous system

1) The CN X - Vagus
2) Phrenic nerve (diaphragm)

Muscle innervation:

1) m. obliguus externus abdominis
2) m. obliguus internus abdominis
3) m. transversus abdominis: –> n. intercostales, n. ilioinguinalis, n. iliohypogastricus
4) m. rectus abdominis: –> nn. intercostales (ThVIThXII), n. iliohypogastricus (ThXII-ThXII)

Skin innervation:

1) r.cutanei anterior et lateralis nn. intercostalium et iliohypogastrici

Circulatory system

1) m. obliguus externus abdominis: –> a. circumflexia ilica superficialis, a. intercostales posteriors, a. thoracica lateralis
2) m. obliguus internus abdominis
3) m. transversus abdominis:–> a. intercostales posteriors, aa. epigastricae superior et inferior, a. musculophrenica
4) m. rectus abdominis: –> aa.epigastricae superior et inferior, aa. intercostales posteriores

Lymphatic system

Surface vessels flowing in:

1) m. obliguus externus abdominis: –> n.lymphatici axillaris, epigastrici et inguinalis superficialis

Deep level of vessels flowing in:

n. lymphatici epigastrici, mediastinalis anterior et thoracicus

Respiratory system

When contracted, these muscles raise the ribs and sternum; they serve as additional muscles participating and helping in the action of breathing. They also activate the CN X - Vagus and the Phrenic nerve functions.

Digestive system:

1) m. obliguus externus abdominis
2) m. obliguus internus abdominis
3) m. transversus abdominis
4) m. rectus abdominis: –> participate in small and large intestine work and organize their peristaltic movements and elimination processes.

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6. ARMS – Butterfly Wings Activation

Goals:

- Release the negative effect of the Tendon Guard response in the upper section of body.
- Activate the dynamics of upper limbs reflex patterns such as: Hands Pulling, Hands Supporting, Hands Grasp, Head Righting, Babinski, Trunk Extension, and Tonic Labyrinthine.
- Stimulate the dynamic of upper gate (regarding construction referred to in Life Long Reflexes course) for work with Thomas Automatic Gait and Matured Gait.
- Activate coordination between homologous, homolateral, and cross-lateral links of the limbs.
- Activate coordination of core-limb pronation/supination abilities and rotation.
- Activate rotation and mobility of shoulders.
- Increase nerve innervation and blood circulation in muscles serving the upper limbs reflex patterns.
- Activate lymph and oxygen supply to muscles serving the upper limbs reflex patterns.
- Activate automatisms and self-regulating processes in the brain stem and balance mechanisms in the cerebellum.

Location: *Regio brachii anterior et posteriors*

Bones: *humerus*

Muscles:

Exercise # 9

1) Biceps m. (m. biceps brachii)
2) Coracobrachialis m. (m. coracobrachialis)
3) Brachial m. (m. brachialis)
4) Triceps m. (m. triceps brachii)

Also the next secondary muscles are involved:

6) Supraspinatus m. (m. supraspinatus)
7) Infraspinatus m. (m. infraspinatus)
8) Teres Minor m. (m. teres minor)
9) Teres Major m. (m. teres major)
10) Subscapularis m. (m. subscapularis)

It also effects the next muscles:

– *Anterior Serratus*
– *Latissimus Dorsi*
– *Trapezius*
– *Brachioradialus*
– *Pectoralis Minor*
– *Deltoids*

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6.1. Butterfly Wings Activation

Procedure

Position of the client: Lying down supine (on back, also on side). Client can be sitting/standing if more comfortable. The arm is raised up and abducted above the head opening the armpit area. The elbow is flexed comfortably.

Technique: Deep tactile-proprioceptive activation in the armpit and triceps area (*m. triceps*). Find the external lateral side of the scapula bone and midline of the connection of the front and back part of the body (thoracic lateral line = midline of side of the rib cage). Project and work with two lines around the armpit:

1) Line-1 is close to the edge of the scapula (posterior boundary of the armpit)

2) Line-2 is on the front side of the thoracic (anterior boundary of the armpit)

Touch firmly and delicately at the beginning of the lines – at the end of the scapula. Stretch these lines parallel way away from the armpit following with sequential touch/pressure in the direction of the triceps origin. Work, not touching the central area of the armpit. Hold 7-10 seconds for every touch along the line. Increase the pressure very lightly.

This technique allows the regulation of the muscular- tendinous tone in the shoulder, armpit, thoracic area, muscles around the neck, larynx, pharynx, and shoulder blades. This exercise regulates the links and differentiation of the biomechanics of all reflexes based on flexion and extension – Core Tendon Guard, Moro, Fear Paralysis, Trunk Extension, and Tonic Labyrinthine. It also works with all upper limbs reflex patterns activation.

For this exercise use deep and slow pressure with fingers/hands (touch # 2, 3).

Position of specialist: Stand close to the open shoulder area where the technique is applied; facing the center of the client's body.

Position of the hand/palm: Fingers of other hand are on one line, and other hand - on other line. Palms are turned away from the armpit ready for stretch of the lines away from the area of the armpit.

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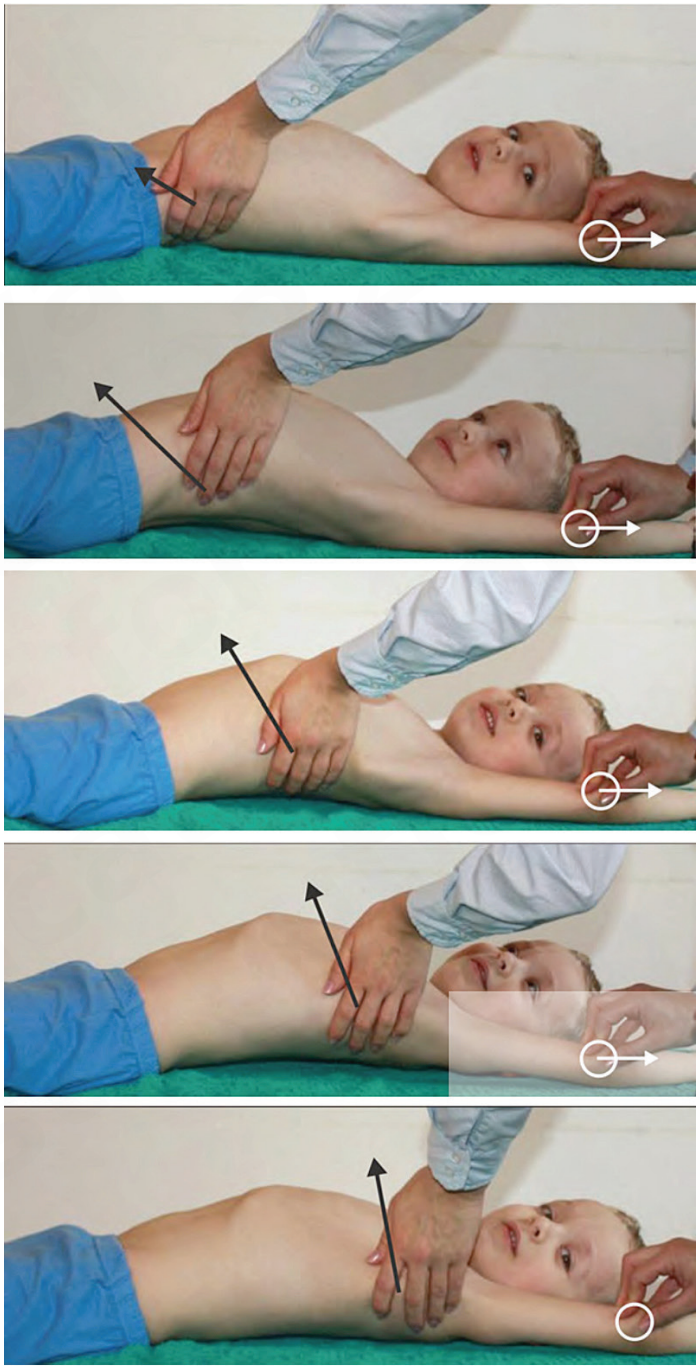
FRONT: Butterfly Wings

Action:

Raise one arm abducted above the head opening the armpit area with the elbow flexed in comfortable range over the head. Follow the next steps:

1) Introductory stretches:

- **Long stretches.** Hold the arm with three fingers at the origin of the *tricep* muscle, and press by other hand in the area above the pelvic bone on the side of the body, and do slow, long stretch for 8-14 seconds moving the side of the body down (inferiorly) and forward (anteriorly). Repeat 2-3 times.
- **Short stretches.** Do the same as above. One hand holds at the origin of the *tricep* muscle stably, and the other hand moves step by step closer to upper part of the chest. Repeat the same as above.



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FRONT: Butterfly Wings

6.1. Butterfly wings (upper arms)

2-1) Work with Golgi tendons and spindle cell aiming at relaxing and lengthening the tendons in areas around armpit. Begin with the anterior *serratus* and intercostals muscles. Move gently around the armpit area and reach the upper part of arm. Press/stretch during 7-10 seconds and repeat 2-3 times.

2-2) Work with triceps muscles in its tendons areas. Hold and press on tendons firmly and delicate way against of the origin and the insertion points. Hold 7-10 seconds and repeat 2-3 times.

3) Traction bringing arm back to core. Bring the arm back adducting it to the side of the body.

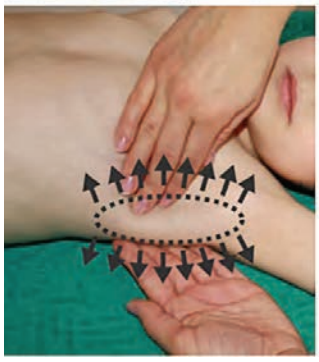
4) Cha-Cha on anterior deltoid. Work with anterior deltoid using fascia activation technique (90° into the muscle)

5) Bracioradialis muscle activation. Activate the muscle using "the stretch muscle away from the bone" technique. Sequentially stretch the muscle following from origin to insertion. Stretching is done against of fascia as you 'stretch the muscle away from the bone'.

Direction of applying the technique: At-a-Glance

1) Armpit area. Press and stretch the lower and upper lines simultaneously point by point very slowly with straightened fingers of both hands directed to anterior and posterior surface of the thoracic area.









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FRONT: Butterfly Wings

2) Triceps muscle tendons. Lengthen the muscle by the tendons at the intermediate zone of the origin and the insertion.

3) Anterior deltoid. Work using fascia activation technique - 90 degrees against of the muscle.

4) Bracioradialis muscle. Using the fascia activating technique stretch the muscle away from the bone by small sequential steps following from origin to insertion.





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Exercise Effects: Butterfly Wings

Nervous system

Muscles

1) m. biceps brachii
2) m. coracobrachialis
3) m. brachialis: –> n. musculocutaneus
4) m. triceps brachii: –> n. radialis
Also for muscles (Exercise #10):
5) m. supraspinatus
6) m. infraspinatus: –> n. suprascapularis
7) m. teres minor: –> n. axillaris
8) m. teres major
9) m. subscapularis: –> n. subscapularis

Circulatory system

1) m. biceps brachii: –> aa. collateralis ulnares superior et inferior, a. brachialis, a. reccurens radialis
2) m. coracobrachialis: –> aa. circumflexa anterior et posterior humeri
3) m. brachialis: –> aa. collaterale ulnares superior et inferior, a. brachialis, a. reccurens radialis
4) m. triceps brachii: –> a. circumflexa humeri posterior, a. profunda brachii, a collateralis ulnares superior et inferior
Also for secondary muscles:
5) m. supraspinatus
6) m. infraspinatus: –> a. suprascapularis, a. circumflexa scapulae
7) m. teres minor: –> a. circumflexa scapulae
8) m. teres major
9) m. subscapularis: –> a. subscapularis

Lymphatic system

Lymphatic vessels flow into armpit, neck and subscapular areas.

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7. Lower Limbs (anterior surface) Quadriceps, Lower Abdominal Sleep Posture, Fascia Lata, Peroneus

Goals:

- Release the negative effect of the Tendon Guard Reflex.
- Activate the muscular-tendinous system involved into: Leg Cross Flexion/Extension, Thomas Automatic Gait, Trunk Extension, lower limb part of ATNR, Moro, Tonic Labyrinthine, Fear Paralysis, Segmental Rolling, Spinning, Foot Tendon Guard, Babinski, and Foot Grasp Reflexes.
- Activate coordination of homologous, homolateral, and cross-lateral motions of the lower limbs.
- Activate coordination of core-limb links for biomechanics of flexion/extension, adduction/abduction and rotation in joints of the lower limbs.
- Activate gravity, grounding, stability, and dynamics of the lower limbs.
- Increase peripheral nerve innervations of muscle system serving reflex patterns activation.
- Strengthen and mature automatisms of the lower limb reflex patterns.
- Activate self-regulating processes in the brain stem and balance mechanisms in the cerebellum.
- Stimulate lymph and blood circulation, and oxygen supply for muscles serving for reflex patterns activation.



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7. 1. Quadriceps

Location: *Regio femoralis facies anterior et medialis*

Bones: *Femur*

Muscles:

1) m. quadriceps femoris: vastus intermedius, lateralis, medialis et rectus femoris
2) m. sartorius
3) m. pectineus
4) m. iliopsoas
5) m. adductor longus
6) m. gracilis
7) m. tensor fasciae latae

Also involves muscles:

Gracilus
Adductors

Procedures

Position of the client: Lying down supine (on back). Client can be sitting/standing if more comfortable.

Technique: Deep tactile-proprioceptive activation of the *quadriceps* muscle (*m. quadriceps)* and other muscles around it. Find the origin of the *quadriceps* from the lateral side of the leg and follow the work in next steps:

1) Preparation. Stroke along the muscle starting from the third of it (below the origin) in both directions – (up and down – proximally and distally) reach the spots close to the origin and insertion of the muscle.

2) Activation of muscle close to origin and insertion by pulling them away from the bone in a pleasant way. Use intermediate space between muscle fibers and tendons, where there are more Golgi and ligament receptors, and less of free nerve ends receptors (less protection),

3) Activation of the whole muscle. Move on using this technique along the whole muscle step by step working sequentially distally (down).

This technique allows regulation of the muscular-tendinous tone in the lower limb for such reflex patterns, as Automatic Gait, Leg Cross Flexion/Extension, Trunk Extension, Segmental Rolling, and Spinning. This exercise also regulates the links and differentiation of the biomechanics of all lower reflex patterns based on flexion and extension. It also works for coordination with core and upper limbs reflex patterns activation.

Position of specialist: Stand at the side close to the legs where the technique is applied; facing the center of the client's body.

Position of the hand/palm: Fingers and palms of both hands are lifting the muscle up and away from the bone in attempt to lengthen it and activate the peripheral nerve system of the upper part of the leg. For this exercise use the deep and slow pressure with fingers/palms (touch # 1, 2, 3 and palms).

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FRONT: QUADRICEPS

Action:

1) ***Preparation.*** Stroke along the muscle starting one-quarter of upper leg from the origin in both directions – up and down (proximally and distally) reach the spots close to the origin and insertion of the muscle,

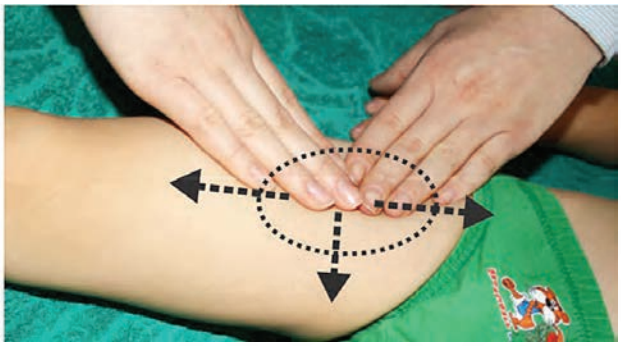
2) ***Activation of origin/insertion of the muscle.*** Hold the muscle in the area below the origin and above the insertion lifting the ends of the muscle away from the hip bone in a pleasant way for 7-10 seconds

3) ***Activation of the whole muscle.*** Move on using this technique along the whole muscle step by step working sequentially distally (down) in knee direction pausing at each step for 7-10 seconds. Work in a safe way pausing for longer time when sensitivity is hyper.

Direction of applying the technique:

1) ***Preparation.*** Stroke along the muscle proximally and distally, and reach the spots close to the origin and insertion of the muscle.





2) ***Activation of origin/insertion of the muscle.*** Hold the muscle in the area below the origin and above the insertion, lifting the ends of the muscle away from the hip bone.

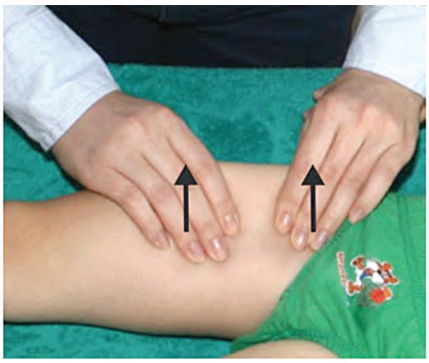


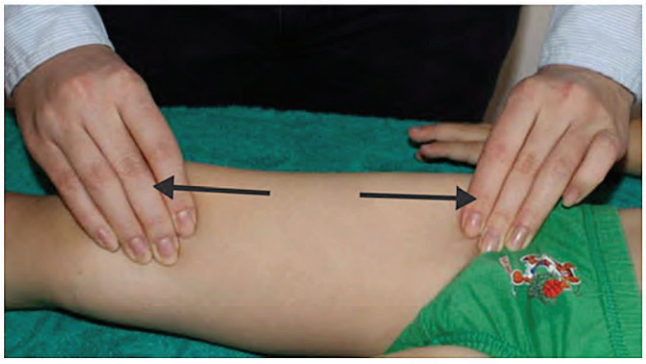


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3) Activation of the whole muscle. Move on, lifting the muscle away from the bone going distally (down) in knee direction.



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Exercise Effects: Quadriceps

Nervous system:

Muscle innervation:

1) *m. quadriceps femoris*
2) *m. sartorius: –> n. femoralis*
3) *m. pectineus*
4) *m. iliopsoas: –> plexus lumbalis*
5) *m. adductor longus*
6) *m. gracilis: –> n. obturatorius*
7) *m. tensor fasciae latae: –> n. gluteus superior*

Skin innervation:

rr. cutanei n. femoralis et n. obturatorius

Circulatory system

1) *m. quadriceps femoris: –> a. femoralis, a. profunda femoris*
2) *m. Sartorius: –> a. femoralis, a. circumflexa femoris lateralis*
3) *m. pectineus: –> a. obturatoria, a. pudenda externa, a. femoralis*
4) *m. iliopsoas: –> a. iliolumbalis, a. circumflexa ilium profunda*
5) *m. adductor longus*
6) *m. gracilis: –> a. obturatoria, a. pudenda externa, a. profunda femoris*
7) *m. tensor fasciae latae: –> a. glutea superior, a. circumflexa femoris lateralis*

Lymphatic system

1) *m. quadriceps femoris: –> nodi lymphatici inguinales superficiales inferiores*
2) *m. sartorius: –> nodi lymphatici inguinales superficiales superomediales et superolaterale*

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7.2. Lower Abdominal Sleep Posture Points Activation

Location: *Regio femoralis facies posterior et medialis*

Bones: *Femur*

Muscles:

1) m. biceps femoris: a) caput longum b) caput breve
2) m. semitendinosus
3) m. semimembranosus
4) m. adductor magnus
5) m. aductor brevis
6) m. gracilis

Procedures

Position of the client: Lying down supine (on back). Client can be reclined sitting/standing if more comfortable.

Technique: Deep tactile-proprioceptive activation of the three points of the Abdominal Sleep

Posture reflex pattern:

1) Point-1. Located at lower third line on *femur* above the knee on the *fascia lata line* (lateral).

2) Point-2 - Located at lower third line on femur above the knee on the inside of the leg across the adductor *brevis* and close to *gracilis* muscles (this point is parallel to Point-1).

3) Point-3 – Located at upper third of the *tibia* below the knee on the lateral side of the tibia on level of *peroneus longus m.* (this point is on lateral side only).

This technique activates the lower part of the Abdominal Sleep Posture Reflex pattern for regulation of the adrenalin and cortisol hormones, to calm down and release stress in the body. Work with these points effects the ATNR (lower part), Thomas Automatic Gait, Leg Cross Flexion-Extension, Babinski, Foot Tendon Guard, and Moro (lower part) Reflexes.

Position of specialist: Stand at the side close to the leg where the technique is applied; facing the center of the client's body.

Position of the hand/palm: Point 1 and 2 - Fingers and palms of both hands are pressing on the sides of the leg flatly (touch # 2, 3 and palms) and give circular motions upward toward the head.

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FRONT: Abdominal Sleep

Action

1) Point-1. Touch and press on the point-area above the knee on the fascia lata line with a deep, flat, and stable touch. Do 3-5 circular motions upward toward the head.

2) Point-2. Touch and press on the point-area above the knee on the inside of the leg across the adductor brevis muscle (this point is parallel to Point-1). Do 3-5 circular motions upward toward the head together with Point-1 simultaneously.

3) Point-3. Touch and press on the point-area, a third of the way below the knee on the lateral side of the tibia bone on level of peroneus longus m. (this point is on lateral side only). Do 3-5 circular motions upward toward the head.

Direction of applying the technique: Activate all the points with proprioceptive touch (deep, flat, stable). Do circular motions upward toward the head; reaching the anterior surface go in an upward direction (superior), and reaching the posterior surface – go in a down (inferior) direction.

Note: Pull the place of each point in upward – head direction, as an alternative.





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Exercise Effects: Lower Abdominal Sleep Posture Points Activation

Nervous system

Muscle innervation:

1) m. biceps femoris:

a) caput longum: –> n. tibialis

b) caput breve –> n.fibularis communis

2) m. semitendinosus

3) m. semimembranosus: –> n. tibialis

4) m. adductor magnus: –> n. obturatorius, n. ishiadicus

5) m. aductor brevis: –> n. obturatorius

6) m. gracilis: –> n. obturatorius

Skin Innervation:

rr. cutanei n. femoralis et n. obturatorius

Skin innervation:

rr. cutanei n. femoralis et n. obturatorius

Circulatory system

1) m. biceps femoris: a) caput longum: –> a. circumflexa femoris medialis, aa. perforantes

2) m. semitendinosus: –> aa. perforantes

3) m. semimembranosus: –> a. circumflexa femoris medialis, aa. perforantes, a. 68opliteal

4) m. adductor magnus

5) m. aductor brevis: –> a.obturatoria, a. perforantes

6) m. gracilis: –> a. obturatoria, a. femoralis, a. profunda femoris

Lymphatic system

1) m. biceps femoris: a) caput longum: –> nodi lymphatici inguinales superficiales inferiores

2) m. semitendinosus: –> nodi lymphatici inguinales superficiales superomediales et superolaterales

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7.3. Fascia Lata and Adductors

Location: R*egio femoralis facies posterior et medialis (et lateralis)*

Bones: *Femur*

Muscles:

1) m. quadriceps femoris: vastus intermedius, lateralis, medialis et rectus femoris
2) m. sartorius
3) m. pectineus
4) m. iliopsoas
5) m. adductor longus
6) m. gracilis
7) m. tensor fasciae latae

Procedures

Position of the client: Lying down supine (on back, on side). Client can be reclining/sitting/standing if more comfortable.

Technique: Deep tactile-proprioceptive activation of the lateral and inside median line of the *femoral* bone along the *fascia lata* and muscles in the adductor muscle area.

1) Use short touch-pressure with quick release technique going along the leg down (distally) in the direction of the knee joint.

2) When finished do stroking movements along the sides of the upper leg.

3) Additional Note: When the goal is to work with lymphatic drainage of the legs – go in opposite direction – proximally – from knee to hip direction.

This technique allows regulation of the muscular-tendinous tone in the lower limb for the following reflex patterns: Automatic Gait, Leg Cross Flexion/Extension, ATNR (lower part), Trunk Extension, Segmental Rolling, and Spinning. This exercise also regulates the links and differentiation of the biomechanics of all lower reflex patterns based on flexion and extension.

It also works for coordination with core and upper limbs reflex patterns activation.

Position of specialist: Stand at the side close to the leg where the technique is applied; facing the center of the client's body.

Position of the hand/palm: Fingers and palm of both hands are closed. Hold your hands pressing on the lateral and internal median sides of the upper leg, and using the quick release technique, go along the leg down (distally) in the direction of the knee joint. When finished, do stroking movements along the sides of the upper leg.

Note:

1) You can do circular motions in the head direction.

2) When the goal is to work with lymphatic drainage of the legs – go in opposite direction – proximally – from knee to hip direction and activate the peripheral nerve system of the upper part of the leg. For this exercise use the deep and slow pressure with fingers/palms (touch # 1, 2, 3 and palms).

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FRONT: Fascia Lata

Action

1) Quick release technique. Hold your hands on the lateral and internal median sides of the upper leg and with short touch-pressure using the quick release technique, go down (distally) in the direction of the knee joint. In case of hypersensitivity pause for longer time each time. Note: Circular motions can also be used– repeat 2-3 times, and move down step by step.

2) Stroking. Do stroking movements along the sides of the upper leg. In case of hypersensitivity pause for longer time each time.

3) Additional Note: Lymphatic drainage. When the work concerns the lymphatic drainage of the legs – go in opposite direction – proximally – from knee to hip direction. Do not go higher that the lateral origin of the adductor longus on the internal side of the upper leg (not to trigger the negative protection).

Direction of applying the technique:

1) Quick release technique. Apply the technique on the lateral and internal median sides of the upper leg going down (distally) in the direction of the knee joint.

Note: Circular motions can also be used– repeat 2-3 times, and move the same way - down.

2) Stroking. Next, do stroking movements along the sides of the upper leg.

3) Additional Note: Lymphatic drainage. When working with lymphatic drainage of the leg – go in opposite direction – proximally – from knee to hip direction.

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FRONT: Fascia Lata

Exercise Effects: Fascia Lata and Adductors

Nervous system

Muscle innervation:

1) m. quadriceps femoris
2) m. sartorius: –> n. femoralis
3) m. pectineus
4) m. iliopsoas: –> plexus lumbalis
5) m. adductor longus
6) m. gracilis: –> n. obturatorius
7) m. tensor fasciae latae: –> n. gluteus superior

Skin innervation:

rr. cutanei n. femoralis et n. obturatorius

Circulatory system

1) m. quadriceps femoris: –> a. femoralis, a. profunda femoris
2) m. sartorius: –> a. femoralis, a. circumflexa femoris lateralis
3) m. pectineus: –> a. obturatoria, a. pudenda externa, a. femoralis
4) m. iliopsoas: –> a. iliolumbalis, a. circumflexa ilium profunda
5) m. adductor longus
6) m. gracilis: –> a. obturatoria, a. pudenda externa, a. profunda femoris
7) m. tensor fasciae latae: –> a. glutea superior, a. circumflexa femoris lateralis

Lymphatic system

1) m. quadriceps femoris: –> nodi lymphatici inguinales superficiales inferiores
2) m. sartorius: –> nodi lymphatici inguinales superficiales superomediales et superolaterales

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7.4. Peroneus Longus Muscle Activation

Additional Goals

- Activate the dynamics of lower limb reflex patterns: Babinski, Foot Tendon Guard, Foot Grasp, Leg Cross Flexion/Extension, Automatic Gait, Trunk Extension, Head Righting, and Babkin Palmomental.
- Activate rotation and mobility of ankles (prevents twisting the ankle joints).

Location: *Regio cruris lateralis*

Bones: *Fibula*

Muscles:

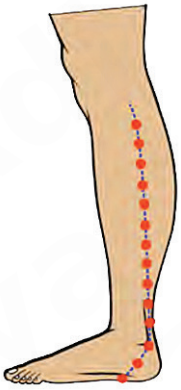
Lateral group of muscles of *tibia:*

1) m. peroneus longus
2) m. peroneus brevis

Procedures

Position of the client: Lying down supine (on back, or on side). Client can be sitting/standing if more comfortable.

Technique: Deep tactile-proprioceptive activation in the area of peroneus muscle (*peroneus longus and brevis*). Find the midline of the lateral side of the *tibia.* Project and work with the line along the peroneus muscle to half of the Achilles tendons distally:



1) The peroneus muscle fascia activation. Pressure straight against the fascia of this muscle is used. After pressure the quick release technique is applied.

2) Note: For hypertonic (too tense) peroneus muscle the same type of work oriented on elongating the muscle is used. The pressure is directed to the center of the muscle fibers increase the pressure. This technique is used when there is no pathological response. In cases of pathological response use the fascia activation technique (# 1).

3) Note: For hypotonic (too loose) peroneus muscle the same type of work oriented on contracting the muscle is used. The pressure is directed from the center of the muscle fibers and should be increased. This technique is used when there is no pathological response only. In cases of pathological response use the fascia activation technique (#1).

This technique allows regulation of the muscular-tendinous tone in the peroneus muscle, and other muscles around the tibia bone, balances links with triceps muscle also. This exercise also regulates the links and differentiation of the biomechanics of the lower limb reflex patterns based on inversion, rotation, flexion and extension. For this exercise use the deep and slow pressure with fingers/hands (touch # 2, 3).

Position of specialist: Stand at the side close to the lower leg area where the technique is applied; facing the client.

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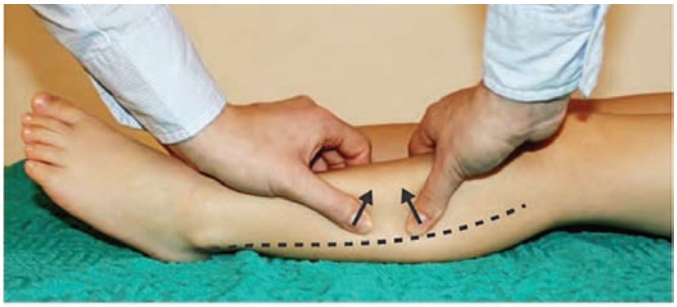
FRONT: Peroneus Longus

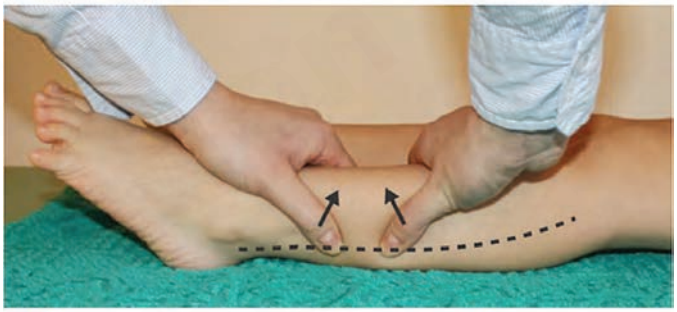
Position of the hand/palm:

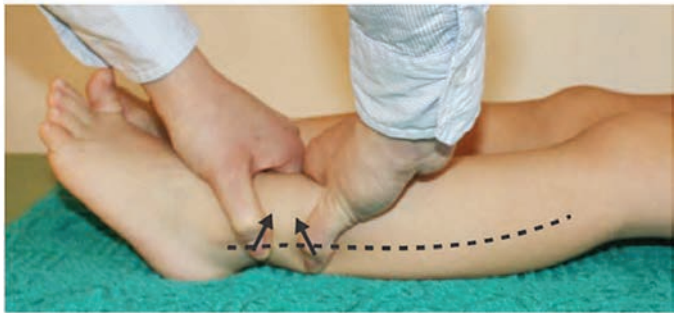
Fingers of both hands are on one line moving along the *peroneus* muscle. Hold your palms/fingers turned to each other by internal lateral surface. Fingers press along the muscle in specific direction indicated above.

Action

Find the midline of the lateral side of the *tibia*. Project and work with the line along the peroneus muscle to half way to the Achilles tendon distally.







1) Preparation. Stroke along the peroneus muscle 1-3 times distally (in direction of the foot) and stretch by pressing firmer on the origin and at the end on the level of half of the Achilles tendon for 5-7 seconds.

2) The peroneus muscle fascia activation. Using your thumbs or several fingers touch deeply and flat and press straight against the *fascia* of the *peroneus* muscle; use zone touch (no pointed touch to keep protection safe). Press along the muscle covering it by pressure within short distances – fingers are at a half-inch distance for children, and 1 inch for adults. After every pressure use slow release technique. With every touch increase the pressure sequentially lightly.

a. *Note: For hypertonic (too tense) peroneus muscle* you can use the same type of work contracting the muscle. Press toward the center of the muscle fibers. With every touch increase the pressure sequentially lightly. Do not use the quick release technique.

b. *Note: For hypotonic (poor) peroneus muscle y*ou can use the extension technique. Press with your fingers away from the center of the muscle fibers. With every touch increase the pressure sequentially lightly. Do not use the quick release technique.

c. *Note: For cases of dysfunctional* or pathological Babinski, Foot Tendon Guard, and Foot Grasp, push straight ahead against the facia of the peroneus muscle. With every touch increase the pressure sequentially. Use the quick release technique when hypotonic response is present.

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FRONT: Peroneus Longus

Direction of applying the technique:

1) Preparation. Stroke along the peroneus muscle (in direction of the foot) and stretch its by pressing firmer on the origin and at the end on the level of half way of the Achilles tendon against the center of the muscle. Do the stretch during 5-7 seconds.

2) The peroneus muscle fascia activation. Do the pressures straight against the fascia of the *peroneus* muscle against the fibula and tibia bone. After every pressure use quick release technique. Work going in down direction distally.

a. Note: For hypertonic (too tense) peroneus muscle use muscle contraction technique pressing in the direction of the center of the muscle fibers. Do not use quick release technique in this case.

b. Note: For hypotonic (poor) peroneus muscle use muscle extension technique pressing in the direction away from the center of the muscle fibers. Do not use quick release technique in this case.

c. Note: For cases of dysfunctional or pathological Babinski, Foot Tendon Guard, and Foot Grasp, push straight ahead against the facia of the peroneus muscle. With every touch increase the pressure sequentially. Use the quick release technique when hypotonic response is present.

Exercise Effects: Peroneus Longus Muscle Activation

Nervous system

Muscle innervation:

1) m. peroneus longus
2) m. peroneus brevis: –> n. peroneus superficialis

Skin innervation:

1) m. peroneus longus: –> n. cutaneous surae medialis
2) m. peroneus brevis: –> n. cutaneous surae lateralis pers distalis: n. suralis
3) n. saphenus from n. femoralis

Circulatory system

1) m. peroneus longus:–> a. inferior lateralis genus, a. fibularis
2) m. peroneus brevis: –> a. peronea

Lymphatic system

Vessels flowing to:

1) m. peroneus longus:–> nodi lymphatici genus capsularis, arteriosus et subfascialis

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8. Foot Reflex Patterns - Foot Tendon Guard, Babinski, Plantar Flexion/Extension, and Foot Grasp

Location: *Regio plantae*

Bones: *Ossa pedis*

Muscles:

Median:

1) m. abductor hallucis
2) m. flexor hallucis brevis
3) m. adductor halluces

Lateral:

4) m. abductor digiti minimi
5) m. flexor digiti minimi
6) m. opponens digiti minimi

Midline group:

7) m. flexor digitorum brevis
8) m. quadratus plantae
9) m. iumbricales
10) m. interossei plantares et dorsales

Procedures

Position of the client: Lying down supine (on the back). Client can be sitting/standing if more comfortable.

Technique: Activation of the reflex patterns and deep tactile-proprioceptive stimulation of the muscles serving the functioning:

1. Foot Tendon Guard

2. Babinski and rotation of the ankle joint (trigger for the rotation of the hip joint)

3. Plantar Flexion/Extension and Foot Grasp

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8.1 Foot Tendon Guard (FTG)

Position of specialist: Stand close to the foot where the technique is applied; facing the center of the client's body. If possible, client's leg is straightened, and foot is in its axis plane.

Position of the hand/palm:

A) Sensory-proprioceptive activation of the FTG points.

Pressure on the FTG Points at the location of group of muscles in the internal plantar surface of the foot: *m. abductor halluces, m. flexor hallucis brevis,* and stretch this plantar muscle.

B) Flexion of big toe and foot

One hand does light traction holding the foot maximum in its axis plane. The other hand is positioned to display the flexion of different parts of the foot.

Action

A) Sensory-proprioceptive activation of the FTG Points.

Touch and press the FTG Points by 2-3 fingers or thumbs of both hands at the location of the origin and insertion of group of muscles in the internal plantar surface of the foot: *m. abductor halluces, m. flexor hallucis brevis*. Stretch these plantar muscles away from their center in attempt to lengthen and raise the foot arch more. Use deep, flat, and slow pressure stretch (touch # 2, 3) for 7-10 seconds.

Repeat this exercise 7-8 times. This exercise must be deep, proprioceptive, and comfortable at the same time.

B) Motor activation of FTG reflex pattern by steps:

- Big toe; flex the big toe superiorly (towards the head)
- Foot; superiorly (towards the head)
- Big toe and foot simultaneously superiorly (towards the head)

One hand does light traction holding the foot by the heel bone at the maximum in its axis plane. The other hand is positioned to display the flexion of different parts of the foot as given above. Repeat every step seven times on one foot, and next on other. For children you can sequence the exercise doing 2-3 repetitions on one foot then on other. Do total 7-8 cycles of activation.

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FRONT: Foot – Foot Tendon Guard

Direction of applying the technique:

A) Sensory-proprioceptive activation of the FTG points. Touch and press the FTG Points and stretch the foot arch raising the foot arch more.

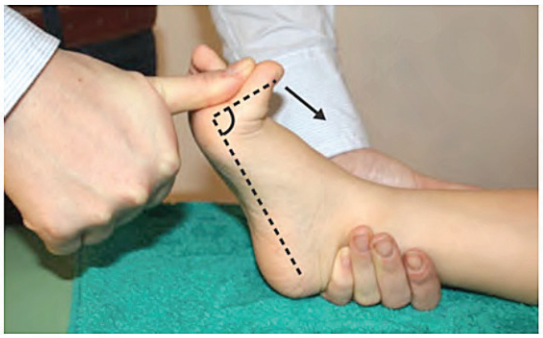
B) Motor activation of FTG reflex pattern:

- ***Big toe***
- ***Foot***
- ***Big toe and foot simultaneously***
- ***Big toe and foot simultaneously + knee***

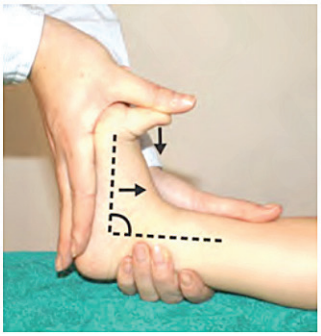
One hand does light traction holding the foot by the heel bone at maximum in its axis plane.

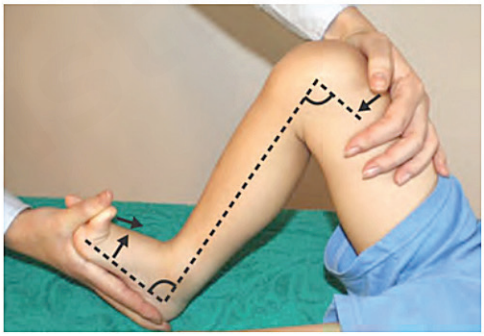
The other hand is positioned to display the flexion of different parts of the foot.











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8.2. Babinski

Position of specialist: Stand close to the foot where the technique is applied; facing the center of the client's body. If possible, the client's leg is straight, and foot is in its axis plane.

Position of the hand/palm:

A) Sensory-proprioceptive activation of the Babinski line. Slow deep pressure on the external surface of the foot for Babinski pattern trigger (touch # 1, 2).

B) Motor activation of Babinski pattern. Activation of the motor pattern for Babinski.

Action:

A) Sensory activation of the Babinski line. Stroke with slow and deep pressure on the external surface of the foot at the plantar line where the receptors for Babinski are located (touch # 1, 2). Repeat 5-7 times.

Provide stimulation for 7–21 seconds.

B) Motor activation of Babinski pattern. Bring the foot into the Babinski position:

1) Give light traction of the foot in the ankle joint.

2) Provide the inward rotation of the foot.

3) Stretch the digits in the metatarsis area fanning them and the toes. Hold the foot in this position firmly, give slow stretch at the *metatarsis* area, keeping the light traction of the ankle joint simultaneously (touch # 2, 3, and palm). Hold the position for 7-10 seconds. This exercise must be deep, proprioceptive, and comfortable and pleasant at the same time. For children, sequence the exercise doing 2-3 repetitions on one foot then on other. Do a total of 7-8 cycles of activation.

Direction of applying the technique:

A) Sensory activation of the Babinski line. Stroke along the external plantar surface of the foot from ***the heel to small toe (distally).***

B) Motor activation of Babinski pattern. Bring the foot into the Babinski position:

1) Light traction of the foot in the ankle joint.

2) Inward rotation of the foot (Note: it is not a plantar supination).

3) Stretch the digits in the *metatarsis* area fanning them and the toes. Hold the foot in this position.

Additional Note: Use the rotation of the feet in joints of *metatarsis* area and ankle joint with light traction. Do exercises for one foot, then repeat on other foot. Do slow circular motions four times inward and four times outwards (laterally) on three levels holding stably:

1) All toes of the foot

2) *Metatrsis* of the foot

3) Ankle joint





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8.3. Plantar Flexion/Extension & Foot Grasp

Position of specialist: Stand close to the foot where the technique is applied; facing the center of the client's body. Leg can be in any comfortable position.

Position of the hand/palm:

A) ***Metatarsis area activation.*** Fingers and hands are positioned to stretch the *metatarsis* area in the space between digits and toes across the horizontal plane of the foot. Stretching is done by two basic techniques:

1) Lateral stretch

2) Circular-lateral stretch (activating anterior and posterior planes of the foot) on three levels of toes, *metatarsis area (os metatarsale),* and closer to the ankle joint - *os cuneiforme and os suboideum.*

Action:

A) Metatarsis area activation. Touch firmly in every space/line between the digits of the *metatarsis* and toes and stretch across the horizontal plane of the foot. Stretch using these techniques:

1) Lateral stretch. Stretch in between each digit activating the ligaments of the *metatarsis and plantar* area. Work deeply and supplying feeling of comfort.

2) Circular-lateral stretch (activating anterior and posterior planes of the foot while stretch). Gently stretch the tendons moving up (superiorly) between the digits of the *metatarsis.* Repeat procedure in the spaces between each toe working outwards (touch # 2, and 3). Work slowly and deeply, aimed at activation of the proprioception, creating the feeling of comfort. Do 2-3 cycles of activation. For children you can sequence the exercise doing it on one foot then on other.

Work on all levels of bone connections of the foot:

- Toes
- *Metatarsis area (os metatarsale)*
- *Medial cuneiform and cuboid*

Direction of applying the technique:

A) Metatarsis area activation. Touch firmly in every space line between the digits of the *metatarsis* and toes and stretch it across the horizontal plane of the foot – one hand leads the stretch in direction of the median plane of the body and the other in the lateral direction using:

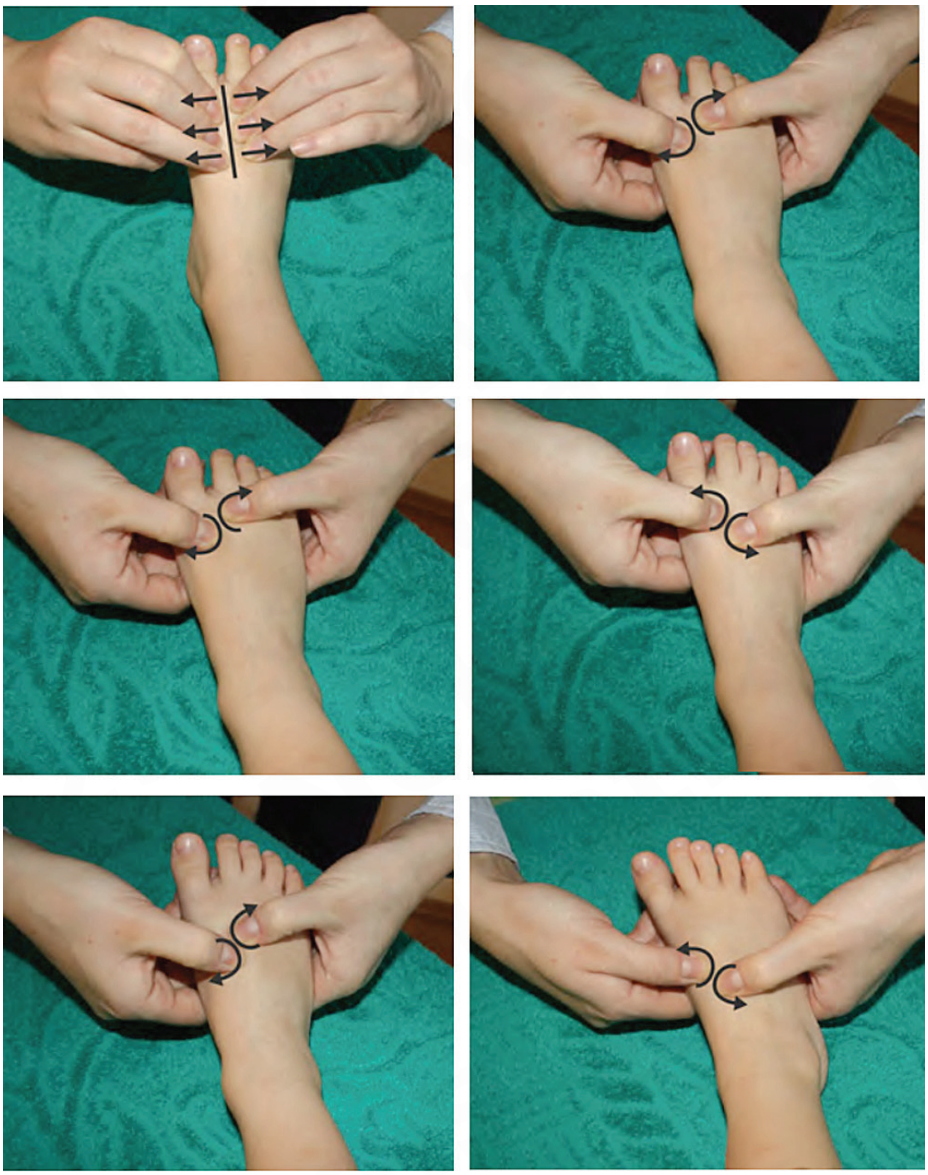
1) Lateral stretch
2) Circular-lateral stretch

Work on levels: toes, next – *metatarsis* area *(os metatarsale), os cuneiforme and os suboideum.*

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FRONT: Foot – Plantar Flexion/Extension & Foot Grasp



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8.4. Plantar Flexion for Foot Grasp

Instructions: The person is lying on their back or sitting. Work with one foot at a time. Invite them for passive activation (no resistance) for 5-7 seconds while exhaling. Repeat each exercise 3–7 times, allowing rest for the neurological system to absorb changes.





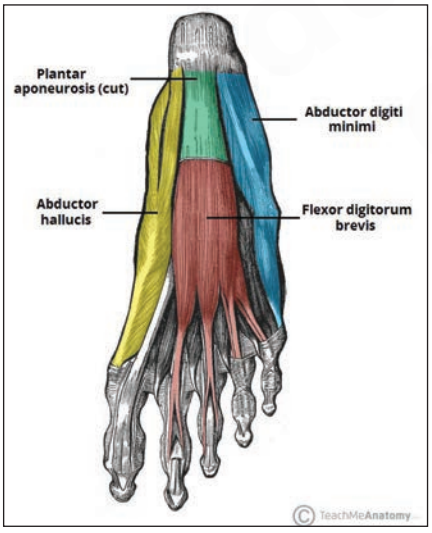
1. Press below the ball of the foot (pic. 1)

Sensory activation for Foot Grasp to activate the Foot Grasp pattern:

A. Pressure at point-zone below the ball of the foot. Press firmly (with no pain) on plantar in the area of the ball of the foot where the foot flexors are located *(tendo m. flexoris halluces longi, flexor halluces brevis, tendines m. flexoris, digitorum longi, tendo m. flexoris digitorum longi, mm. lumbricales).*

2. Plantar flexion for Foot Grasp (pic. 2)

Touch and hold firmly your two fingers of one hand under the ball of the foot. Flex the toes (1-4) in plantar direction delicate way with your other hand to keep the position of the foot grasp. At the end of this activation extend the whole foot and ankle joint slightly (direction is plantar and down) giving a light stretch (it also protects against of cramps).



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Exercises Effects: Foot Reflex Patterns

Nervous system

1) m. abductor halluces: –> n. plantaris medialis
2) m. flexor hallucis brevis: –> n. plantaris medialis et lateralis
3) m. adductor halluces
4) m. abductor digiti minimi
5) m. flexor digiti minimi
6) m. opponens digiti minimi: –> n. plantaris lateralis
7) m. flexor digitorum brevis: –> n. plantaris medialis
8) m. quadratus plantae: –> n. plantaris lateralis
9) m. iumbricales: –> n. plantaris medialis et laterali
10) m. interossei plantares et dorsales: –> n. plantaris lateralis

Circulatory system

1) m. abductor halluces: –> a.plantaris medialis
2) m. flexor hallucis brevis: –> a.plantaris medialis, arcus plantaris profundus
3) m. adductor halluces: –> arcus plantaris profundus, aa. metatarsales plantares
4) m. abductor digiti minimi
5) m. flexor digiti minimi
6) m. opponens digiti minimi: –> a. plantaris lateralis
7) m. flexor digitorum brevis: –> a. plantaris lateralis et medialis
8) m. quadratus plantae: –> a. plantaris lateralis
9) m. iumbricales: –> a. plantaris lateralis et medialis
10) m. interossei plantares et dorsales: –> arcus plantaris profundus, aa. metatarsales plantares

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BACK: GREETING – TACTILE ACTIVATION



9. Tactile-Proprioceptive Greeting for Back of the Body

Procedure

Greeting - Tactile Activation

Location:

1) Infraclavicularis
2) Deltoidea
3) Regio brachii anterior et
4) Posterior
5) Regio cubiti anterior et
6) Posterior
7) Regio antebrachii anterior et posterior

Bones: *Clavivula, humer, radius, ulna*

Exercise Effects

Nervous system

Skin innervation:

1) n. supraclavicularis, n. intercostales
2) n. cutaneus brachii lateralis superficialis nn. supraclaviculares
3) n. cutaneous brachii medialis
4) n. cutaneus brachii lateralis superior et inferior, n. cutaneus brachii posterior
5) n. cutaneus anterbrachii medialis et lateralis

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9.1. Tactile-Proprioceptive Greeting for Back of the Body

Procedure

Position of the client: Lying down prone (on stomach). Client can be sitting/standing if more comfortable.

Technique: Deep tactile-proprioceptive stroking along the limbs and back (posterior plane of the body) from the upper part of the back to down – arms and legs – distally.

Follow next two steps:

1) Upper limbs

a) Stroking along the arms. Put both your hands symmetrically above the shoulder blades, press firmly and giving the feeling of comfort, next move along both arms distally stroking them simultaneously with firm pressure (360 - 800gr/sm2);

b) Completing by arms traction. When reaching the wrists complete stroking with light traction of the arms.

c) Pressure on joints. While repeating the second and third time, press with a pleasant and embracing the joints (shoulder, elbows, wrists). Repeat B each time.

2) Back and lower limbs

a) Stroking along the back and legs. Put both your hands symmetrically above the shoulder blades, press firmly and giving the feeling of comfort, next move along the sides of the back (sacrospinalis group of muscles) and both legs distally stroking them simultaneously with firm pressure (360–800gr/sm^2);

b) Completing by legs traction. When reaching the ankle complete stroking with light traction of the legs.

c) Pressure on joints. While repeating the second and third time, press with a pleasant pressure on L5 and embracing the joints (hip, knee, ankle). Repeat B each time.

This technique allows to prepare the body to receive the next techniques for regulation of the muscular-tendoneous tone in the upper and lower limbs and trunk for functioning of these reflex patterns: Trunk Extension, Spinal Galant, Spinal Perez, Segmental Rolling, Spinning, Automatic Gait, Leg Cross Flexion/ Extension, and Crawling (lower part). This exercise also regulates the links and differentiation of the biomechanics of reflex patterns serving for flexion and extension, coordination of the trunk motions with upper and lower limb reflex patterns, and activates the peripheral nerve system of the posterior plane of the upper and lower parts of the body.

Position of specialist: Stand close to the upper body where the technique application starts, and moving down the body when work is applied on the lower part of the body; faced to the center of the body of the client.

Position of the hand/palm: Palms and fingers of both hands are touching and stroking the posterior plane of the body (palms are open and fingers more together). For this exercise use the deep and slow pressure and pulling with fingers/palms (touch # 2, 3 and palms).

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Back & Upper Limbs



1A

Pic. 1A) Light pressure above shoulder blades and move along shoulder.



1B

Pic. 1B) Close view.



2A

Pic. 2A) Continue to stroke along the arms.



2B

Pic. 2B) Close view.



3A

Pic. 3A) Continue to stroke along wrists and do light traction.



3B

Pic. 3B) Close view.

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Back and Lower Limbs

4A 

Pic. 5A) Light pressure on shoulders and move to sides of back.

4B 

Pic. 5B) Close view.

5A 

Pic. 6A) Continue stroking sides of back.

5B 

Pic. 6B) Close view.

6A 

Pic. 6A) Continue stroking along sides of hips.

6B 

Pic. 6B) Close view.

7A 

Pic. 7A) Continue stroking along the upper legs.

7B 

Pic. 7B) Close view.

8A 

Pic. 8A) Continue stroking to the knees.

8B 

Pic. 8B) Close view.

9 

Pic. 9) Continue stroking to the ankles.

10 

Pic. 10) Hold ankle joints and give light traction for 2-3 seconds.

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BACK: Spinal Perez

According to Reflex Pattern

1. Sensory stimulation of the Spinal Perez Reflex Pattern

Move your fingers firmly and very slowly along their spine area and sacrospinalis, upwards from the sacrum to the neck. Repeat several times slowly. This exercise is especially important for cases of hypersensitivity, hyperactivity, and tendency for emotional instability.



2. Sacrum and upper back activation

Apply pressure on their back in the T1-2 area and on the sacrum area. Invite your partner to push against your hands.



3. Sacrum activation

Using both hands, press on the upper sacrum area at a 45° angle toward the feet (inferior/ anterior direction). Ask your partner to move into your hands.



4. Upper back and legs activation

Apply pressure simultaneously on the upper back at T1-2 and the back of both ankles. Ask your partner to activate the reflex pattern by coming into the Spinal Perez position by moving into your hands. You can also do this exercise for each leg individually.



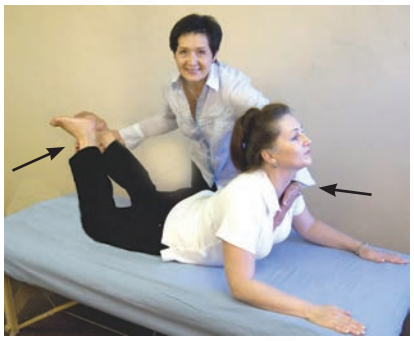
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Against Reflex Pattern

5. Activation of Perez from body front

Place one hand on the upper chest (upper sternum) and other hand under front of ankles. Invite your partner to activate the reflex pattern by moving into your hands.



You can also do this exercise for each leg individually.

Variants of Reflex Patterns

6. Sacrum and crossed ankles

Invite your partner to bend their legs at 90° or more at the knees and cross their ankles. Place one hand on the front of ankles and the other hand on the sacrum at a 45° angle. Invite your partner to move into an anterior pelvic tilt while applying leverage on the front of their ankles.

7. Sacrum and shoulder activation

Apply pressure on the sacrum and on the back of one shoulder. Invite them to move into your hands.



8. Hip and opposite shoulder activation

Press on the back of one shoulder and opposite hip. Invite them to press into your hands.



9. Leg and opposite shoulder activation from the back and then front

Simultaneously, apply pressure on one ankle and the opposite shoulder. Invite your partner to activate the reflex pattern by moving into your hands. Repeat for other leg and opposite shoulder. Do both combinations from two directions: First from the back and pushing downward (body front); then, from the body front and pushing upward (back of the body).



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10. Activation of front and back of the body simultaneously

Simultaneously, apply pressure on the upper chest and on the back of one ankle. Invite your partner to activate the reflex pattern by moving into your hands. Repeat with opposite leg.



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10. Perez - S2 Points Activation, Sacrum Triangle, Sacrum Squeeze

Goals

- Regulate the tone of the tendons and ligaments around the sacrum area and postural control of the trunk.
- Activate the proprioceptive system and body muscle-skeleton structure links for optimizing the automatisms of the reflex patterns concerned with the work of the sacrum – Spinal Perez, Spinal Galant, Trunk Extension, STNR, lower part of Moro and ATNR, Leg Cross Flexion/Extension.
- Optimize the functioning of Spinal Perez for activation of detoxifying mechanisms.
- Activate the receptors of deep touch and pressure.
- Stimulate cerebra-spinal fluid pumping mechanism.
- Stimulate the functioning of the whole body symmetry mechanism.
- Optimize the regulation of stress hormones letting go adrenalin and cortisol.
- Increase peripheral nerve innervations, lymphatic and blood circulation.
- Activate Spinal Perez Reflex pattern to increase the functioning of spine flexibility and proper spine development.
- Optimize the Spinal Galant Reflex pattern to activate the Beta and Alpha waves in the brain wave spectrum.
- Activate the Trunk Extension and STNR reflex patterns for balance of the visual converge and divergence.
- Activate spinal reflex patterns to prevent scoliosis, lordosis, and kyphosis pathologies.

Position of the client: In all exercises below the position is lying down prone (on stomach) or on the side. Client can be half-sitting/sitting leaned more forward (in cases of refusal to lie down or for reason of physical condition), or even lying down supine.

Technique: All the exercises below are tactile-proprioceptive activation in the area of the sacrum using deeper pressure, circular stimulation, stretch and compression-traction. These techniques allow regulation of the muscular-tendoneous tone in the sacrum, mobility of the spine and sacrum area, and structural links of reflex patterns concerned by lower and upper part of the body to function properly. It is aimed at activation of the receptors specialized for deep touch and pressure, and proprioceptors and ligament receptors.

Position of specialist: Stand at the side of the lower part of the body closer to the sacrum, where the technique is applied; faced to the center of the client's body.

Position of the hands: Fingers of both hands touch the certain are of the sacrum deeply and evenly. Use the techniques slowly with increasing pressure or stretch (touch # 2-3).

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10.1. Perez – S2 points activation

Goals

Location: *Regio sacralis*

Bones: *Os sacrum, os ilium*

Muscles:

1) m. latissimus dorsi

2) m. erector spinae: a) m. iliocostalis lumborum; b) m. longissimus thoracis

3) m. transversospinalis: a) m. multifidi lumborum; b) m. intertransversarii lumborum; c) m. rotatores lumborum

Action:

Place fingers in sacrum area at points S-2 (where gravity center, body weight and gravity line and body midline come together) and using a firm and flat touch (must be pleasant). Do 3-4 circular motions to one side, and then to other side. Move the circles up, over to the side of the body and down. Repeat 2-3 times. Use touch # 2 and 3.



Exercise Effects: Perez – S2 points activation

Nervous system

1) m. latissimus dorsi: –> n. thoracodorsalis

2) m. erector spinae:

a) m. iliocostalis lumborum,

b) ramus posterior nervi spinalis lumborum

3) m. transversospinalis: a) m. multifidi lumborum; b) m. intertransversarii lumborum; c) m. rotatores lumborum: –> ramus posterior nervi spinalis

Circulatory system

1) m. latissimus dorsi: –> a. thoracodorsalis, a. circumflexa humeri posterior a. intercostales posteriores

2) m. erector spinae:

a) m. iliocostalis lumborum: –> aa. lumbales a. intercostales posteriors;

b) m. longissimus thoracis:–> a. intercostales posteriores, a. lumbales

3) m. transversospinalis: a) m. multifidi lumborum; b) m. intertransversarii lumborum; c) m. rotatores lumborum: –> a. lumbales, a. intercostales posteriors

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10.2. Perez – Sacrum Triangle

Location: *Regio sacralis*

Bones: *Os sacrum, os ilium*

Muscles:

1) m. latissimus dorsi
2) m. erector spinae: a) m. iliocostalis lumborum
3) m. gluteus maximus
4) m. gluteus minimus
5) m. gluteus medius
6) m. piriformis
7) m. gemelli superior et inferior

Procedure

Action: Slow deep circular motions in sacrum triangle (in the area of the connection of sacrum bone and pelvic bone).

Exercise Effects: Perez – S2 points activation

A) Triangle of the sacrum. Start with one side of the triangle. Touch the line on the boundary of sacrum and do three circular motions moving your hands away from the center of the sacrum – to the sides of the body (12 o'clock at the sacrum bone).

B) Slow deep stretch: Continue with stretch in area of the line you worked in previous step. Press on the line (middle of the circular motions) and do deep and micro stretch in direction: inward – anterior, and lateral (90 degrees away from the sacrum bone) for 7-10 seconds. Repeat A and B sequentially 2-3 times on one side, then on other.

C) Top of the triangle. Do the same as in A) and B) – on the top of the sacrum triangle – the connection place of the sacrum and spine/back of the body.

Note: Do three circular motions sequentially in the direction of one side, and then another. Next do stretch in this area, holding the fingers (touch #1 and 2) between LV and S1 in an inferior direction towards the sacrum.



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Exercise Effects: Perez – Sacrum Triangle

Nervous system

1) m. latissimus dorsi: –> n. thoracodorsalis
2) m. erector spinae: a) m. iliocostalis lumborum: –>ramus posterior nervi spinalis lumborum
3) m. gluteus maximus: –> n. gluteus inferior
4) m. gluteus minimus,
5) m. gluteus medius: –> n. gluteus superior
6) m. piriformis: –> plexus sacralis SI -SIII
7) m. gemelli superior et inferior: –> plexus sacralis LIV-LV, SI-SIII

Circulatory system

1) m. latissimus dorsi: –> a. thoracodorsalis, a. circumflexa humeri posterior, a. intercostales posteriores
2) m. erector spinae: a) m. iliocostalis lumborum: –> aa. lumbales a. intercostales posteriores
3) m. gluteus maximus: –> a. glutea superior et inferior, a. circumlexa femoris medialis
4) m. gluteus minimus: –> a. glutea superior, a. circumflexa femoris lateralis
5) m. gluteus medius: –> a. glutea superior, a. circumflexa femoris lateralis
6) m. piriformis: –> a. glutea superior et inferior
7) m. gemelli superior et inferior: –> a. glutea inferior, a. pudenda interna, a. obturatoria

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10.3. Perez – Sacrum Squeeze, Pressure on Sacrum

Goals

Location: *Regio sacralis*

Bones: *Os sacrum, os ilium*

Muscles:

1) m. latissimus dorsi
2) m. erector spinae: a) m. iliocostalis lumborum
3) m. gluteus maximus
4) m. gluteus minimus
5) m. gluteus medius
6) m. piriformis
7) m. gemelli superior et inferior

Procedure

Action:

1) Sacrum squeeze. Place fingers of one hand in the area of both sides of the sacrum between S4 and S5 and fingers of the other hand in between S1 and S2, press anteriorly, and squeeze in 45° angle in the direction of S2 (where Gravity Center, Body Weight and Gravity line and body midline come together). Press using a firm, flat, and pleasant touch (touch #2 and 3). Repeat 2-3 times. This motion will allow a micro- stretch of the ligaments of the sacrum area and activate spine dynamic for Spinal Perez, Spinal Galant, Trunk Extension, and other reflex patterns.

2) Pressure on sacrum. Place the edge of the palm of one hand in the area of the sacrum between L_V-S_I and the other hand on the top of the first hand, press anteriorly and down - inferiorly 45° angle (Gravity Center, Body Weight and Gravity line and Body Midline come together). Press using a firm, flat, pleasant touch, and do a micro-stretch of the ligaments of the sacrum area. Repeat 2-3 times. This pressure will allow activation of the spine dynamics for Perez, Galant, Trunk Extension, and other reflex patterns.



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Exercise Effects: Perez – Sacrum Squeeze, Pressure on Sacrum

Circulatory system

1) m. latissimus dorsi: –> a. thoracodorsalis, a. circumflexa humeri posterior, a. intercostales posteriores

2) m. erector spinae: a) m. iliocostalis lumborum:–> aa. lumbales a. intercostales posteriores

3) m. gluteus maximus:–> a. glutea superior et inferior, a. circumlexa femoris medialis

4) m. gluteus minimus:–> a. glutea superior, a. circumflexa femoris lateralis

5) m. gluteus medius:–> a. glutea superior, a. circumflexa femoris lateralis

6) m. piriformis:–> a. glutea superior et inferior

7) m. gemelli superior et inferior:–> a. glutea inferior, a. pudenda interna, a. obturatoria

Nervous system

1) m. latissimus dorsi:–> n. thoracodorsalis

2) m. erector spinae: a) m. iliocostalis lumborum:–> ramus posterior nervi spinalis lumborum

3) m. gluteus maximus:–> n. gluteus inferior

4) m. gluteus minimus, 5) m. gluteus medius:–> n. gluteus superior

6) m. piriformis:–> plexus sacralis S_I-S_{III}

7) m. gemelli superior et inferior:–> plexus sacralis L_{IV}-L_V S_I-S_{III}

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According to Reflex Pattern

1. Sensory stimulation of the Spinal Galant Reflex pattern

With one hand based at T1, slowly stroke with your fingers of your other hand along the side of your partner's spine from the base of the neck to S1 at the top of the sacrum. Hold both points for seven seconds. Repeat this exercise 3–7 times. Repeat this exercise on the other side of the body.



Optional: Repeat sensory stimulation before each of the following steps.

2. Downward activation of shoulder and hip (passive/active)

Apply pressure on your partner's shoulder and hip downward in the direction of the body center, and invite them to activate the reflex pattern by holding the position and/or moving into your hand. Repeat this exercise on the other side of the body.



3. Flexion-Contraction of the side of the body

A. Passive. Contract or squeeze the side of the body by pushing hip and shoulder together. Ask your partner to stay in a relaxed state and passively experience the movement. Repeat this exercise on the other side of the body.



B. Active. Attempt to lengthen the side of the body by pushing the shoulder and hip away from the horizontal plane of the body. Ask your partner to move into your hands. Repeat this exercise on the other side of the body.



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Against Reflex Patterns

4. Extension of the side of the body

A. Passive. Lengthen the side of the body by pushing hip and shoulder away from the horizontal plane of the body. Ask your partner to stay in a relaxed state and passively experience the movement. Repeat this exercise on the other side of the body.



B. Active. Contract or squeeze the side of the body by pushing hip and shoulder together. Ask your partner to move into your hands. Repeat this exercise on the other side of the body.

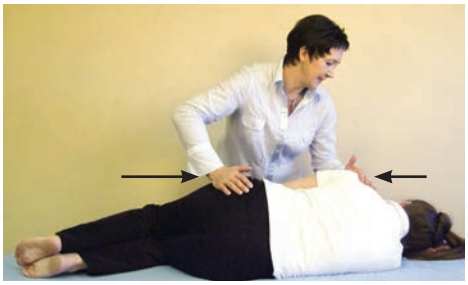


Alternative With/Against Patterns

5. Contraction-Extension of the side of the body sequentially (passive and active)

Perform exercises sequentially. Ask your partner to:

A. Passive. Stay in a relaxed state and passively experience the movement.





B. Active. Move into your hands.





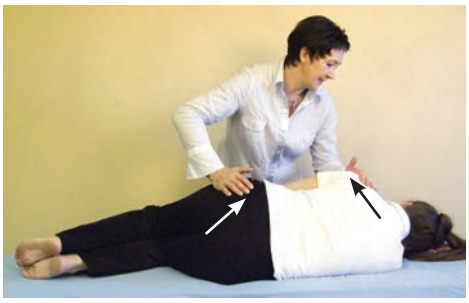
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Variants Supporting Reflex Pattern

6. Attempt to turn the body to stomach (passive and active)

Apply pressure at an inward 45-degree angle on the back shoulder area and hip as if trying to turn your partner onto their stomach. Ask them to activate the reflex pattern by holding the position and/or moving into your hand. Repeat this exercise on the other side of the body.



7. Attempt to turn the body to back (passive and active)

Apply pressure at an outward 45° angle on the shoulder area and hip from the front of the body, as if attempting to turn them onto their back. Ask your partner to activate the reflex pattern by holding the position and/or moving into your hand. Repeat this exercise on the other side of the body.



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11. Galant - Spinal Dominos, Spinal Squeeze

Position of specialist: Standing at the side of the body closer to the spine, where the technique is applied; faced to the center of the client's body.

Position of the hands: Fingers of both hands touch the specific areas along the side of the spine deeply and evenly. Use the techniques slowly with increasing pressure and stretch (touch # 1, 2-3).

11.1. Galant - Spinal Dominos
11.2. Spinal Squeeze

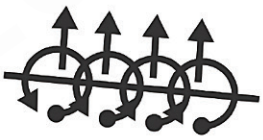
Action

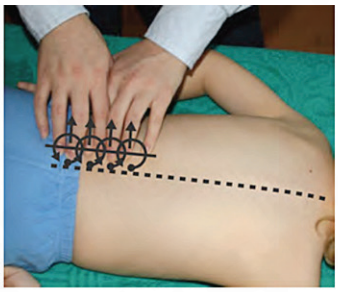
Slow deep circular motions along the spine column (in the area of the connection of spine column and lower part of the trunk, and with rib cage).

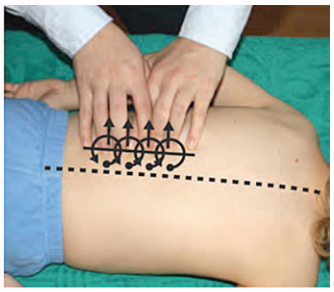
A) Circular activation. Start with one side of the spine column. Touch the line on the boundary between spine and sacrospinalis muscle, do three circular motions away from the spine column (12 o'clock at the spine vertebrae) starting at the lumber area and moving up (superiorly – neck direction) to the T_{I-II} area.

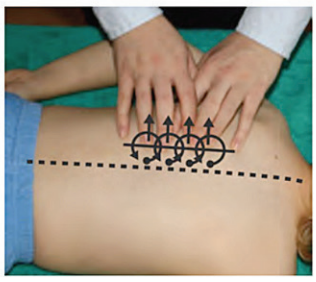
B) Slow deep stretch. Continue with stretch in area of the line you worked in previous step. Press on the line (middle of the circular motions) and do deep pressure to inward – anteriorly, and micro stretch laterally (90 degrees away from the spinal column) during 7-10 seconds (touch #1 and 2). Repeat A and B sequentially 2-3 times on one side, then on other.

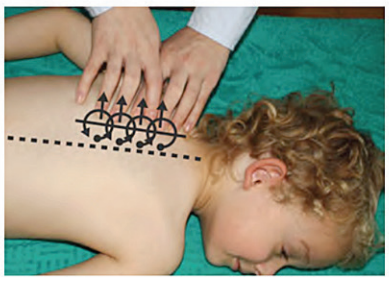
Move in section of the spine. While working, give some pause for creating the ease in the perception of the depth of the pressure. Usually this exercise feels comfortable and releases negative tendency of the guard response and somatic stress.









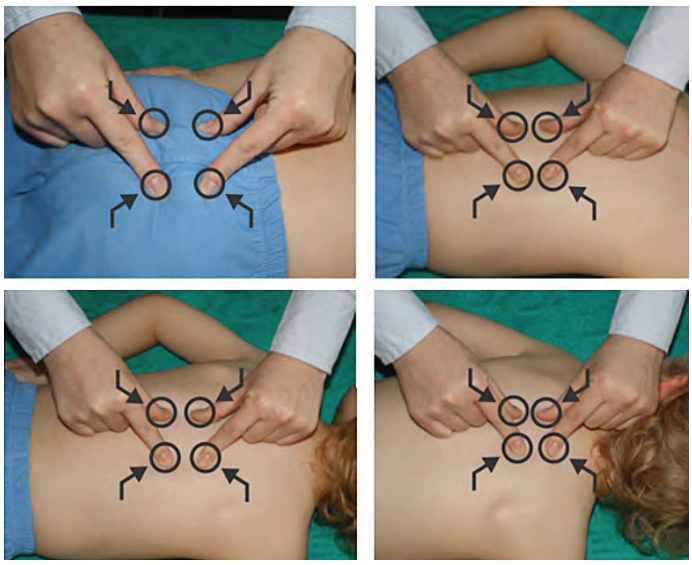


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BACK: Spinal Squeeze

Note. You can work between every vertebrae on both sides of the spine. Use your thumb and first 1-2 fingers doing circular movements on both sides of the spine column simultaneously. Work following the levels of the spinal column.



Levels

I. Lumbar area

Location: *Columna vertabralis, regio lumbalis,*

Bones: *columna vertabralis*

Muscles:

1) m. latissimus dorsi
2) m. erector spinae: a) m. longissimus thoracic
3) m. transversospinalis:
a) m. multifidi
b) m. rotatores lumborum
c) m. interspinales lumborum
d) m. intertransversarii iumborum
4) m. serratus posterior inferior

Exercise Effects

Nervous system

1) m. latissimus dorsi: –> n. thoracodorsalis
2) m. erector spinae
3) m. transversospinalis: –> ramus posterior nervi spinalis lumborum
4) m. serratus posterior inferior: –> nn. intercostales (Th_{IX}-Th_{XII})

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BACK: Spinal Squeeze

Circulatory system

1) m. latissimus dorsi: –> a. thoracodorsalis, a. circumflexa humeri posterior, a. intercostales posteriores
2) m. erector spinae
3) m. transversospinalis: –> aa. lumbales a. intercostales posteriores
4) m. intertransversarii iumborum: –> a. intercostales posterior

II. Thoracic area

Location: *Regio thoracicus*

Bones: *Columna vertabralis, costae*

Muscles:

1) m. latissimus dorsi
2) m. serratus posterior inferior
3) m. romboidei major et minor
4) m. serratus posterior superior
5) m. splenius capitis et cervicis
6) m. erector spinae: m. iliocostalis thoracis, m. longissimus cervicis, m. longissimus capitis, m. spinalis thoracis et cervicis
7) m. transversospinalis: m. semispinalis thoracis, m. multifidi, m. rotatores thoracis, m. interspinalis, m. intertransversarii thoracis

Work sequentially: part-1 – L_V-L_I; part-2 – T_{XII}-TH_{VI}; part-3 – T_V-T_I. For tall clients, divide the spine into more levels.

Exercise Effects

Nervous system

1) m. latissimus dorsi: –> n. thoracodorsalis
2) m. serratus posterior inferior: –> nn. intercostales (ThIX-ThXII)
3) m. romboidei major et minor: –> n. dorsalis scapulae
4) m. serratus posterior superior: –> nn. intercostales (ThI-ThIV)
5) m. splenius capitis et cervicis
6) m. erector spinae
7) m. transversospinalis: –> posterior nervi spinalis

Circulatory system

1) m. latissimus dorsi: –> a. thoracodorsalis, a. circumflexa humeri posterior, a. intercostales posteriores
2) m. serratus posterior inferior: –> a. intercostales posterior
3) m. romboidei major et minor: –> a. intercostales posterior, a. suprascapularis, a. transversa cervicis
4) m. serratus posterior superior: –> a. intercostales posterior, a. cervicalis profunda
5) m. splenius capitis et cervicis –> a. occipitalis, a. cervicalis profunda
6) m. erector spinae
7) m. transversospinalis: –> a. intercostales posterior, a. cervicalis profunda, a. lumbales

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BACK: Galant – Spinal Squeeze

III. Cervical area

Location: *Regio cervicis*

Bones: *Columna vertebralis*

Muscles:

1) m. trapezius
2) m. splenii capitis et cervicis
3) m. serratus posterior superior
4) m. erector spinae: m. iliocostalis cervicis, m. longissimus capitis, m. spinalis capitis et cervicis
5) m. transversospinalis: m. semispinalis cervicis et capitis, m. multifidi, m. rotatores cervicis, m. interspinalis cervicis, m. intertransversarii cervicis

Exercise Effects

Nervous system

1) m. trapezius: –> n. accessories, plexus cervicalis
2) m. splenii capitis et cervicis
3) m. serratus posterior superior: –> nn. intercostales (ThI-ThIV)
4) m. erector spinae
5) m. transversospinalis: –> r. posterior nervi spinalis

Circulatory system

1) m. trapezius: –> a. transversa cervicis, a. occipitalis, a. suprascapularis, aa. intercostales posteriores
2) m. splenii capitis et cervicis: –> a. occipitalis, a. cervicalis profunda
3) m. serratus posterior superior: –> a. intercostales posterior, a. cervicalis profunda
4) m. erector spinae
5) m. transversospinalis: –> a. intercostales posterior, a. cervicalis profunda, a. lumbales

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12. Scapular Area Mobilization – Abdominal Sleep Posture Reflex Pattern

Goals

- Regulate the work of deep touch and pressure receptors, tone of the tendons and ligaments around the scapular area, and postural control of the thoracic and other levels of the spine.
- Activate the proprioceptive system and body muscle-skeleton structure links for optimizing the automatisms of the reflex patterns concerned with the work of the arms and whole body: Hands Supporting, Hands Pulling, Trunk Extension, STNR, Spinal Galant, the upper part of Moro and ATNR, and Core Tendon Guard.
- Stimulate functioning of the upper body symmetry mechanism.
- Optimize the regulation of the Abdominal Sleep Posture Reflex for stress hormones (adrenalin and cortisol) release.
- Increase peripheral nerve innervations, lymphatic and blood circulation.
- Activate the functioning of spine flexibility and proper spine development.
- Optimize the work of reflex patterns triggering convergence and divergence.

12.1. Circles and Scapular Stretch

Location: *Regio scapularis*

Bones: *Scapulae*

Muscles:

1) m. supraspinatus
2) m. infraspinatus
3) m. teres minor
4) m. teres major
5) m. subscapularis

Action:

Circular activation of the median side of the scapular. Start with median plane of the scapular bone. Touch the line on the boundary of scapular and do three circular motions moving your hands in upward direction and away from the spine, next to the sides of the body (12 o'clock at the scapular bone).

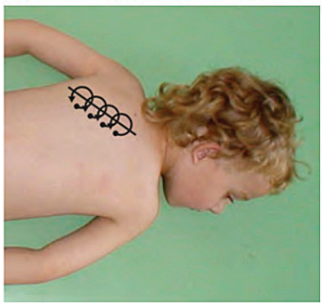
A) Slow deep stretch

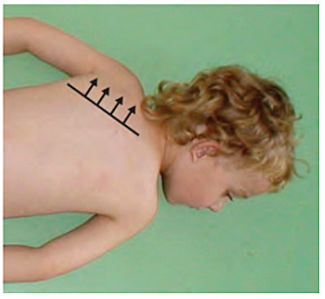
Continue with stretch in area of the line you worked in previous step. Press on the line (middle of the circular motions) and do deep and micro stretch in direction: inward – anteriorly, and laterally (90 degrees away from the scapular bone) for 7-10 seconds. Repeat A and B sequentially 2-3 times. Do the same on other scapular.

Note: Usually a deeper, and flat pressure is effective. Use the pressure appropriate for clients needs.









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12.2 Scapular Points Activation

Activate the points in the middle of the scapular, and half on the median and lateral sides of scapular bone. Do 5-7 circular motions at each point. Repeat 2-3 times. Do the same on other scapular.

Note: These points can be activated one by one, all together, or symmetrically on both scapula simultaneously (touch #1 and 2). While working, give some pauses.

Exercise Effects

Nervous System

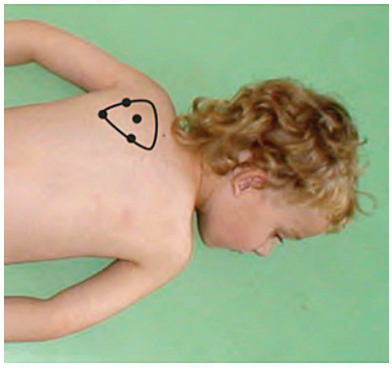
1) m. supraspinatus
2) m. infraspinatus: –> n. suprascapularis
3) m. teres minor: –> n. axillaris
4) m. teres major
5) m. subscapularis: –> n. subscapularis

Circulatory system

1) m. supraspinatus
2) m. infraspinatus: –> a. suprascapularis, a. circumflexa scapulae
3) m. teres minor: –> a. circumflexa scapulae
4) m. teres major
5) m. subscapularis: –> a. subscapularis

Lymphatic system

Lymphatic vessels flowing in to armpit, thoracic, neck and scapular nodes.



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13. Abdominal Sleep Posture (upper part)

Procedure

Step -1 Quadratus Lumborum Point Activation

Location: *Regio lumbalis*

Bones: *Columna vertebralis, os ilium*

Muscles:

1) m. latissimus dorsi
2) m. erector spinae: a) m. iliocostalis lumborum
3) m. transversospinalis: a) m. multifidi, b) m. rotatores lumborum, c) m. interspinales lumborum, d) m. intertransversarii iumborum
4) m. serratus posterior inferior
5) m. quadratus lumborum

13.1. Abdominal Sleep Posture (upper part)

Action

1A) Circular activation of the Abdominal Point-1 (m Quadratus Lumborum)

Touch the point-zone in the third of the *Quadratus Lumborum* muscle, and do three circular motions moving your hands in the direction away from the spine (12 o'clock is the direction of the spine column).

1B) Slow deep stretch

Continue with stretch in area of the point worked in previous step. Press on the point (middle of the circular motions) and do deep stretch in direction of the growth of the fibers: inward – anteriorly, inferiorly and laterally for 7-10 seconds. Repeat A and B sequentially three times: three circular motions and one stretch. Next, work with Rhomboid muscle area (close to scapular).

Note: Usually a deeper, and flat pressure is effective. Use the pressure appropriate for client's needs.

Step-2. Romboideus point activation

Location: *Regio dorsalis*

Bones: *Scapula, costae*

Muscles:

1) m. trapezius
2) m. romboideus major et minor
3) m. erector spinae: m. iliocostales cervicis et capitis, m. longissimus cervicis et capitis
4) m. splenius cervicis
5) m. serratus posterior superior
6) m. transversospinalis: m. semispinalis cervicis, mm. multifidi, m. rotatores cervicis, m. interspinalis et intertransversarii cervicis

Action

1A) Circular activation of the Rhomboid Point-3 *(m. romboideus)* Touch the point-zone in the half of the *m. romboideus* (the level of the middle of the muscle), and do three circular motions moving your hands in the direction away from the spine (12 o'clock is the direction of the spine column).

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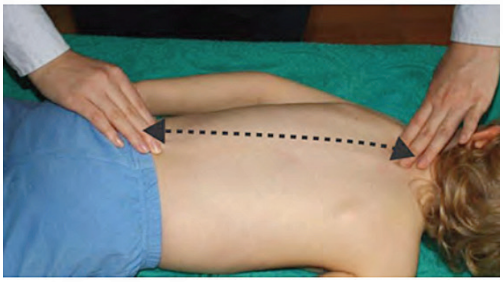
1B) Slow deep stretch

Continue with stretch in area of the point worked in previous step. Press on the point (middle of the circular motions) and do deep stretch in direction of the growth of the fibers: inward – anteriorly, inferiorly and laterally for 7-10 seconds. Repeat A and B sequentially three times: three circular motions and one stretch. Next, work with *Rhomboid* muscle area (close to scapular).

Note: Usually a deeper, and flat pressure is effective. Use the pressure appropriate for client's needs.







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BACK: Abdominal Sleep

Exercise – Step-1 Effects

Nervous system

1) m. latissimus dorsi: –> n. thoracodorsalis
2) m. erector spinae: a) m. iliocostalis lumborum: –> ramus posterior nervi spinalis lumborum
3) m. transversospinalis: a) m. multifidi: –> ramus posterior nervi spinalis lumborum
4) m. serratus posterior inferior: –> nn. intercostales (Th_{IX}-Th_{XII})
5) m. quadratus lumborum –> plexus lumbalis (Th_{XII}, L_I-L_{II})

Circulatory system

1) m. latissimus dorsi: –> a. thoracodorsalis, a. circumflexa humeri posterior a. intercostales posteriores
2) m. erector spinae: a) m. iliocostalis lumborum: –> aa. lumbales a. intercostales posteriores
3) m. transversospinalis: –> aa. lumbales a. intercostales posteriores
4) m. serratus posterior inferior: –> a. intercostales posterior
5) m. quadratus lumborum: –> a.subcostalis, aa. lumbales, a. iliolumbalis

Exercise - Step 2 Effects

Nervous system

1) m. trapezius: –> n. accessories, plexus cervicalis
2) m. romboideus major et minor: –> n. dorsalis scapulae
3) m. erector spinae
4) m. splenius cervicis
5) m. serratus posterior superior: –> nn. intercostales (Th_I-Th_{IV})
6) m. transversospinalis: –> r. posterior nervi spinalis

Circulatory system

1) m. trapezius: –> a. transversa cervicis, a. occipitalis, a. suprascapularis, aa. intercostales posteriores
2) m. romboideus major et minor: –> a. transversa cervicis, a. suprascapularis, aa. intercostales posteriores
3) m. erector spinae
4) m. splenius cervicis: –> a. occipitalis, a. cervicalis profunda
5) m. serratus posterior superior: –> a. intercostales posterior, a. cervicalis profunda
6) m. transversospinalis: –> a. intercostales posterior, a. cervicalis profunda, a. lumbales

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14. LOWER LIMBS: BACK OF LEGS

Lower Limbs (posterior surface) - Hamstring, Calf Muscles, Achilles Tendons

Goals

- Release the negative effect of the Tendon Guard Reflex.
- Activate the muscular-tendinous system for: Leg Cross Flexion/Extension, Thomas Automatic Gait, Trunk Extension, lower limb part of ATNR, Moro, TLR, Fear Paralysis, Segmental rolling and Spinning, also Foot Tendon Guard, Babinski, Foot Grasp.
- Activate coordination of homologous, homolateral and cross-lateral motions of the lower limbs.
- Activate coordination of core-limb links for biomechanics of flexion/extension, adduction/abduction and rotation of joints of the lower limbs.
- Activate gravity, grounding, stability, and dynamics of the lower limbs.
- Increase peripheral nerve innervations of muscle system serving for reflex pattern activation.
- Strengthen and mature automatisms of the lower limb reflex patterns.
- Activate self-regulating processes of the brain stem and balance mechanisms in the cerebellum.
- Stimulate lymph and blood circulation, and oxygen supply for muscles serving for reflex patterns activation.

Position of the client for all the exercises: Lying down prone (on stomach). Client can be on side (in cases of refusal to lie down on stomach or for reason of physical condition, feeding tube discomfort).

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14.1. Hamstring Muscles Activation

Location: *Regio femoralis facies posterior et medialis*

Bones: *Femur*

Muscles:

1) m. biceps femoris: a) caput longum b) caput breve
2) m. semitendinosus
3) m. semimembranosus
4) m. adductor magnus
5) m. aductor brevis
6) m. gracilis

Procedure

Technique: Deep tactile-proprioceptive activation of the hamstring muscle (m. biceps femoris) and other muscles around it. Find the origin of this muscle and go in the direction of the insertion – distally, following:

1) Preparation. Stroking along the muscle starting from the third of it (below the origin) in up and down (proximally and distally).

2) Activation of muscle close to origin and insertion by pulling up away from the bone.

3) Activation of the whole muscle. Use of the above technique (# 2) along the whole muscle sequentially distally (down).

This technique allows to regulate the muscular-tendinous tone in the upper lower limb for the reflex patterns: Automatic Gait, Leg Cross Flexion/Extension, Trunk Extension, Segmental Rolling, Spinning, and Crawling (lower part). This exercise also regulates the links and differentiation of the biomechanics of all lower reflex patterns based on flexion and extension. It also works for coordination with core and upper limbs reflex patterns activation.

Position of specialist: Stand close to the upper leg where the technique is applied; facing the center of the client's body.

Position of the hand/palm: Fingers and palms of both hands are pulling the muscle away from the bone in attempt to lengthen it and activate the peripheral nerve system of the upper part of the leg. Use deep and slow pressure, pulling with fingers/palms (touch # 2, 3 and palms).

Action:

1) Preparation. Stroke along the muscle starting from the third of it (below the origin) in both directions simultaneously – up and down (proximally and distally) reach the spots close to the origin and insertion of the muscle.

2) Activation of origin/insertion of the muscle. Hold the muscle in the area below the origin and above the insertion lifting the ends of the muscle away from the hip bone pleasant way for 7-10 seconds. Use intermediate space between muscle fibers and tendons, where there are more Golgi and ligament receptors, and less free nerve ends receptors (less of protection).

3) Activation of the whole muscle. Move on using this technique along the whole muscle step by step working sequentially distally (down) in knee direction pausing at each step for 7-10 seconds. Work safe way pausing for longer time when sensitivity is hyper.

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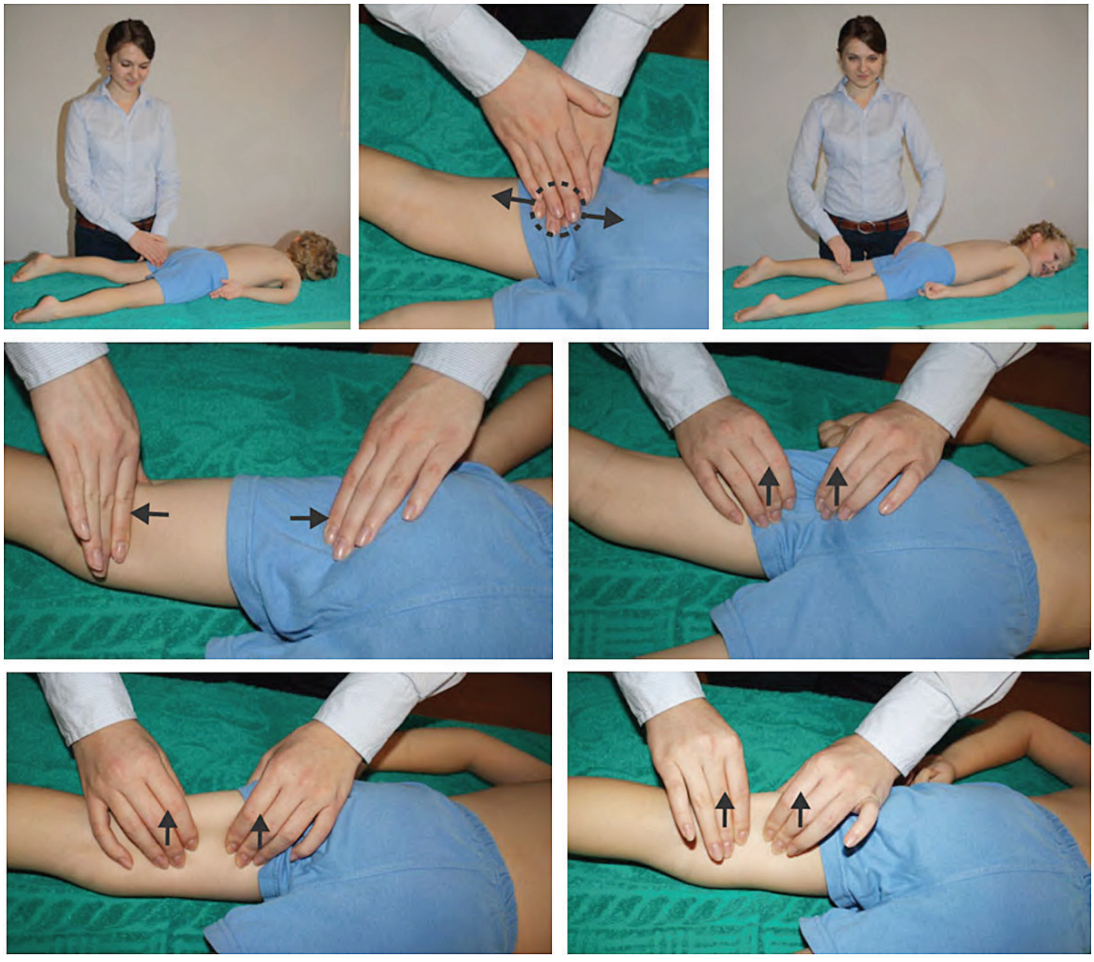
BACK: HAMSTRING

Direction of applying the technique:

1) Preparation. Stroke along the muscle proximally and distally simultaneously, and reach the spots close to the origin and insertion of the muscle.

2) Activation of origin/insertion of the muscle. Hold the muscle in the area below the origin and above the insertion, pulling the ends of the muscle away from the hip bone.

3) Activation of the whole muscle. Move on pulling the muscle away from the bone going distally (down) toward the knee.



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Exercise Effects: Hamstring Muscles Activation

Nervous system

Muscle innervation:

1) m. biceps femoris: a) caput longum: –> n. tibialis, b) caput breve: –> n.fibularis communis
2) m. semitendinosus
3) m. semimembranosus: –> n. tibialis
4) m. adductor magnus: –> n. obturatorius, n. ishiadicus
5) m. aductor brevis: –> n. obturatorius
6) m. gracilis: –> n. obturatorius

Skin innervation:

rr. cutanei n. femoralis et n. obturatorius

Circulatory system

1) m. biceps femoris: –> a. circumflexa femoris medialis, aa. perforantes
2) m. semitendinosus: –> aa. perforantes
3) m. semimembranosus: –> a. circumflexa femoris medialis, aa. perforantes, a. poplitea
4) m. adductor magnus
5) m. aductor brevis: –> a.obturatoria, a. perforantes
6) m. gracilis: –> a. obturatoria, a. femoralis, a. profunda femoris

Lymphatic system

1) m. biceps femoris: –> nodi lymphatici inguinales superficiales inferiores
2) m. semitendinosus: –> nodi lymphatici inguinales superficiales superomediales et superolaterales

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14.2. Calf Muscles Activation

Location: *Regio cruris posterior et lateralis*

Bones: *Tibia. Et fibula*

Muscles:

Surface posterior muscle of the tibia area:

1) m. triceps surae

2) m. plantaris

Deep posterior muscles of the tibia area:

3) m. popliteus

4) m. flexor digitorum longus

5) m. flexor hallicus longus

6) m. tibialis posterior

Lateral group of muscles of the tibia area:

7) m. peroneus longus

8) m. peroneus brevis

Procedure

Position of the client: Lying down prone (on stomach). Client can be on side (in cases of refusal to lie down on stomach or for reason of physical condition, feeding tube discomfort).

Technique: Deep tactile-proprioceptive activation of the calf group muscles (mm. triceps surae, plantaris, popliteus, and other). Find the origin of triceps muscle and go in the direction of the insertion – distally, follow the next steps:

1) Preparation. Stroking along the muscle starting from the third of it (below the origin) in up and down (proximally and distally).

2) Activation of Golgi system. Activation of the muscle close to origin and insertion by pulling them up and away from the bone.

3) Activation of the whole muscle. Use of the above technique (# 2) along the whole muscle sequentially distally (down).

This technique allows regulation of the muscular-tendinous tone in the upper lower limb for such reflex patterns as: Automatic Gait, Leg Cross Flexion/Extension, Foot Tendon Guard, Crawling (lower part), ATNR (lower part), Trunk Extension, Segmental Rolling, and Spinning. This exercise also regulates the links and differentiation of the biomechanics of all lower limb reflex patterns based on flexion and extension, also partly adduction/abduction.

Position of specialist: Stand close to the upper leg where the technique is applied; facing the center of the client's body.

Position of the hand/palm: Fingers and palms of both hands are pulling the muscle away from the bone in attempt to lengthen it and activate the peripheral nerve system of the upper part of the leg. For this exercise use deep, slow pressure and pulling with fingers/ palms (touch # 2, 3 and palms).

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BACK: CALF

Action:

1) Preparation. Stroke along the muscle starting from the third of it (below the origin) in both directions simultaneously – up and down (proximally and distally) reach the spots close to the origin and insertion of the muscle.

2) Activation of Golgi system. Hold the muscle in the area below the origin and above the insertion lifting the ends of the muscle away from the hip bone pleasant way for 7-10 seconds. Use intermediate space between muscle fibers and tendons, where there are more Golgi and ligament receptors, and less of free nerve ends receptors (less of protection).

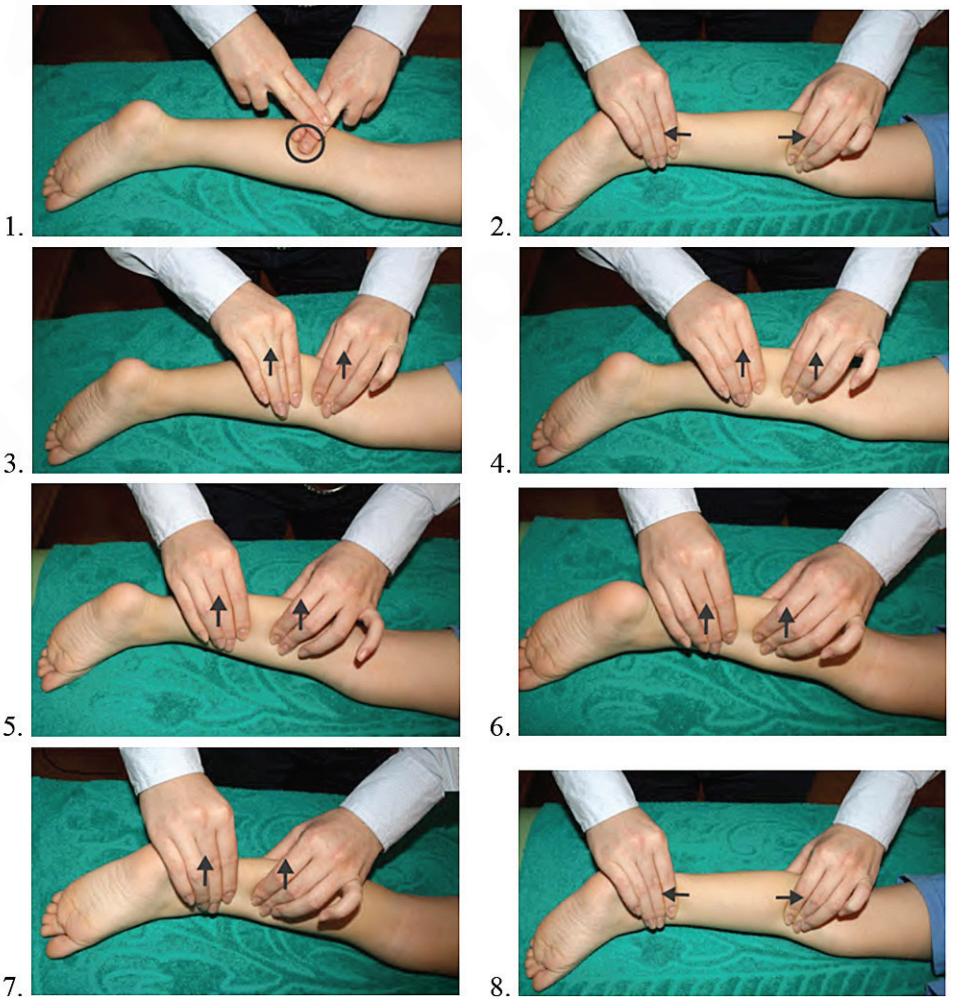
3) Activation of the whole muscle. Move on using this technique along the whole muscle step by step working sequentially distally (down) in ankle direction pausing at each step for 7-10 seconds. Work safe way pausing for longer time when sensitivity is hyper.

Direction of applying the technique:

1) Preparation. Stroke along the muscle proximally and distally simultaneously, and reach the spots close to the origin and insertion of the muscle.

2) Activation of origin/insertion of the muscle. Hold the muscle in the area below the origin and above the insertion, pulling the ends of the muscle away from the hip bone.

3) Activation of the whole muscle. Move on pulling the muscle away from the bone going distally (down) in ankle direction.



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Calf Muscles Activation

Nervous system

Muscle innervation:

1) m. triceps surae
2) m. plantaris
3) m. popliteus
4) m. flexor digitorum longus
5) m. flexor hallicus longus
6) m. tibialis posterior: –> n. tibialis
7) m. peroneus longus
8) m. peroneus brevis: –> n. peroneus superficialis

Skin innervation:

1) m. triceps surae: –> n. cutaneous surae medialis
2) m. plantaris: –> n. cutaneous surae lateralis Pers distalis: n. suralis
3) m. popliteus: –> n. saphenus from n. femoralis

Circulatory system

1) m. triceps surae: –> a. tibialis posterior
2) m. plantaris
3) m. popliteus: –> a. poplitea
4) m. flexor digitorum longus
5) m. flexor hallicus longus
6) m. tibialis posterior: –> a. tibialis posterior
7) m. peroneus longus: –> a. inferior lateralis genus, a. fibularis
8) m. peroneus brevis: –> a. peronea

Lymphatic system

Vessels flowing in:

1) m. triceps surae: –> nodi lymphatici genus capsularis, arteriosus et subfascialis

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14.3. Achilles Tendon Activation
(Tendo Achillaris)

Location: *Regio cruris posterior*

Bones: *Os calcanei, fibulae tibia*

Muscles:

1) m. triceps surae: m. gastrocnemius lateralis, m. gastrocnemius medialis, m. soleus

Procedure

Technique: Deep tactile-proprioceptive activation of the Achilles Tendon *(Tendo Achillaris)*. Work with the insertion of the triceps muscle *(m. triceps surae)* proximally slightly moving it in the direction of the growth of muscle fibers (in knee direction; touch # 2, 3).

This technique allows regulation of the muscular-tendinous tone in the lower limb for such reflex patterns as: Foot Tendon Guard, Babinski, Foot Grasp, Automatic Gait, Leg Cross Flexion/Extension, Trunk Extension, Segmental Rolling, Spinning, lower part of Moro and Fear Paralysis, and Crawling. This exercise regulates the links and differentiation of the biomechanics of all lower reflex patterns based on flexion and extension. It also works for coordination with core and upper limbs reflex patterns activation.

Position of specialist: Stand close to the leg with knee bent 90° where the technique is applied; facing the center of the client's body.

Position of the hand/palm: Fingers and palms of both hands are pushing the muscle in the direction of the bone in attempt to lengthen the tendon against of muscle fibers. For this exercise use the deep and slow pressure-stretch (pushing) with fingers/palms (touch # 2, 3 and palms).

Action:

1) Bend the knee 90° so foot is raised with bottom of foot facing up. Flex the big toe with the thumb of your one hand, and next, flex the foot with your palm.

2) Touch and hold with your other hand firmly and stably the area above the Achilles tendon and slightly move in the direction of the growth of muscle fibers proximally (in knee direction; touch # 2, 3) for 7-10 seconds. Use intermediate space between muscle fibers and tendons, where there are more Golgi and ligament receptors, and less of free nerve ends receptors (less of protection). Work safely pausing for longer time when sensitivity is hyper.





Direction of applying the technique:

1) Bend the knee 90° flex the big toe and foot of your client dorsally (in head direction).

2) Stretch Achilles tendon proximally and in the direction of the *tibia* bone for 60° superiorly and anteriorly.

Exercise Effects: Achilles Tendon Activation

Circulatory system

1) m. triceps surae: –> a. tibialis posterior

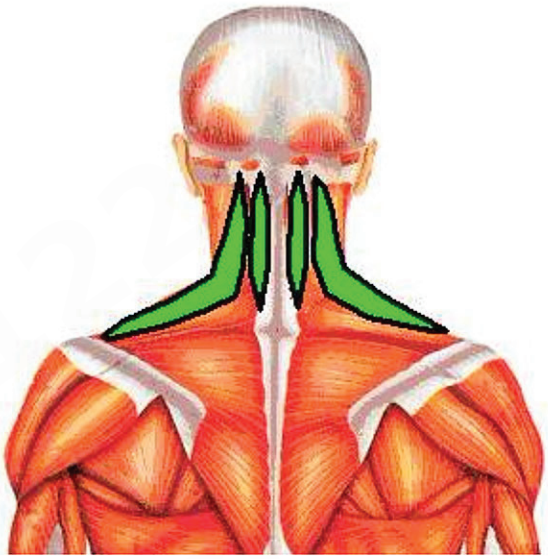
Nervous system

1) m. triceps surae: –> n. tibialis posterior

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15. NECK (Advanced Technique)



Goals

- Release negative effect of the Tendon Guard Reflex - Green Light (extension) and Red Light (flexion).
- Activate reflex patterns: Upper Tendon Guard - Green Light, Head Righting, Head Extension, Head Tilting, Tonic Labyrinthine in Flexion and Extension (cranial nerves X, XI, XII, XIII).

Group-1

Location: *Regio cervicalis anterior*

Bones: *Columna vertebralis cervicis*

Muscles:

1) m. trapezius
2) m. sternocleidomastoideus
3) m. splenius capitis
4) m. transversospinalis: m. semispinalis capitis
5) m. erector spinae: m. longissimus capitis, m. spinalis capitis

Group-2:

Location: *Regio colli anterior*

Bones: *Columna vertebralis cervicis*

Muscles:

1) m. stylohyoideus
2) m. omohyoideus
3) m. digastricus: a) venter posterior, b) venter anterior

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15.1. Neck Area Guard Regulation

Action

A) Preparation. Circular motions at occiput protuberances *(protuberantia occipitale externa).* Start with deep, proprioceptive and stable touch at occiput protuberances closer to tendons of the muscles in the area of the occiput *(m. occipitofrontalis, venter occipitalis; m. longus capitis, m. lomgus colli, m. trapezius pars descendens, m. rectus capitis posterior major, m. obloquus capitis superior, m. obloquus capitis inferior, also scalenus anterior and medius).* Do small circular motions around the occiput protuberances moving laterally (from the center to the sides) to release the negative effect of the upper Tendon Guard Reflex (tension in neck extensors) or to regulate the muscles mentioned above.

B) Stretch at the base of the occiput. Next, hold the head with fingers of both your hands in the area of occiput firmly, and start light increasing stretch for activation of the neck tendons in the following directions:

- Straight ahead in cranial (superior)
- To the sides
- Into the base of the occiput area in direction of sacrospinalis muscles (SIII) and along the sterno-cleido-masdoides muscle. The chin in this exercise moves forward to the center along the in the direction of the sternum-clavicle-mastoids muscle.

C) Stretch of the neck area on the following planes:

- *The back*. Hold with one your hand the long neck muscle pulling it delicately away from the cervical vertebrae and touching the base of the occiput by last fingers stretching the neck superiorly. The fingers of the other hand hold the head at the basis of the occiput and pull superiorly. The effect of the pulling by both hands must create simultaneous and stable micro-motion in superior direction.

Additionally: Bring the head forward and stretch the long neck muscle (scalenus anterior and medius), holding the head at the occiput and sacrospinalis muscles on both sides of the spine on level TIII-IV.

- *The sides*. Move the head very slowly from side to side to lengthen the neck muscles. Stretch the group of muscles on the sides of the neck - holding the head at the side of the base of the occiput and at shoulder at the insertion of the upper trapezius muscles. Do exercise slowly, holding the head at every position 7-10 seconds or longer.

Note: The head moves into shoulder, and in position lying down the nose is directed to the ceiling.

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Exercise Effects

Group-1

Nervous system

1) m. trapezius: => n. accessories, plexus cervicalis
2) m. sternocleidomastoideus: => n. accessories
3) m. splenius capitis: => r. posterior nervi spinalis

Circulatory system

1) m. trapezius: =>a. transversa cervicis, a. occipitalis, a. suprascapularis, aa. intercostales posteriores
2) m. sternocleidomastoideus: => a. occipitalis rr sternocleidomastoideus
3) m. splenius capitis: => a. occipitalis, a. cervicalis profunda
4) m. transversospinalis: m. semispinalis capitis
5) m. erector spinae: m. longissimus capitis, m. spinalis capitis: => a. intercostales posterior, a. cervicalis profunda, a. lumbales

Group-2

Nervous system

1) m. stylohyoideus: => n. facialis,
2) m. omohyoideus: => ansa cervicalis;
3) m. digastricus: a) venter posterior => n. facialis; b) venter anterior: => n. mylohyoideus

Circulatory system

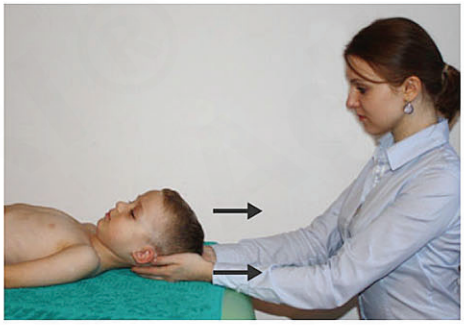
1) m. stylohyoideus: => a. occipitalis, a. facialis
2) m. omohyoideus: => a. thyroidea inferior, a. transversa cervicis
3) m. digastricus: a) a) venter posterior: => a. submentalis r. digastricus, b) venter anterior: => a. occipitalis, a. auricularis posterior

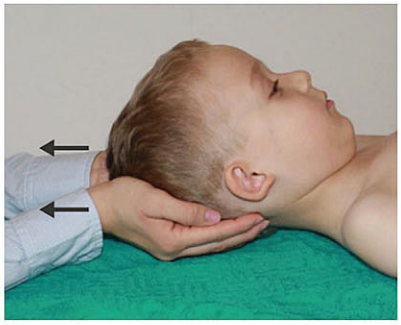
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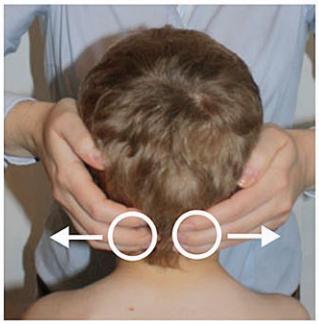
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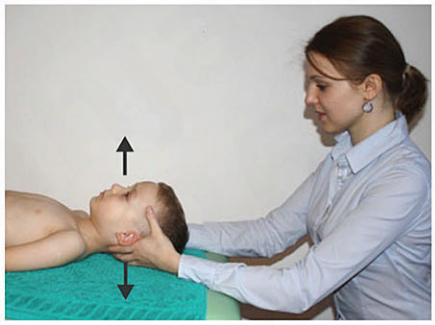
BACK: Neck

1A. 

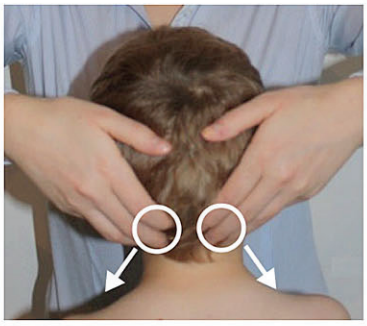
1B. 

1C. 

2A. 

2B. 

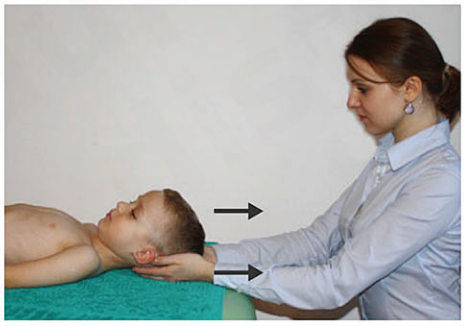
2C. 

3A. 

3B. 

3C. 

4A. 

4B. 

4C. 

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15.2. Platysma

Location: *Regio colli anterior*

Bones: *Clavicula, ramus mandibularis*

Muscles:

1) m. platysma

Also muscles:

- *Sternocleidomastoideus*
- *Sternohyoideus*
- *Omohyoideus*
- *Sternohyoideus*
- *Hyrohyoideus*

Action:

Work with the front of the neck. Stretch the platysma muscle. Turn the head very slowly to the side to lengthen this muscle on one side. Stretch it and other muscles around holding with one your hand at the lower jaw - ramus of the mandibular bone, and other hand below the clavicle bone on the same side against of the pectoralis major muscle. Do all these exercises slowly, holding the head at every position 7-10 seconds or longer, repeat 2-3 times.

Exercise Effects: Platysma

Circulatory system

m. platysma: => a. facialis, a. transversa cervicis

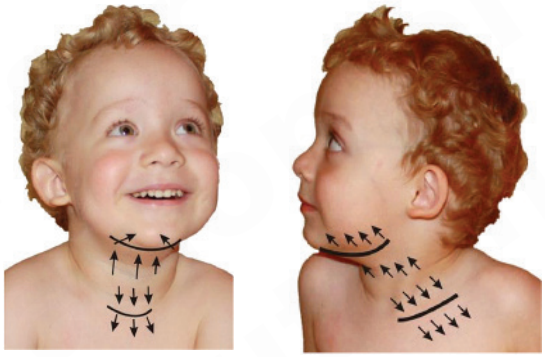
Nervous system

Muscle innervation:

m. platysma: => R. colli nervi facialis

Skin innervation:

m. platysma: => ramus cutaneus nervi facialis





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16. Completion

16.1. Lazy Lion & Symmetry Check

Note: This exercise can be done only if physical condition of the client allows.

Procedure

Position of the client: Lying down supine (on back) or on side (if physical condition allows). The client flexes the leg in the knee joint and brings the leg on other side crossing the midline; the other leg is straight; the head is turned to the opposite side to extend the stretch even more.

Technique: Stretch - deep tactile-proprioceptive activation of the whole body side when the spine is rotated.

This technique allows to coordinate the upper and lower parts of the body for functioning of such reflex patterns: Segmental Rolling, Spinning, ATNR, Automatic Gait, Leg Cross Flexion/Extension, Crawling, and Trunk Extension. This exercise also regulates the links and differentiation of the biomechanics of core/trunk extension, also adduction/abduction.

Position of specialist: Stand close to the horizontal midline of the body where the technique is applied; facing the client.

Position of the hand/palm: Palms of both hands hold and press at the anterior shoulder area, and opposite hip in attempt to stretch the side of the body lengthening muscle of the whole core and trunk. This activates coordination of the upper and lower parts of the body, trains the freedom of the movements of the lower part of the body, and activates the peripheral nerve system of the core/trunk.

Action:

Invite your client to allow their pelvis to follow in the direction of their bent knee, and move their shoulder and arm on the top side in the opposite direction.

A) Active stretch. Press gently on the shoulder and hip or leg to rotate and elongate the core in a cross pattern, as much as possible.

B) Passive self-stretch. Invite your client to lie down and relax in this position for 1-2 minutes more. Repeat on the other side.

Direction of applying the technique:

1) Palms of both hands are holding and press at the anterior shoulder area – to open the arm, and opposite hip – to cross the midline, in attempt to stretch the side of the body lengthening muscle of the whole core and trunk. Use stable and slow pressure (touch by palms or forearms).



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PART 3. ADDENDUM

Noraxon Printout

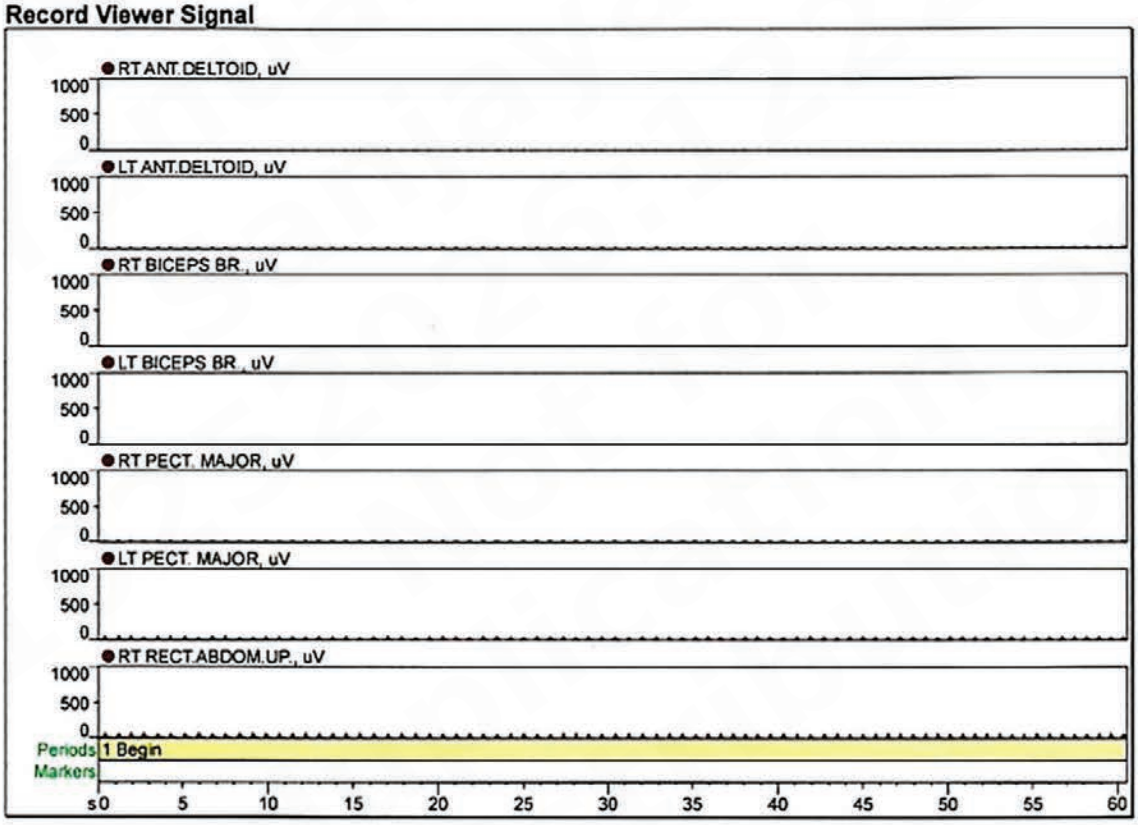
Patient

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First Name	D
Last Name	M
Sex	Male
Date of birth	

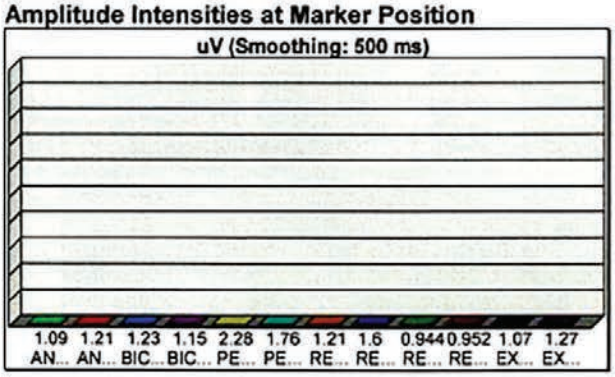
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Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:31

Record Viewer Signal



Amplitude Intensities at Marker Position



Subject Comments

Record Comments

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ADDENDUM

Noraxon Printout

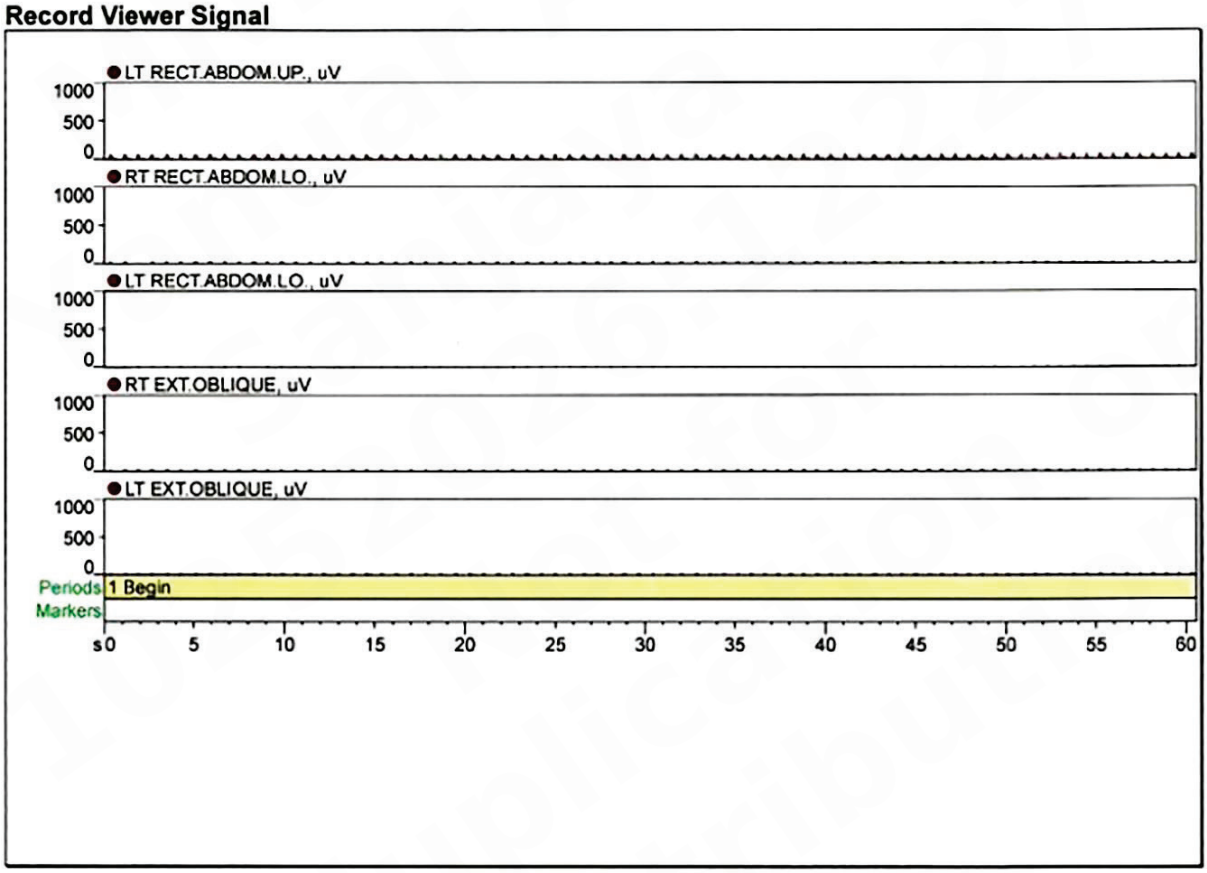
Patient

Project	Neuro-pendulum-1
First Name	D
Last Name	M
Sex	Male
Date of birth	

Record

Name	8 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:31

Record Viewer Signal



Subject Comments

Record Comments

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ADDENDUM

Noraxon Standard EMG Analysis

Subject:

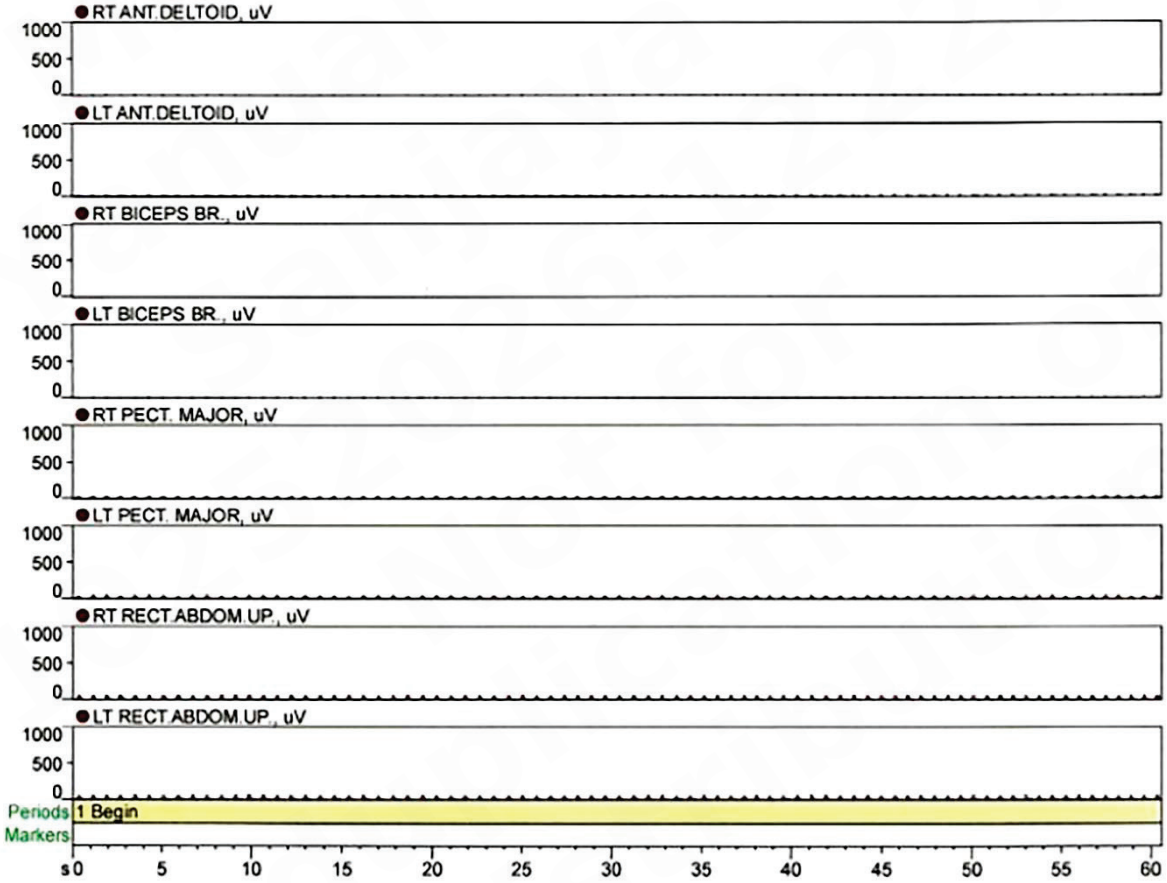
First Name	D
Last Name	M
Date of birth	
Diagnosis	

Record:

Name	8 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:31
Number of periods	1

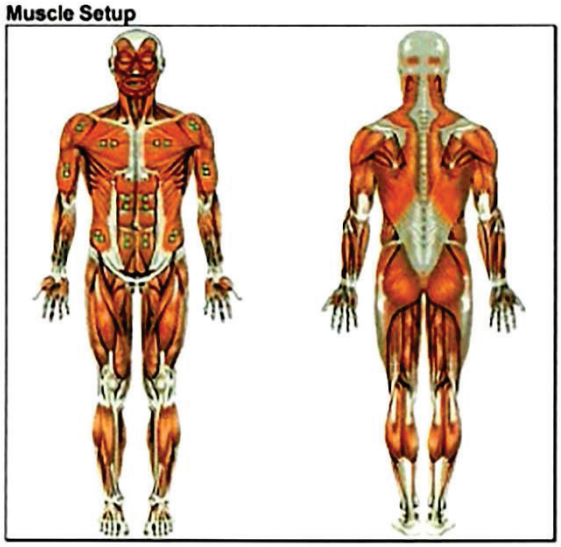
Standard Analysis

Analyzed Signals / Periods



Subject Comments

Muscle Setup



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Noraxon Standard EMG Analysis

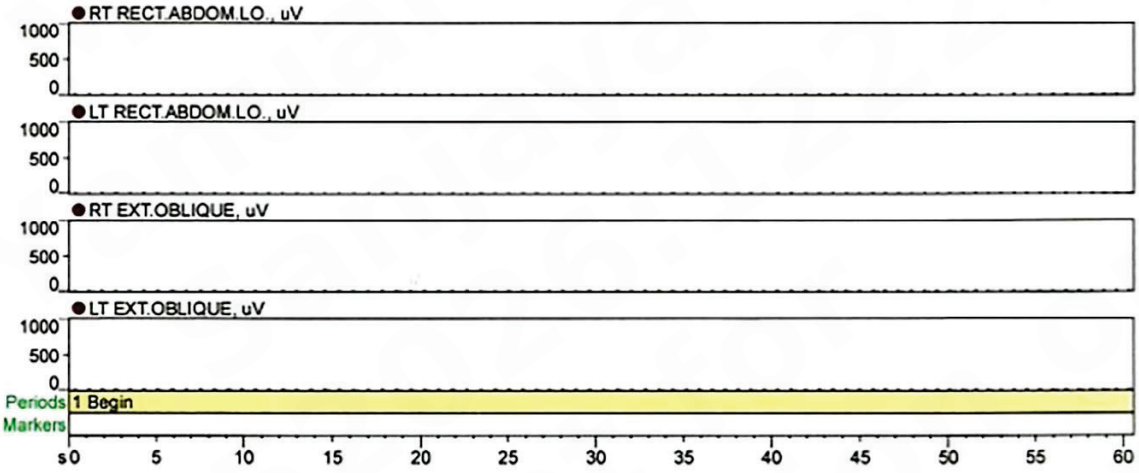
Subject:

First Name	D
Last Name	M
Date of birth	
Diagnosis	

Record:

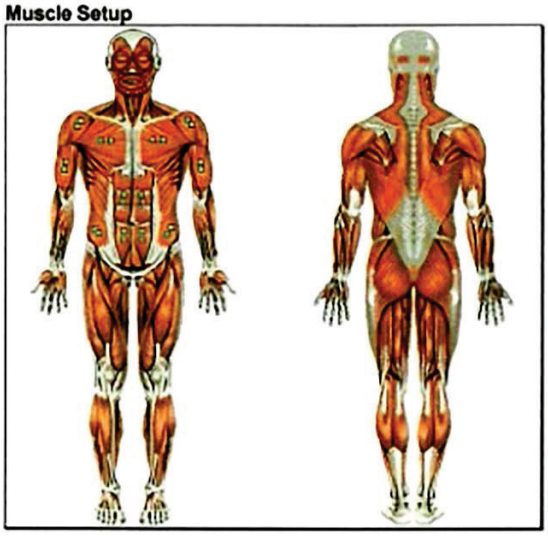
Name	8 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:31
Number of periods	1

Analyzed Signals / Periods



Subject Comments

Muscle Setup



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ADDENDUM

Noraxon Standard EMG Analysis

Subject

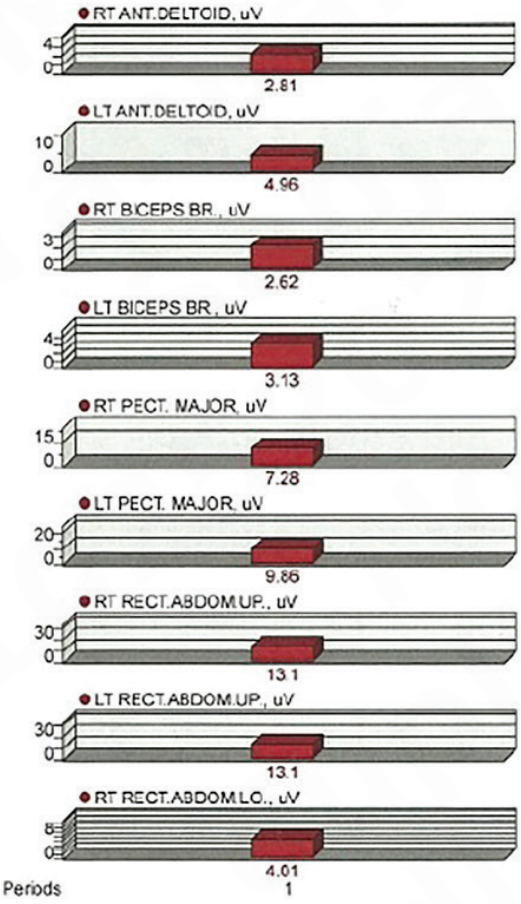
First Name	D
Last Name	M
Date of birth	
Diagnosis	

Record

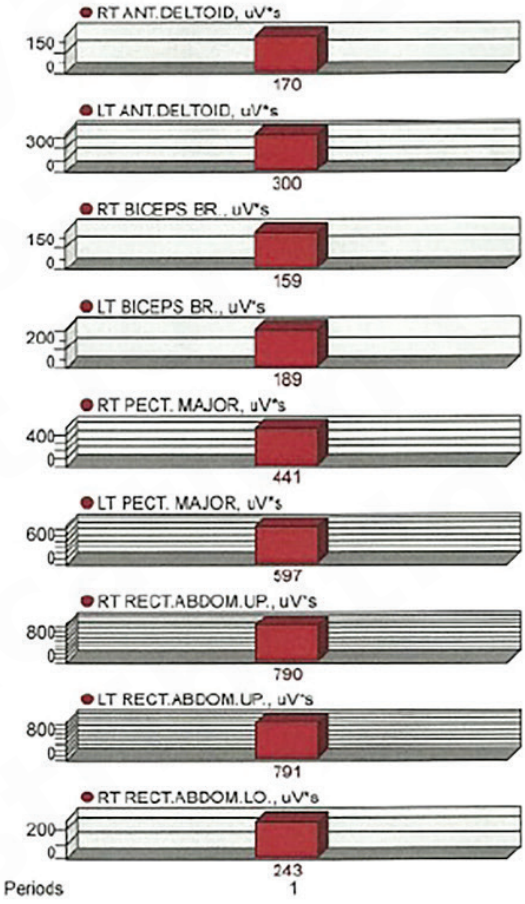
Name	8 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:31
Number of periods	1

Standard Analysis

Mean of Each Period



Area of Each Period



Averaged Mean Amplitude of All Periods

Channel	Mean, uV
RT ANT.DELTOID, uV	2.81±0
LT ANT.DELTOID	4.96±0
RT BICEPS BR.	2.62±0
LT BICEPS BR.	3.13±0
RT PECT. MAJOR	7.28±0
LT PECT. MAJOR	9.86±0
RT RECT.ABDOM.UP.	13.1±0
LT RECT.ABDOM.UP.	13.1±0
RT RECT.ABDOM.LO.	4.01±0
LT RECT.ABDOM.LO.	3.16±0

Averaged Area of All Periods

Channel	Mean, uV*s
RT ANT.DELTOID, uV*s	170±0
LT ANT.DELTOID	300±0
RT BICEPS BR.	159±0
LT BICEPS BR.	189±0
RT PECT. MAJOR	441±0
LT PECT. MAJOR	597±0
RT RECT.ABDOM.UP.	790±0
LT RECT.ABDOM.UP.	791±0
RT RECT.ABDOM.LO.	243±0
LT RECT.ABDOM.LO.	191±0

Record Comments

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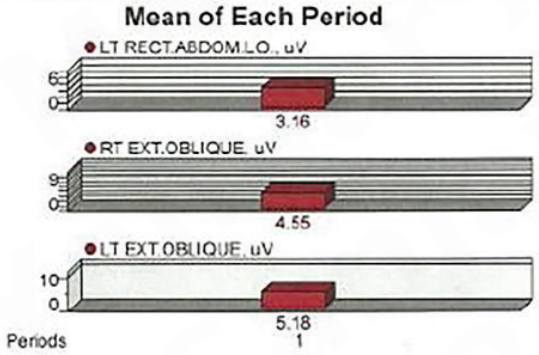
ADDENDUM

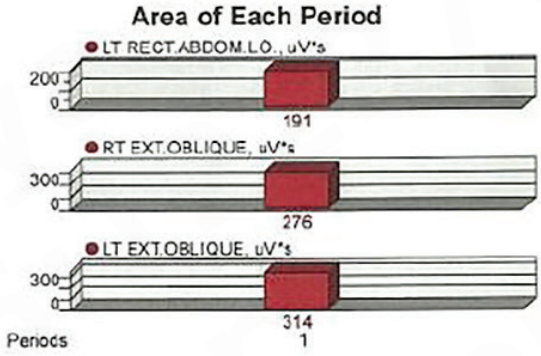
Noraxon Standard EMG Analysis

Subject

Record







Averaged Mean Amplitude of All Periods

Channel	Mean, uV
RT EXT.OBLIQUE	4.55±0
LT EXT.OBLIQUE	5.18±0

Averaged Area of All Periods

Channel	Mean, uV*s
RT EXT.OBLIQUE	276±0
LT EXT.OBLIQUE	314±0

Record Comments

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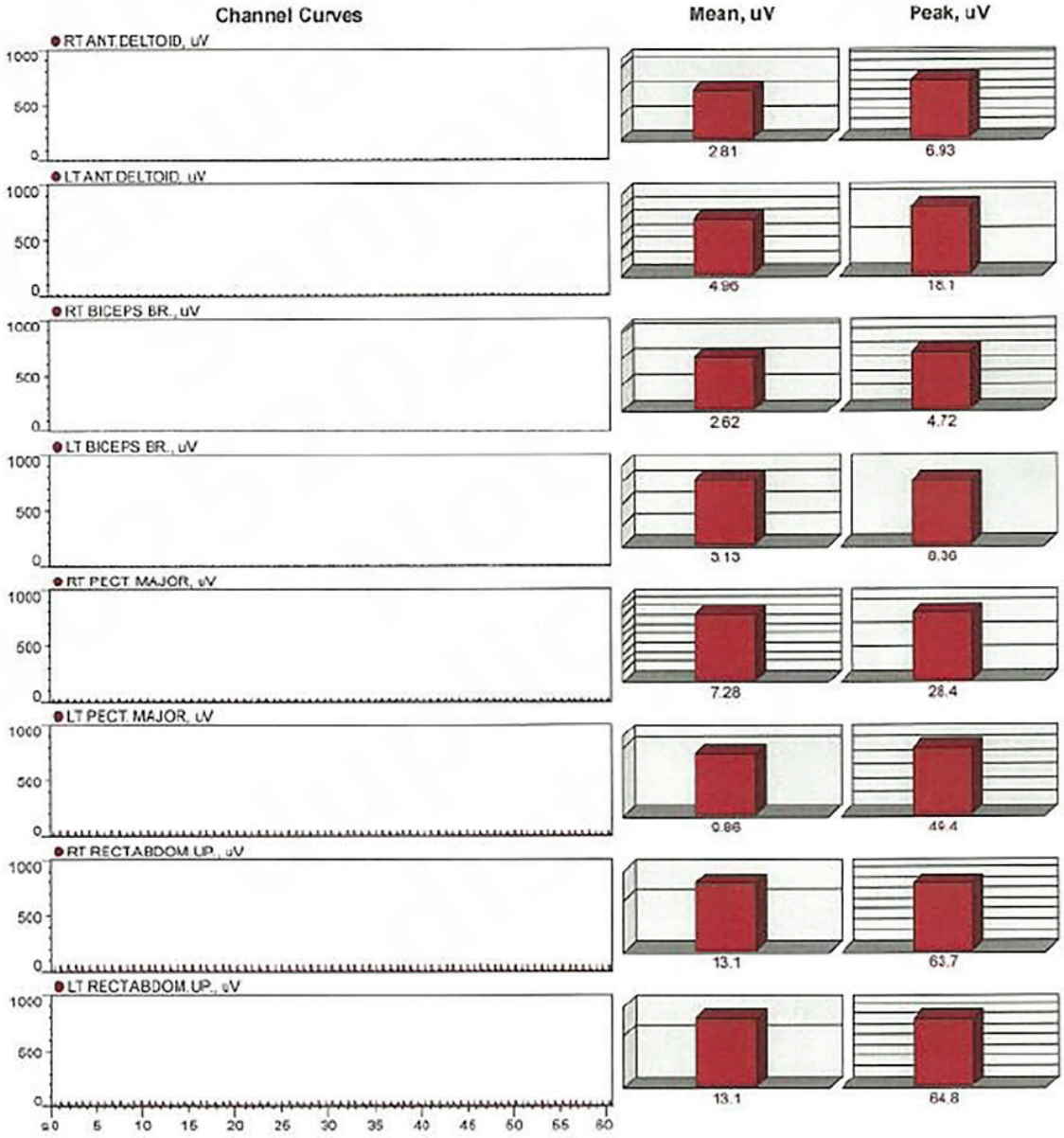
Noraxon Quick Analysis

Patient

Project	Neuro-pendulum-1
First Name	Denis
Last Name	Masgutov
Sex	Male
Date of birth	

Record

Name	8 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:31



Subject Comments

Record Comments

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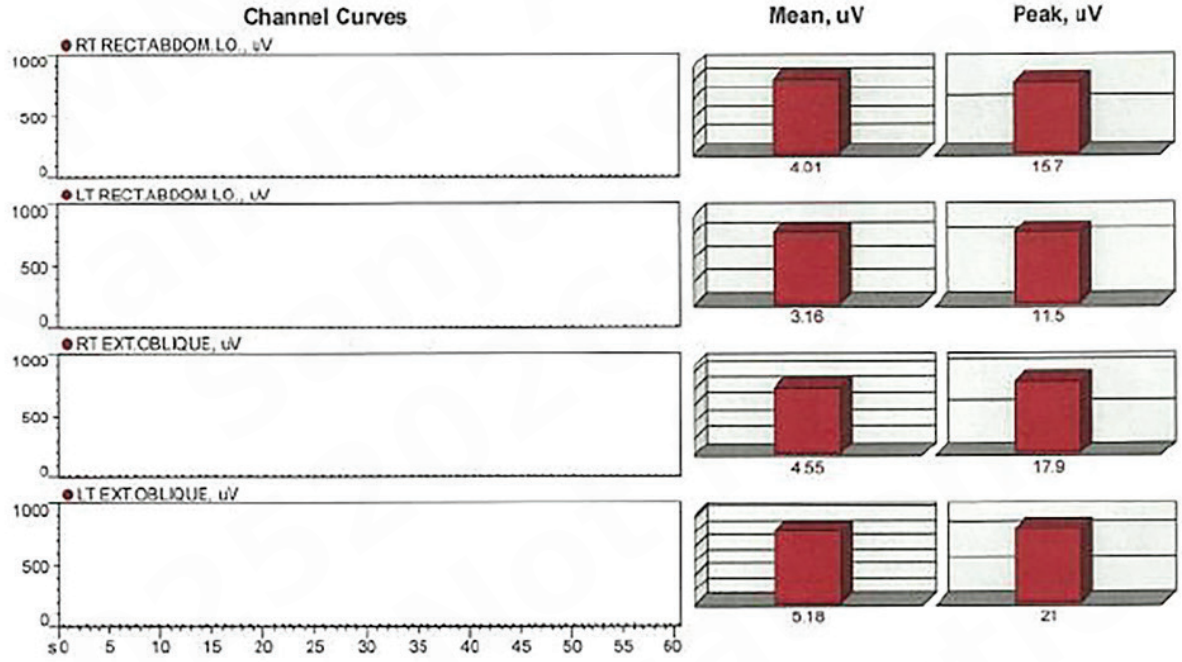
Noraxon Quick Analysis

Patient

Project	Neuro-pendulum-1
First Name	Denis
Last Name	Masgutov
Sex	Male
Date of birth	

Record

Name	8 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:31



Subject Comments

Record Comments

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Noraxon Printout

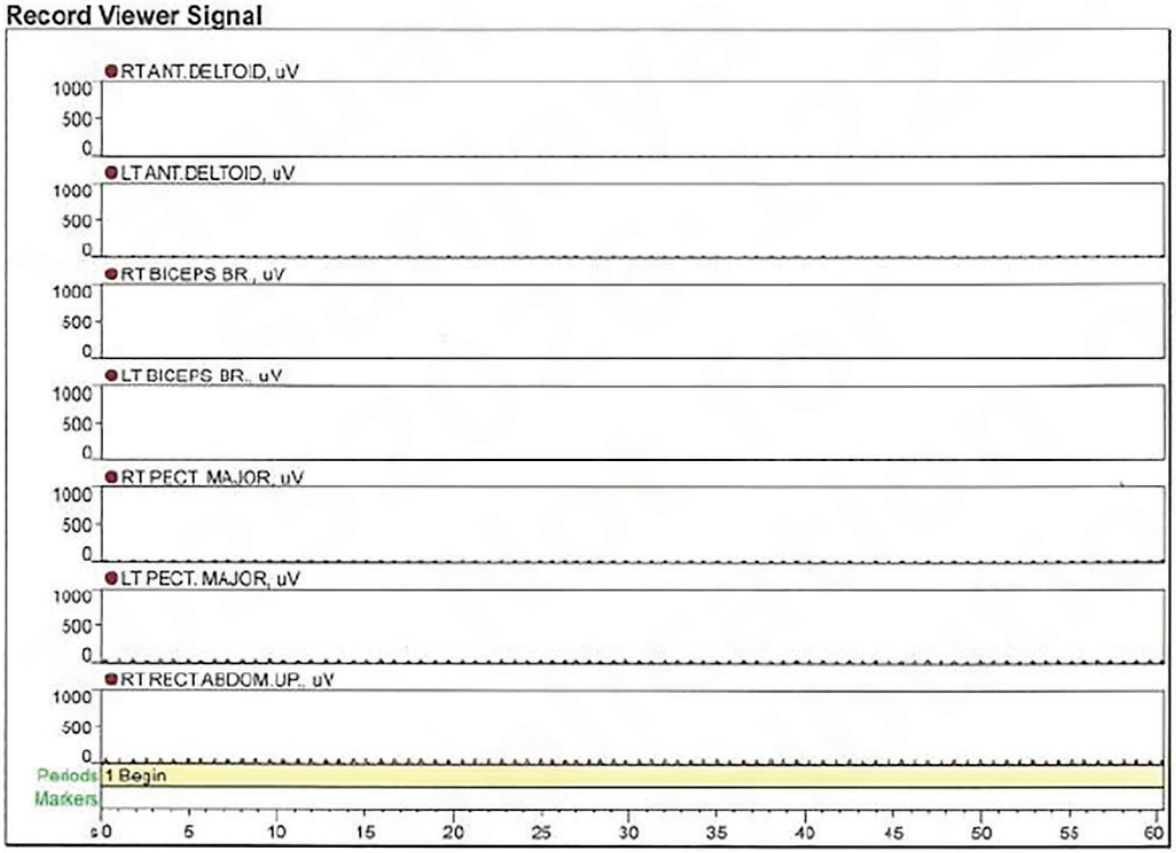
Patient

Project	Neuro-core-3-2
First Name	D
Last Name	M
Sex	Male
Date of birth	

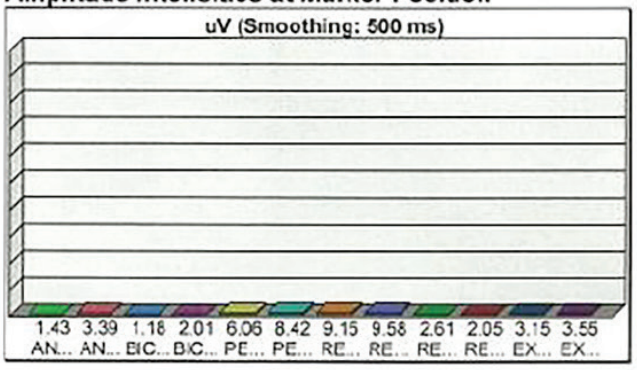
Record

Name	8 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:27

Record Viewer Signal



Amplitude Intensities at Marker Position



Subject Comments

Record Comments

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Noraxon Printout

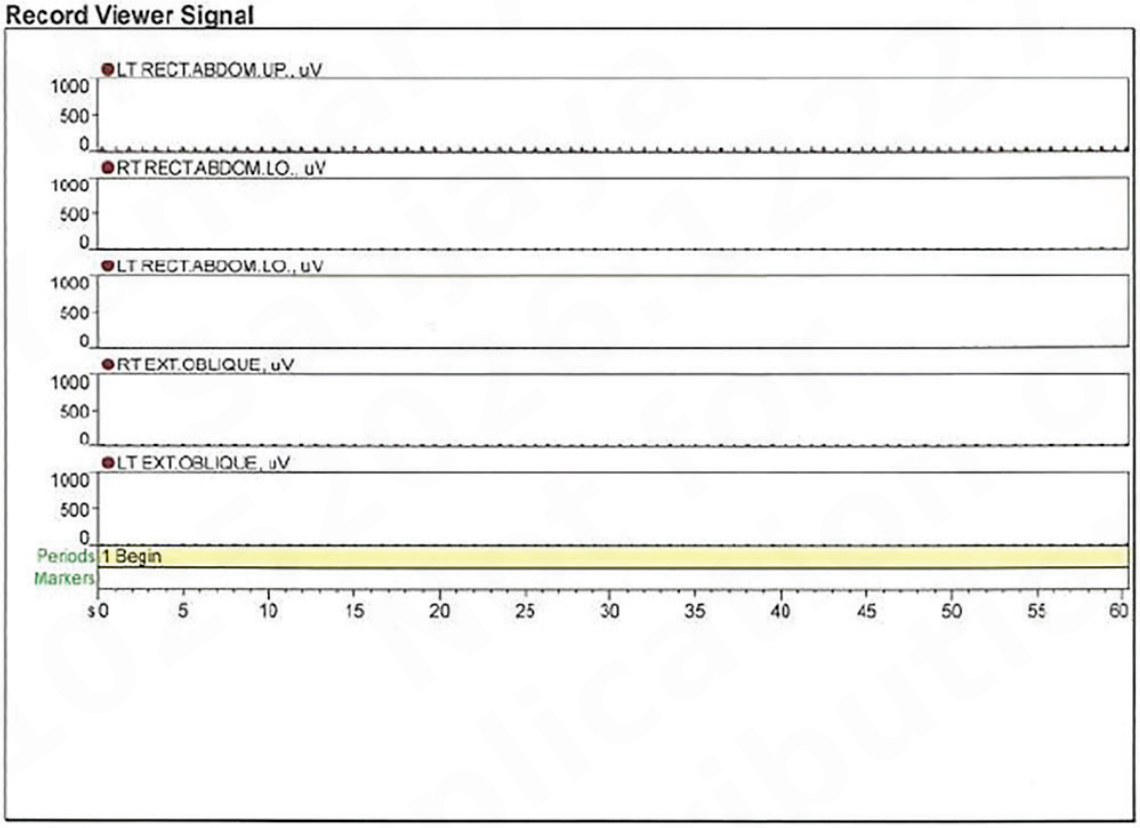
Patient

Project	Neuro-core-3-2
First Name	D
Last Name	M
Sex	Male
Date of birth	

Record

Name	8 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:27

Record Viewer Signal



Subject Comments

Record Comments

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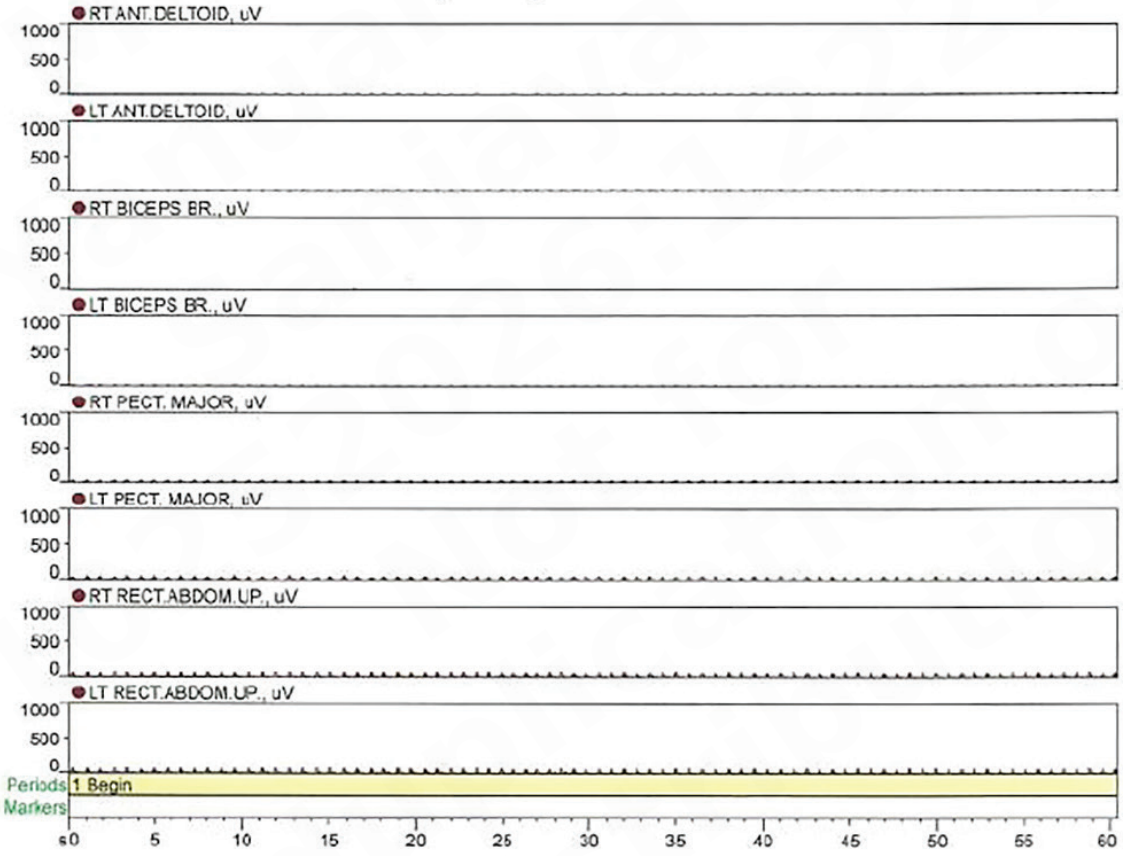
ADDENDUM

Noraxon Standard EMG Analysis

Subject:		Record:	
First Name	D	Name	8 CHANNELS
Last Name	M	Record type	A1. Standard EMG Analysis
Date of birth		Date Measured	2011-11-01 22:27
Diagnosis		Number of periods	1

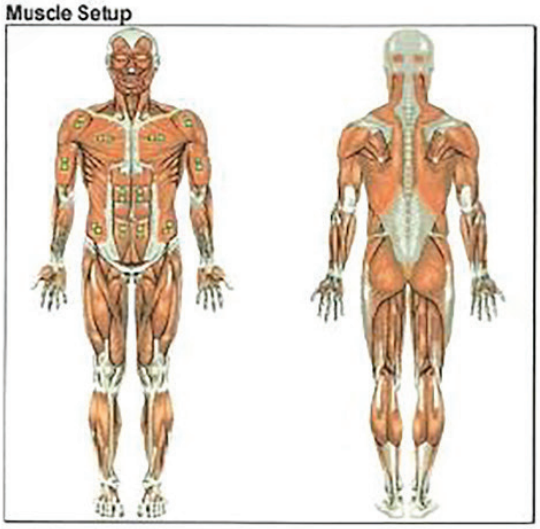
Standard Analysis

Analyzed Signals / Periods



Subject Comments

Muscle Setup



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ADDENDUM

Noraxon Standard EMG Analysis

Subject:

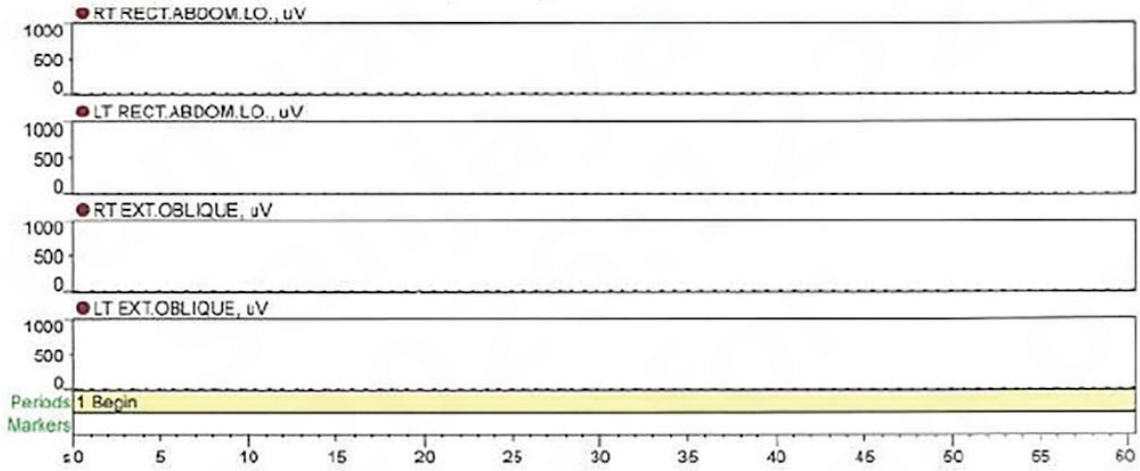
First Name	D
Last Name	M
Date of birth	
Diagnosis	

Record:

Name	8 CHANNELS
Record type	A1, Standard EMG Analysis
Date Measured	2011-11-01 22:27
Number of periods	1

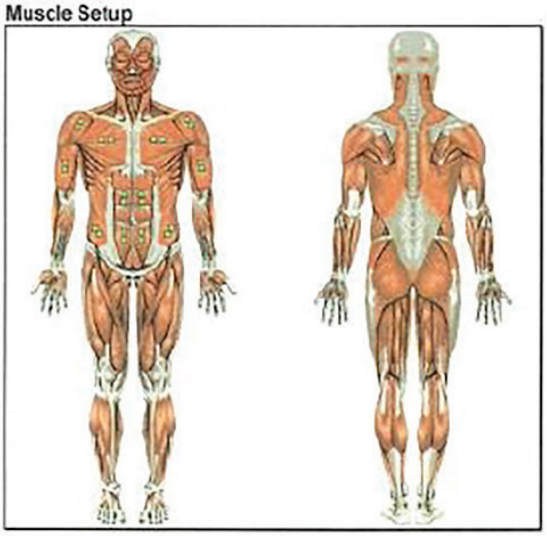


Analyzed Signals / Periods



Subject Comments

Muscle Setup



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ADDENDUM

Noraxon Standard EMG Analysis

Subject

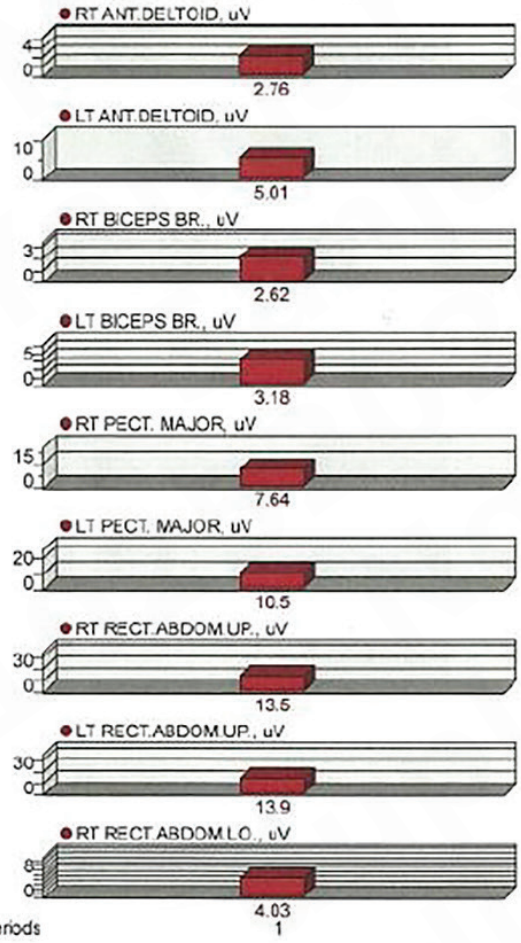
First Name	D
Last Name	M
Date of birth	
Diagnosis	

Record

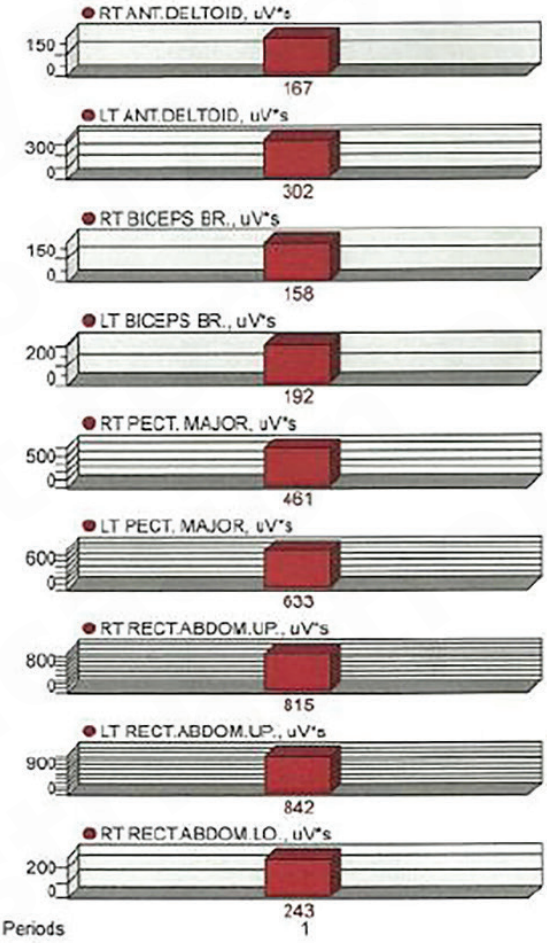
Name	8 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:27
Number of periods	1

Standard Analysis

Mean of Each Period



Area of Each Period



Averaged Mean Amplitude of All Periods

Channel	Mean, uV
RT ANT.DELTOID, uV	2.76±0
LT ANT.DELTOID	5.01±0
RT BICEPS BR.	2.62±0
LT BICEPS BR.	3.18±0
RT PECT. MAJOR	7.64±0
LT PECT. MAJOR	10.5±0
RT RECT.ABDOM.UP.	13.5±0
LT RECT.ABDOM.UP.	13.9±0
RT RECT.ABDOM.LO.	4.03±0
LT RECT.ABDOM.LO.	3.13±0

Averaged Area of All Periods

Channel	Mean, uV*s
RT ANT.DELTOID, uV*s	167±0
LT ANT.DELTOID	302±0
RT BICEPS BR.	158±0
LT BICEPS BR.	192±0
RT PECT. MAJOR	461±0
LT PECT. MAJOR	633±0
RT RECT.ABDOM.UP.	815±0
LT RECT.ABDOM.UP.	842±0
RT RECT.ABDOM.LO.	243±0
LT RECT.ABDOM.LO.	169±0

Record Comments

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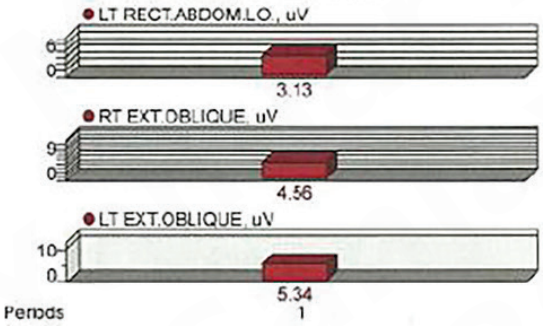
Noraxon Standard EMG Analysis

Subject

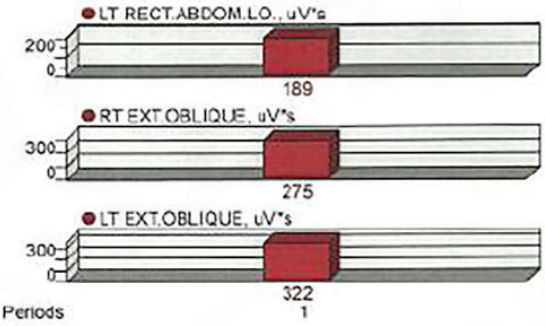
Record

Standard Analysis

Mean of Each Period



Area of Each Period



Averaged Mean Amplitude of All Periods

Channel	Mean, uV
RT EXT.OBLIQUE	4.56±0
LT EXT.OBLIQUE	5.34±0

Averaged Area of All Periods

Channel	Mean, uV*s
RT EXT.OBLIQUE	275±0
LT EXT.OBLIQUE	322±0

Record Comments

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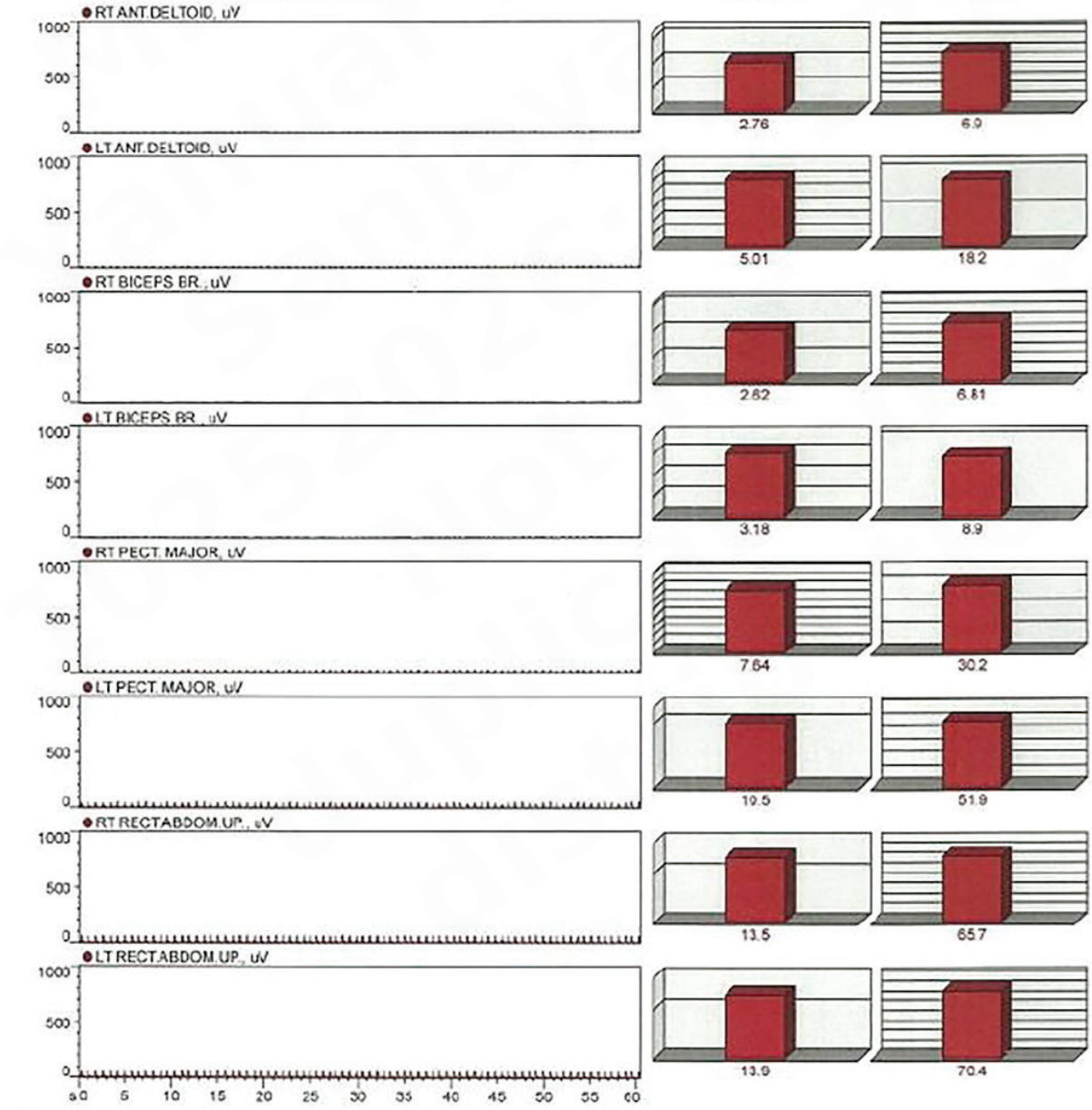
Noraxon Quick Analysis

Patient

Project	Neuro-ocre-3-2
First Name	D
Last Name	M
Sex	Male
Date of birth	

Record

Name	6 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:27



Subject Comments

Record Comments

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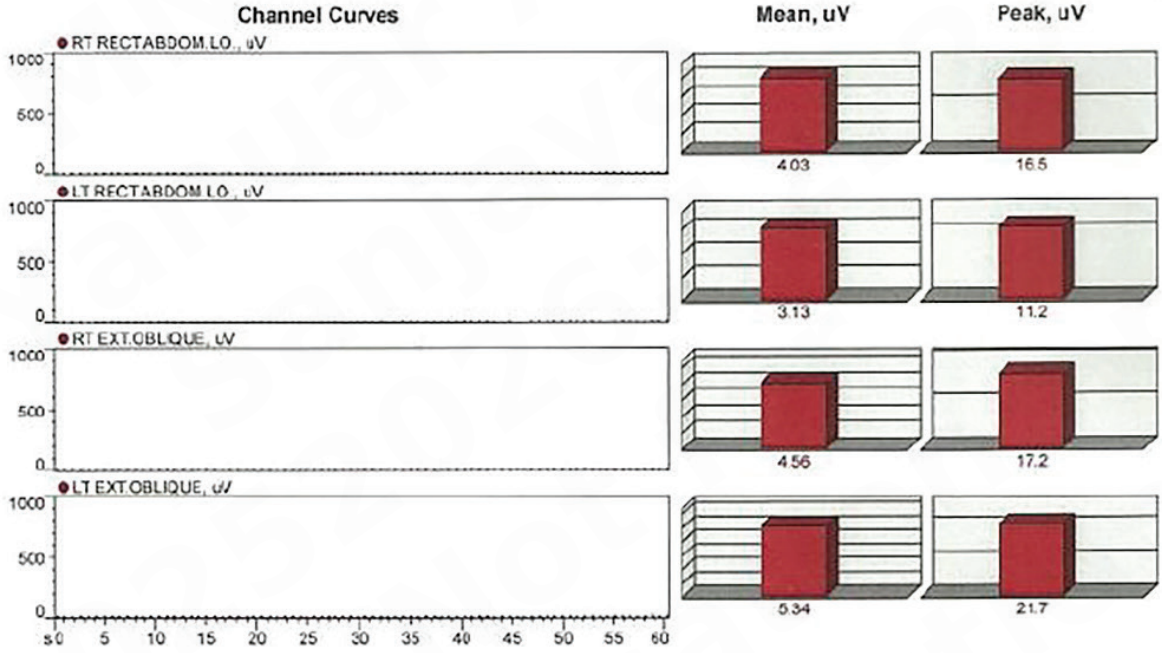
Noraxon Quick Analysis

Patient

Project	Neuro-core-3-2
First Name	D
Last Name	M
Sex	Male
Date of birth	

Record

Name	8 CHANNELS
Record type	A1. Standard EMG Analysis
Date Measured	2011-11-01 22:27



Subject Comments

Record Comments

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BIBLIOGRAPHY

Abrahams P. (2002). *The atlas of the human body.* Bright Star Publishing, Plc. USA.

Alexandrov J. I. (2003). *Introduction to systemic psychophysiology. Psychology of 20th Century*. Per Se. (Russ.). Moscow, Russia.

Amosov N. M. (1978).*Thinking about health.* (Russ.). Moscow. Russia. 16-29.

Andre A., H. Thomas, Y. Chesni, A. Disgassies (1962). *The neurological examination of the infant.* National Spastic Society. London, U.K.

Anokhin P. K. (1968). *Biology and Neurophysiology of a Conditioned Reflex.* Medicina. (Russ.). Moscow, Russia.

Anokhin P. K. (1975). *Introduction to Physiology of Functional Systems.* (Russ.). Moscow, Russia.

Akhmatova N. K., Goncharova S. M., Masgutova S. K. (2000). *Integration of Dynamic and Postural Reflexes into Whole Body Movement System.* (Russ.). Moscow, Russia.

Apgar, V. A. (1953). Proposal for a new method of evaluation of the newborn infant. *Anesthesia and Analgesia, Current Research*, 22, 260-267.

Atlas. *The Human Nervous System. The structure and dysfunctions.* (1977). Addition: B. M. Astapova, J. V. Mikadze. Moscow Institute for Postgraduates in Education. (Russ.). Moscow, Russia.

Aylward E., Reiss A., Reader M., Singer H., Brown J, Denckla M. (1996, March). Basal Ganglia Volumes in Children with Attention-Deficit Disorder. *Journal of Child Neurology* 11 (2). USA. 112-115.

Ayres J. (1971). Characteristics of Types of Sensory Integrative Functions. *The American Journal of Occupational Therapy,* 25(27). USA. 329-334.

Badalian L. O. (1984). *Child neurology.* Medicina. (Russ.). Moscow. Russia.

Barashniev J. I. (2001). *Prenatal neurology.* Triada-X. (Russ.). Moscow, Russia. 640.

Bernstein N. A. (1966). *The Introduction to Physiology of Movement and Physiology of Activity.* (Russ.). Moscow, Russia.

Bernstein N. A. (1997). *Bio-Mechanics and Physiology of the Movement.* (Russ.). Moscow-Voroniez, Russia.

Berthoz A. (2000). *The Brain's sense of Movement*. Harvard University Press London, U.K.

Bobath K. (1972). The normal postural reflex mechanism and this deviation in children with cerebral palsy. *Physiotherapy,* 57, 515.

Bobath B. (1963) Treatment principles and planning in cerebral palsy. */Physiotherapy* April: 1–3.

Blum F., Lezerson A., Hovstedter A. (1988). *The Brain, Mind and Behavior.* World. Moscow.

Brazelton, T. B., Nugent, J.K. (1995). *The Neonatal Behavioral Assessment Scale.* Mac Keith Press, Cambridge.

Campbell St. (2004). *Watch Me Grow! A Unique, 3-Dimensional, Week Look at Baby's Behavior and Development Inside the Womb.* Carroll and Brown Publishers Limited. London, U.K.

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BIBLIOGRAPHY

Caesar P., Dubovittz V. (1985). Neonatal Clinical Neurological Assessment. In Harel S. Anastasion N.J. (eds): *The At-Risk Infant. Psycho/Social/Medical Aspects.* Paul H. Brookes Baltimore. N.Y, USA. 197-229.

Cohen B. B. (1993). *Sensing, Feeling, and Action. The Experiential Anatomy of Body-Mind Centering.* Northampton, M.A. Contact Edition, USA.

Colorado Model Content Standards. (1997). *Physical Education:* adopted November, 13, 1997. (Microfilm). Denver, CO: Colorado State Department of Education. USA.

Commons M. L., Richards F. A. (1984a). A General Model of Stage Theory. In M. L. Commons, F. A. Richards, & C. Armon (Eds.), *Beyond Formal Operations: Vol. 1. Late Adolescent and Adult Cognitive Development* (pp. 120-140). NY: Praeger.

Cook A., Woollacott M. (1997). *Motor Control, Theory and Practical Applications.* Baltimore, Williams and Wilkins. USA.

Danilova N. N., Krilova A. L. (1997). *Physiology of the High Nerve System Activity/Performance.* Uchebnaja literatura. (Russ.). Moscow, Russia.

Delacato C. H. (1974). *The Diagnosis and Treatment of Speech and Reading Problems.* Garden City., New York , Doubleday. USA.

Dennison P. E. Dennison G. E. (2000). *Core Re-patterning.* Ventura. USA.

Dennison P. E., Dennison G. E. (1994). *Educational Kinesiology In-Depth: The Seven Dimensions of Intelligence.* Ventura, C.A. USA.

Duane E. Haines. (2002). *Fundamental Neuroscience,* Second Edition. N.Y. Edinburgh, London, Philadelphia. U.K.

Doman G. (1984). *How to Multiply Your Baby's Intelligence.* Garden City., New York, Doubleday. USA.

Feldenkrais M. (1981). *The Elusive Obvious.* Cupertino, Calif.: Meta Publications, p. 7-9. ISBN 0-916990-09-5

Fidel T. (2000). *Touch Therapy.* Churchill Livingstone. London

Frankenberg W. K., Dodds J. B. (1996). *The Denver Developmental Screening Test.* Denver, CO: University of Colorado Medical Center.

Galant S. (1917). *Der Ruckgratreflex.* Diss. Basel.

Goldberg S. (2007). *Clinical Neuroanatomy made ridiculously simple.* Interactive Adition. MrdMaster Inc. Miami, Fl., USA.

Goldberg E. (2009). *The New Executive Brain. Frontal Lobes in a Complex World.* Oxford University Press. New York.

Gralton T. (2003). *Touch for Health. A practical Guide to Health Improvement based on the Book by John Thie,* D.C. Switzerland.

Gross D. R. (1987). *Psychology. The Science of Mind and Behavior.* Hodder and Stoughton, London. U.K.

Gordejeva N. D., Zinczenko V. P. (1982). *Functional Structure of the Action.* Moscow State University. (Russ.). Moscow. Russia.

Humphrey T. (1964). *Some Correlations Between the Appearance of Homan Fetal Reflexes and the Development of the Nervous System. Progress in Brain Research.* Amsterdam, Elsevier. 4.

Khomskaja E.D. (1987). *Neuropsychology.* Moscow University. (Russ.) Moscow. Russia.

Konorski J. (1969). *Integrating Activity of the Brain,* PZWL. (Pl.). Warsaw.

Kephard N., Godfrey B. (1969). *Movement Patterns and Motor Education.* Educational Series in Health, Physical Education, Physical Therapy and Recreation). N.Y. Appleton-Century-Crofts. USA.

Koltsova M. M. (1973). *The Movement Activity and Brain Functions Development of the High Nervous System Activity of the Child.* Medgiz. (Russ.). Leningrad, Russia.

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BIBLIOGRAPHY

Krasikova I. (2001). *Child Massage and Gym for prophylactics and treatment of spine disorders, scoliosis and flat foot.* (Russ.). Moscow, Russia.

Krefft A. (2007). *Diagnosis of Functions of Invisible Phenomena* (Statistic Mathematic Analysis). Oficyna Wydawnicza Politechniki Wrocławskiej. (Pl.).Wrocław, Poland.

Landau A., Mikschiczek D., Peiper A., Spath L., Hansischer Verlagskontor, Lubeck, Hooker D. (1952). *The prenatal origin of behavior.* University of Kansas Press, Laurence. USA.

Leontiev A. N. (1977). *The Activity, Consciousness, Personality*. (Russ.) Moscow. Russia.

Lesgaft P. F. (1998). *Psychology of the Moral and Physical Training*. Voroniez. (Russ.). Russia.

Levy T., Orlans M. (1995). *Attachment Theory and Assessment*. In Carole A. McElvey (Ed). Give Them Roots, Let Them Fly. Evergreen, CO. Attachment Center at Evergreen City. USA.

Lundy-Ekman L. (2002). *Neuroscience. Fundamentals for Rehabilitation.* 2nd Edition. Pacific University. Forest Grove, Oregon. USA.

Luria A. R. (1969). *High Cerebrum Functions of the Human and their disorders in cases of local damages*. Moscow State University. (Russ.). Moscow, Russia.

Luria A. R. (1978). Functional Brain Organization. // In Book: *Biological-Scientific Basis of Psychology.* / Ed. By A. A. Smirnov, A. R. Luria, V. D. Niebilistsin. Pedagogika. (Russ.). Moscow, Russia. 109-139.

Mahoni F. (1999). *Hypertonic Muscle Release.* Assent. (Russ.). Moscow, Russia.

Marieb E. N. (2006). *Essentials of human anatomy and physiology.* 8th ed. San Francisco, Boston, New York, USA.

Masgutova S. K. (2005). *Integration of Oral-Facial Reflexes in Children with Central Nervous System Damage. International Conference materials. Prophylactics and Therapy of Children and Youth with Disorders in Psychological Health.* Polish National Foundation for Handicapped Children "Promyk Slonca." Wroclaw. December, 9, 2005. (Pl.). Wroclaw, Poland. 57-66.

Masgutova S. K. (2006). *Infant Reflex Integration in Movement Development.* In Materials of Polish National Conference: Modern Methods of Stimulation of Movement Development and Learning in Children with Difficulties in Learning, ADHD and Autism. 29.04. 2006. Scientific edition: Dr. S. Masgutova, Dr. Anna Regner. Prof. M.D. E. Ronin-Walknowska. Maternity Department, Pomorska Medical Academy in Szczecin MINK. (Pl.). Warsaw – Szczecin, Poland. 5-28.

Masgutova S. K. (2006). *Neuro-Kinesiology as the Support for Development and Learning. Children with Learning Challengea and ADHD.* In Materials of Polish National Conference: "Modern Methods of Stimulation of Movement Development and Learning in Children with Difficulties in Learning, ADHD and Autism. 29.04. 2006. Scientific edition: Dr. S. Masgutova, Dr. Anna Regner, Prof. M.D. E. Ronin-Walknowska. Maternity Department, Pomorska Medical Academy in Szczecin. MINK. (Pl.). Warsaw – Szczecin, Poland. 236-253.

Masgutova S. K. (2006). *Facial Reflex Integration and Its Implications for Child Development.* In Materials of Polish National Conference: "Modern Methods of Stimulation of Movement Development and Learning in Children with Difficulties in Learning, ADHD and Autism. 29.04. 2006. Scientific edition: Dr. S. Masgutova, Dr. Anna Regner, Prof. M.D. E. Ronin-Walknowska. Maternity Department, Pomorska Medical Academy in Szczecin. MINK. (Pl.). Warsaw – Szczecin, Poland. 268-275.

Masgutova S. K. (2006). *NeuroKinesiological Basis of the Reflex Integration for Successful Learning of the Child.* In materials of Polish National Conference: "NeuroKinesiology as the Method of Facilitation of Development and Learning for Children and Youth with Dyslexia, ADHD, Autism and Asperger." 08.04.2006. (Pl.). Warsaw, Poland. 118-133.

Masgutova S. K. (2007). *Neuro-Sensory-Motor Development: Visual and Auditory Reflexes Integration.* Facilitation Program

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BIBLIOGRAPHY

of Development and Learning for Children and Adults. Scientific Edition: Prof. N. Akhmatova. MISM. (Pl.). Warsaw, Poland. 86.

Masgutova S. K. (2008). *Neuro-sensory-motor Reflex Integration in Children with Developmental Challenges as the Successful Facilitation for Their Every Day Activities. Diagnosis and Correction Program for Children with Cerebral Palsy, Autism and FAS.* Conference Materials. How to Assist Handicapped Individuals Successfully? (Pl.). Warsaw-Sandomirz, Poland. 26.

Masgutova S. K., Akhmatova N. K., (1998). *Integration of Dynamic and Postural Reflexes into the Whole Body Movement System.* Ascent. (Russ.). Moscow, Russia.

Masgutova S. K., Akhmatova N. K. (2004, 2007). *Integration of Dynamic and Postural Reflexes into the Whole Body Movement System.* Scientific edition: Prof. N. Akhmatova. MINK. (Pl.). Warsaw, Poland.

Masgutova S. K., Akhmatova N. K., Kiselevsky M. V. (2008). Clinic-Immunological Assessment of Effect of the Therapy Program of Neuro-sensory-motor Integration of Reflex Patterns at Chronic Inflammatory Diseases of Respiratory System. *Russian Immunology Journal,* 2008, 2(11),#4. (Russ.). 454–463.

Masgutova S., Curlee P. (2004, 2007). *You are a Winner. Trauma Recovery. A New Choice Through Natural Developmental Movements.* Revised Edition. 1st World Publishing. Iowa, USA. 105.

Masgutova S., Kowal J. (2005). *NeuroKinesiology Tactile Therapy by Dr. S. Masgutova*. International Conference Materials: Modern Methods of Stimulation of Motor and Language Development." International Kinesiology Rehabilitation Camp for Children with Challenges of Dr. S. Masgutova Institute, MINK. (Pl.). Warsaw, Poland. 96-107.

Masgutova S. K., Kowal J., Mazur G., Masgutov D. P., (2005). *NeuroKinesiology Tactile Therapy™ by Dr S. Masgutova*, Scientific Edition: Dr. A. Regner, J. Szymczak. MINK. (Pl.). Warszawa, Poland. 133.

Masgutova S. K., Masgutov D. P. (2002). *My Inner Child: Integration of Lifelong Reflexes into Movement Development.* MINK. (Pl.). Warsaw, Poland. 140.

Masgutova S. K., Masgutov D.P. (2004). *Dr. Svetlana Masgutova Method of Facial Reflex Integration. Techniques for Facilitation of Motor and Speech Development.* 2003. MINK. (Pl.). Poland. 85.

Masgutova S. K., Regner A. (2007). *Language Development Using Senso-Motor Integration Approach.* Scientific edition: Prof. T. Galkowski, Dr. B. Dolyk. MISM, Continulo. (Pl.), Warsaw, Poland. 196.

Middleton, Frank A. and Strick P.L. (1994). Anatomical Evidence for Cerebral and Basal Ganglia Involvement in Higher Cognitive Function. *Science,* October 21, 1994 (vol. 266). USA.

Nelson K. B., Elleberg J. H. (1979). Neonatal Signs as Predictors of Cerebral Palsy. *Pediatrics,* 64-68, 225-231.

Neurologia Dziecięca (2000) *Child Neurology*. Wydawnictwo Medyczne - Urban & Partner. (Pl.). Wrocław, Poland.

Nikitin B. P., Nikitina L. A. (2004). *We and Our Children.* Medicina. (Russ.). Moscow, Russia.

Ornstien R. E., Sobel D. (1987). *The Healing Brain. Breakthrough Discoveries about How the Brain Keeps Us Healthy.* New York. Simon and Schuster. USA.

Pavlov I. P. (1927). *Conditioned Reflexes: An Investigation of the Physiological Activity of the Cerebral Cortex.* Translated and edited by G.V. Anrep. London: Oxford University. Press. U.K.

Pavlov I. P. (1960). *Conditioned Reflexes: An Investigation of the Physiological Activity of the Cerebral Cortex.* (Anrep G.V., D. Sc. Trans., 1960). New York. Dover Publications Inc. USA.

Penfield W., Rasmussen T. (1950). *The Cerebral Cortex of Man: A Clinical Study of localization of Function.* Macmillian, New York, USA.

Piaget, J.(1954). *The Construction of Reality in the Child.* (M. Cook, Trans.). New York: Ballantine Books. New York, USA.

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BIBLIOGRAPHY

Piaget, J., Inhelder, B., Sinclair-de Zwart, H. (1973). *Memory and Intelligence.* A Pomerans (trans.).New York: Basic Books. New York, USA.

Piaget, J. (1976). *The Grasp of Consciousness: Action and Concept in the Young Child.* Cambridge. M.A. Harvard. USA.

Polakov G. I. (1965). *About the Principles of Functional Brain Organization.* (Russ.). Moscow, Russia.

Provine R. R., Westerman J. A. (1979). Crossing the Midline: Limits of Early Eye-Hand Behavior. *Child Development,* 50. 437-441.

Rutter M. (1980). Attachment and the Development of social relationships. In: Rutter, M (ed). *Scientific Foundations of Developmental psychiatry.* Heinmann Medical. London. U.K.

Sadowska L. (1998). Neurologiczne uzasadnienie wczesnej kinezjologicznej diagnostyki i rehabilitacji metodą Vojty. [*Neurological Basis of Early Kinesiological Diagnosis and Rehabilitation Using Vojta Method]*. Medycyna Manualna, 2. (Pl.). Wrocław, Poland.

Setchenov I. M. (1960). *Selected physiological and psychological works* (S. Belsky, Trans., K. Koschtoyants, Ed.). Foreign Languages Publishing House. (Original work published 1935). (Russ.). Moscow, Russia.

Setchenov I. M. (1961). *Reflexes of the Brain.* (Russ.). Moscow. Russia.

Semionov E. V. (1998). *The Atlas of Human Anatomy.* SEV-Press. (Russ.). Moscow, Russia.

Semionov E. V. (2000). *Physiology and Anatomy.* Russian State Medical University. (Russ.). Moscow, Russia.

Semionova K. A. (1999). *Rehabilitation Treatment of the CP Patients in Residual State.* (Russ.). Moscow, Russia.

Simernickaja E. G. (1985). *The Human Brain and Mental Processes in Ontogenesis.* (Russ.). Moscow, Russia.

Sinelnikov P. D. (1942). *Atlas of Human Anatomy.* Book – V. Medgiz, (Russ.). Moscow-Leningrad, Russia.

Sherrington C. (1947). *The Integrative Action of the Nervous System.* Cambridge University. Press, London, U.K.

Springer S. P. Deutsch G. (1981). *Left Brain, Right Brain. Perspective from Cognitive Neuroscience.* Fifth Edition. Worth Publishers, New York and Basingstoke. USA.

Touch for Health. (1998). *A practical guide to health improvement based on the Book Touch for Health by John Thie,* D.C. Books – 1-4. International Kinesiology College of Zurich. Zurich, Zwitzerland.

Turlough FitzGerald M. J, Gruener G., Mtui E. (2007). *Clinical Neuroanatomy and Neuroscience.* Fifth Edition. Elsevier Saunders.

Ukhtomsky A. A. (1950-1952). *The Study of Dominance* // Collected works in 6 books. Book 6. (Russ.). Leningrad, Russia.

Van DeGraaff K. M. (1984). *Human Anatomy.* Dubuque, I.A., Wm. C. Brawn Publishers.

Vygotsky L. S. (1986). *The Child Psychology. The Problems of Child Development.* In 6 Books. Book – 4. Pedagogika. (Russ.). Moscow, Russian.

Wolf-Heidegger's Atlas of Human Anatomy. Systemic Anatomy, Body Wall, Upper and Lower Lims. Karger. London, U.K., New York, USA. 5.

William J. L. (2001). *Human Embryology.* Third Edition. Philadelphia. USA.

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